

SOLVED PRACTICE WORKBOOK

General Science: Biology and Physiology - Solved Practice Workbook

UPSC complete learning session | Foundation to advanced | Concepts, evidence, practice and answer writing

General Science: Biology and Physiology	
REFRESHED TEACHING NAVIGATION — BASIC/CORE FIRST	
01 FOUNDATION — CELL TYPES ORGANELLES BIOMOLECULES AND ENZYMES Read Cell types organelles biomolecules and enzymes as a sequence:	02 FOUNDATION — MEMBRANES DIFFUSION OSMOSIS ACTIVE TRANSPORT AND TONICITY Read Membranes diffusion osmosis active transport and tonicity as a
03 FOUNDATION — MITOSIS MEIOSIS CHROMOSOMES AND BIOLOGICAL VARIATION Read Mitosis meiosis chromosomes and biological variation as a	04 CORE — DNA RNA REPLICATION TRANSCRIPTION TRANSLATION AND Read DNA RNA replication transcription translation and microRNA as
05 CORE — MENDELIAN INHERITANCE MUTATION SEX DETERMINATION AND SEX LINKAGE Mechanism chain: Inheritance mutation and sex linkage Qualified use:	06 CORE — DIGESTION ABSORPTION NUTRIENTS VITAMINS AND DEFICIENCY LOGIC Read Digestion absorption nutrients vitamins and deficiency logic as a
07 CORE — BLOOD CIRCULATION HEART GAS TRANSPORT AND CLOTTING Read Blood circulation heart gas transport and clotting as a sequence:	08 CORE — RESPIRATION EXCRETION NEPHRON AND HOMEOSTASIS Mechanism chain: Kidney nephron and homeostasis Qualified use:
09 CORE — NERVOUS CONTROL REFLEX ARCS ENDOCRINE GLANDS AND FEEDBACK Read Nervous control reflex arcs endocrine glands and feedback as a	10 CORE — INNATE ADAPTIVE HUMORAL CELLULAR AND TOLERANCE PATHWAYS Mechanism chain: Hormones metabolism and internal stability
11 CORE — PATHOGENS TRANSMISSION VECTORS ZOONOSSES AND DISEASE CLASSES Read Pathogens transmission vectors zoonoses and disease classes as	12 CORE — ANTIBIOTICS ANTIVIRALS ANTIMICROBIAL RESISTANCE AND BIOFILMS Read Antibiotics antivirals antimicrobial resistance and biofilms as a
13 CORE SYNTHESIS — MICROBIOME PROBIOTICS EXTREMOPHILES AND MICROBIAL CONTEXT Probiotics are live microorganisms intended to confer a benefit in an	14 CORE SYNTHESIS — PLANT PHYSIOLOGY PHOTOSYNTHESIS TRANSPORT HORMONES Mechanism chain: Plant physiology transport and stress response -
15 CORE SYNTHESIS — ROUTED ORGANISMS CELLULOSE INSTITUTIONS PYQs AND EVIDENCE Read Routed organisms cellulose institutions PYQs and evidence	

Original deterministic study visual; labels are explanatory, not textual quotations.

Source order: repository knowledge -> official current linkage -> clearly marked analysis. No fabricated PYQs or statistics.

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BASIC MCQS / REMEDIATION

Q1. Which statement correctly identifies Cell organisation and organelle boundary?

A. A cell is the basic structural and functional unit of life; prokaryotes lack a membrane-bound nucleus and membrane-bound organelles, whereas eukaryotes possess them. The nucleus stores most genetic material, ribosomes synthesize protein, mitochondria support aerobic ATP generation, the endoplasmic reticulum and Golgi system process and move products, lysosomes contain hydrolytic enzymes, and plant cells additionally possess a cellulose wall, chloroplasts and a large vacuole. B. Mitosis is one division producing two genetically similar daughter cells for growth and repair, while meiosis consists of two divisions producing four haploid products for sexual reproduction; crossing over and independent assortment generate variation. Neither process should be described as gene editing, and chromosome number must be tracked relative to the starting cell. C. Carbohydrates commonly supply or store energy and provide structures such as cellulose, lipids store energy and form membranes, proteins perform structural, transport, receptor and catalytic roles, and nucleic acids store or express information. Enzymes are biological catalysts whose activity depends on substrate fit and conditions such as temperature and pH; a biomolecule class does not establish one universal function. D. Diffusion moves particles down a concentration gradient, osmosis is net water movement across a selectively permeable membrane, and active transport uses cellular energy to move substances against a gradient. Tonicity describes the effect of an external solution on cell water balance, so swelling, plasmolysis and normal volume require the membrane and concentration context.

Answer: A. Explanation: A cell is the basic structural and functional unit of life; prokaryotes lack a membrane-bound nucleus and membrane-bound organelles, whereas eukaryotes possess them. The nucleus stores most genetic material, ribosomes synthesize protein, mitochondria support aerobic ATP generation, the endoplasmic reticulum and Golgi system process and move products, lysosomes contain hydrolytic enzymes, and plant cells additionally possess a cellulose wall, chloroplasts and a large vacuole. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q2. Which option preserves the technical boundary of Cell organisation and organelle boundary?

A. Mitosis is one division producing two genetically similar daughter cells for growth and repair, while meiosis consists of two divisions producing four haploid products for sexual reproduction; crossing over and independent assortment generate variation. Neither process should be described as gene editing, and chromosome number must be tracked relative to the starting cell. B. A cell is the basic structural and functional unit of life; prokaryotes lack a membrane-bound nucleus and membrane-bound organelles, whereas eukaryotes possess them. The nucleus stores most genetic material, ribosomes synthesize protein, mitochondria support aerobic ATP generation, the endoplasmic reticulum and Golgi system process and move products, lysosomes contain hydrolytic enzymes, and plant cells additionally possess a cellulose wall, chloroplasts and a large vacuole. C. DNA replication copies hereditary material, transcription produces RNA from a DNA template, and translation uses messenger RNA, transfer RNA and ribosomes to build protein from codons. RNA also has regulatory roles: microRNA can bind complementary messenger RNA and reduce protein production or promote RNA degradation, so DNA storage, RNA regulation and protein function are distinct stages. D. Diffusion moves particles down a concentration gradient, osmosis is net water movement across a selectively permeable membrane, and active transport uses cellular energy to move substances against a gradient. Tonicity describes the effect of an external solution on cell water balance, so swelling, plasmolysis and normal volume require the membrane and concentration context.

Answer: B. Explanation: A cell is the basic structural and functional unit of life; prokaryotes lack a membrane-bound nucleus and membrane-bound organelles, whereas eukaryotes possess them. The nucleus stores most genetic material, ribosomes synthesize protein, mitochondria support aerobic ATP generation, the endoplasmic reticulum and Golgi system process and move products, lysosomes contain hydrolytic enzymes, and plant cells additionally possess a cellulose wall, chloroplasts and a large vacuole. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q3. Which statement uses Cell organisation and organelle boundary without changing its institution, unit or status?

A. Mitosis is one division producing two genetically similar daughter cells for growth and repair, while meiosis consists of two divisions producing four haploid products for sexual reproduction; crossing over and independent assortment generate variation. Neither process should be described as gene editing, and chromosome number must be tracked relative to the starting cell. B. A gene is a functional hereditary unit on DNA; alleles are alternative forms, segregation separates allele pairs into gametes and independent assortment applies to appropriately unlinked genes. Mutations may be point, frameshift or chromosomal changes and supply variation without guaranteeing benefit; human XX-XY sex determination and X-linked recessive inheritance explain why haemophilia or red-green colour blindness can show sex-biased patterns. C. A cell is the basic structural and functional unit of life; prokaryotes lack a membrane-bound nucleus and membrane-bound organelles, whereas eukaryotes possess them. The nucleus stores most genetic material, ribosomes synthesize protein, mitochondria support aerobic ATP generation, the endoplasmic reticulum and Golgi system process and move products, lysosomes contain hydrolytic enzymes, and plant cells additionally possess a cellulose wall, chloroplasts and a large vacuole. D. DNA replication copies hereditary material, transcription produces RNA from a DNA template, and translation uses messenger RNA, transfer RNA and ribosomes to build protein from codons. RNA also has regulatory roles: microRNA can bind complementary messenger RNA and reduce protein production or promote RNA degradation, so DNA storage, RNA regulation and protein function are distinct stages.

Answer: C. Explanation: A cell is the basic structural and functional unit of life; prokaryotes lack a membrane-bound nucleus and membrane-bound organelles, whereas eukaryotes possess them. The nucleus stores most genetic material, ribosomes synthesize protein, mitochondria support aerobic ATP generation, the endoplasmic reticulum and Golgi system process and move products, lysosomes contain hydrolytic enzymes, and plant cells additionally possess a cellulose wall, chloroplasts and a large vacuole. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q4. Which option avoids the standard UPSC close-option trap about Cell organisation and organelle boundary?

A. A gene is a functional hereditary unit on DNA; alleles are alternative forms, segregation separates allele pairs into gametes and independent assortment applies to appropriately unlinked genes. Mutations may be point, frameshift or chromosomal changes and supply variation without guaranteeing benefit; human XX-XY sex determination and X-linked recessive inheritance explain why haemophilia or red-green colour blindness can show sex-biased patterns. B. DNA replication copies hereditary material, transcription produces RNA from a DNA template, and translation uses messenger RNA, transfer RNA and ribosomes to build protein from codons. RNA also has regulatory roles: microRNA can bind complementary messenger RNA and reduce protein production or promote RNA degradation, so DNA storage, RNA regulation and protein function are distinct stages. C. Digestion converts food into absorbable molecules: salivary amylase begins starch digestion, pepsin works in acidic stomach conditions, bile emulsifies fat without acting as an enzyme, pancreatic and intestinal enzymes continue digestion, and the small intestine is the principal absorption site. Vitamins, minerals, protein and energy have different roles; deficiency, absorption failure and excess are separate nutritional problems. D. A cell is the basic structural and functional unit of life; prokaryotes lack a membrane-bound nucleus and membrane-bound organelles, whereas eukaryotes possess them. The nucleus stores most genetic material, ribosomes synthesize protein, mitochondria support aerobic ATP generation, the endoplasmic reticulum and Golgi system process and move products, lysosomes contain hydrolytic enzymes, and plant cells additionally possess a cellulose wall, chloroplasts and a large vacuole.

Answer: D. Explanation: A cell is the basic structural and functional unit of life; prokaryotes lack a membrane-bound nucleus and membrane-bound organelles, whereas eukaryotes possess them. The nucleus stores most genetic material, ribosomes synthesize protein, mitochondria support aerobic ATP generation, the endoplasmic reticulum and Golgi system process and move products, lysosomes contain hydrolytic enzymes, and plant cells additionally possess a cellulose wall, chloroplasts and a large vacuole. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q5. Which statement correctly identifies Biomolecules and enzyme specificity?

A. Carbohydrates commonly supply or store energy and provide structures such as cellulose, lipids store energy and form membranes, proteins perform structural, transport, receptor and catalytic roles, and nucleic acids store or express information. Enzymes are biological catalysts whose activity depends on substrate fit and conditions such as temperature and pH; a biomolecule class does not establish one universal function. B. Mitosis is one division producing two genetically similar daughter cells for growth and repair, while meiosis consists of two divisions producing four haploid products for sexual reproduction; crossing over and independent assortment generate variation. Neither process should be described as gene editing, and chromosome number must be tracked relative to the starting cell. C. Diffusion moves particles down a concentration gradient, osmosis is net water movement across a selectively permeable membrane, and active transport uses cellular energy to move substances against a gradient. Tonicity describes the effect of an external solution on cell water balance, so swelling, plasmolysis and normal volume require the membrane and concentration context. D. DNA replication copies hereditary material, transcription produces RNA from a DNA template, and translation uses messenger RNA, transfer RNA and ribosomes to build protein from codons. RNA also has regulatory roles: microRNA can bind complementary messenger RNA and reduce protein production or promote RNA degradation, so DNA storage, RNA regulation and protein function are distinct stages.

Answer: A. Explanation: Carbohydrates commonly supply or store energy and provide structures such as cellulose, lipids store energy and form membranes, proteins perform structural, transport, receptor and catalytic roles, and nucleic acids store or express information. Enzymes are biological catalysts whose activity depends on substrate fit and conditions such as temperature and pH; a biomolecule class does not establish one universal function. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q6. Which option preserves the technical boundary of Biomolecules and enzyme specificity?

A. Mitosis is one division producing two genetically similar daughter cells for growth and repair, while meiosis consists of two divisions producing four haploid products for sexual reproduction; crossing over and independent assortment generate variation. Neither process should be described as gene editing, and chromosome number must be tracked relative to the starting cell. B. Carbohydrates commonly supply or store energy and provide structures such as cellulose, lipids store energy and form membranes, proteins perform structural, transport, receptor and catalytic roles, and nucleic acids store or express information. Enzymes are biological catalysts whose activity depends on substrate fit and conditions such as temperature and pH; a biomolecule class does not establish one universal function. C. DNA replication copies hereditary material, transcription produces RNA from a DNA template, and translation uses messenger RNA, transfer RNA and ribosomes to build protein from codons. RNA also has regulatory roles: microRNA can bind complementary messenger RNA and reduce protein production or promote RNA degradation, so DNA storage, RNA regulation and protein function are distinct stages. D. A gene is a functional hereditary unit on DNA; alleles are alternative forms, segregation separates allele pairs into gametes and independent assortment applies to appropriately unlinked genes. Mutations may be point, frameshift or chromosomal changes and supply variation without guaranteeing benefit; human XX-XY sex determination and X-linked recessive inheritance explain why haemophilia or red-green colour blindness can show sex-biased patterns.

Answer: B. Explanation: Carbohydrates commonly supply or store energy and provide structures such as cellulose, lipids store energy and form membranes, proteins perform structural, transport, receptor and catalytic roles, and nucleic acids store or express information. Enzymes are biological catalysts whose activity depends on substrate fit and conditions such as temperature and pH; a biomolecule class does not establish one universal function. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q7. Which statement uses Biomolecules and enzyme specificity without changing its institution, unit or status?

A. DNA replication copies hereditary material, transcription produces RNA from a DNA template, and translation uses messenger RNA, transfer RNA and ribosomes to build protein from codons. RNA also has regulatory roles: microRNA can bind complementary messenger RNA and reduce protein production or promote RNA degradation, so DNA storage, RNA regulation and protein function are distinct stages. B. A gene is a functional hereditary unit on DNA; alleles are alternative forms, segregation separates allele pairs into gametes and independent assortment applies to appropriately unlinked genes. Mutations may be point, frameshift or chromosomal changes and supply variation without guaranteeing benefit; human XX-XY sex determination and X-linked recessive inheritance explain why haemophilia or red-green colour blindness can show sex-biased patterns. C. Carbohydrates commonly supply or store energy and provide structures such as cellulose, lipids store energy and form membranes, proteins perform structural, transport, receptor and catalytic roles, and nucleic acids store or express information. Enzymes are biological catalysts whose activity depends on substrate fit and conditions such as temperature and pH; a biomolecule class does not establish one universal function. D. Digestion converts food into absorbable molecules: salivary amylase begins starch digestion, pepsin works in acidic stomach conditions, bile emulsifies fat without acting as an enzyme, pancreatic and intestinal enzymes continue digestion, and the small intestine is the principal absorption site. Vitamins, minerals, protein and energy have different roles; deficiency, absorption failure and excess are separate nutritional problems.

Answer: C. Explanation: Carbohydrates commonly supply or store energy and provide structures such as cellulose, lipids store energy and form membranes, proteins perform structural, transport, receptor and catalytic roles, and nucleic acids store or express information. Enzymes are biological catalysts whose activity depends on substrate fit and conditions such as temperature and pH; a biomolecule class does not establish one universal function. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q8. Which option avoids the standard UPSC close-option trap about Biomolecules and enzyme specificity?

A. A gene is a functional hereditary unit on DNA; alleles are alternative forms, segregation separates allele pairs into gametes and independent assortment applies to appropriately unlinked genes. Mutations may be point, frameshift or chromosomal changes and supply variation without guaranteeing benefit; human XX-XY sex determination and X-linked recessive inheritance explain why haemophilia or red-green colour blindness can show sex-biased patterns. B. Blood contains plasma, red cells carrying haemoglobin, immune white cells and platelets involved in clotting; the four-chambered human heart supports pulmonary and systemic circuits. Arteries carry blood away from the heart and veins return it, so oxygenation is not the defining rule; ABO and Rh compatibility, gas transport and clotting must be treated as different functions. C. Digestion converts food into absorbable molecules: salivary amylase begins starch digestion, pepsin works in acidic stomach conditions, bile emulsifies fat without acting as an enzyme, pancreatic and intestinal enzymes continue digestion, and the small intestine is the principal absorption site. Vitamins, minerals, protein and energy have different roles; deficiency, absorption failure and excess are separate nutritional problems. D. Carbohydrates commonly supply or store energy and provide structures such as cellulose, lipids store energy and form membranes, proteins perform structural, transport, receptor and catalytic roles, and nucleic acids store or express information. Enzymes are biological catalysts whose activity depends on substrate fit and conditions such as temperature and pH; a biomolecule class does not establish one universal function.

Answer: D. Explanation: Carbohydrates commonly supply or store energy and provide structures such as cellulose, lipids store energy and form membranes, proteins perform structural, transport, receptor and catalytic roles, and nucleic acids store or express information. Enzymes are biological catalysts whose activity depends on substrate fit and conditions such as temperature and pH; a biomolecule class does not establish one universal function. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q9. Which statement correctly identifies Membrane transport and water balance?

A. Diffusion moves particles down a concentration gradient, osmosis is net water movement across a selectively permeable membrane, and active transport uses cellular energy to move substances against a gradient. Tonicity describes the effect of an external solution on cell water balance, so swelling, plasmolysis and normal volume require the membrane and concentration context. B. Mitosis is one division producing two genetically similar daughter cells for growth and repair, while meiosis consists of two divisions producing four haploid products for sexual reproduction; crossing over and independent assortment generate variation. Neither process should be described as gene editing, and chromosome number must be tracked relative to the starting cell. C. A gene is a functional hereditary unit on DNA; alleles are alternative forms, segregation separates allele pairs into gametes and independent assortment applies to appropriately unlinked genes. Mutations may be point, frameshift or chromosomal changes and supply variation without guaranteeing benefit; human XX-XY sex determination and X-linked recessive inheritance explain why haemophilia or red-green colour blindness can show sex-biased patterns. D. DNA replication copies hereditary material, transcription produces RNA from a DNA template, and translation uses messenger RNA, transfer RNA and ribosomes to build protein from codons. RNA also has regulatory roles: microRNA can bind complementary messenger RNA and reduce protein production or promote RNA degradation, so DNA storage, RNA regulation and protein function are distinct stages.

Answer: A. Explanation: Diffusion moves particles down a concentration gradient, osmosis is net water movement across a selectively permeable membrane, and active transport uses cellular energy to move substances against a gradient. Tonicity describes the effect of an external solution on cell water balance, so swelling, plasmolysis and normal volume require the membrane and concentration context. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q10. Which option preserves the technical boundary of Membrane transport and water balance?

A. A gene is a functional hereditary unit on DNA; alleles are alternative forms, segregation separates allele pairs into gametes and independent assortment applies to appropriately unlinked genes. Mutations may be point, frameshift or chromosomal changes and supply variation without guaranteeing benefit; human XX-XY sex determination and X-linked recessive inheritance explain why haemophilia or red-green colour blindness can show sex-biased patterns. B. Diffusion moves particles down a concentration gradient, osmosis is net water movement across a selectively permeable membrane, and active transport uses cellular energy to move substances against a gradient. Tonicity describes the effect of an external solution on cell water balance, so swelling, plasmolysis and normal volume require the membrane and concentration context. C. Digestion converts food into absorbable molecules: salivary amylase begins starch digestion, pepsin works in acidic stomach conditions, bile emulsifies fat without acting as an enzyme, pancreatic and intestinal enzymes continue digestion, and the small intestine is the principal absorption site. Vitamins, minerals, protein and energy have different roles; deficiency, absorption failure and excess are separate nutritional problems. D. DNA replication copies hereditary material, transcription produces RNA from a DNA template, and translation uses messenger RNA, transfer RNA and ribosomes to build protein from codons. RNA also has regulatory roles: microRNA can bind complementary messenger RNA and reduce protein production or promote RNA degradation, so DNA storage, RNA regulation and protein function are distinct stages.

Answer: B. Explanation: Diffusion moves particles down a concentration gradient, osmosis is net water movement across a selectively permeable membrane, and active transport uses cellular energy to move substances against a gradient. Tonicity describes the effect of an external solution on cell water balance, so swelling, plasmolysis and normal volume require the membrane and concentration context. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q11. Which statement uses Membrane transport and water balance without changing its institution, unit or status?

A. A gene is a functional hereditary unit on DNA; alleles are alternative forms, segregation separates allele pairs into gametes and independent assortment applies to appropriately unlinked genes. Mutations may be point, frameshift or chromosomal changes and supply variation without guaranteeing benefit; human XX-XY sex determination and X-linked recessive inheritance explain why haemophilia or red-green colour blindness can show sex-biased patterns. B. Blood contains plasma, red cells carrying haemoglobin, immune white cells and platelets involved in clotting; the four-chambered human heart supports pulmonary and systemic circuits. Arteries carry blood away from the heart and veins return it, so oxygenation is not the defining rule; ABO and Rh compatibility, gas transport and clotting must be treated as different functions. C. Diffusion moves particles down a concentration gradient, osmosis is net water movement across a selectively permeable membrane, and active transport uses cellular energy to move substances against a gradient. Tonicity describes the effect of an external solution on cell water balance, so swelling, plasmolysis and normal volume require the membrane and concentration context. D. Digestion converts food into absorbable molecules: salivary amylase begins starch digestion, pepsin works in acidic stomach conditions, bile emulsifies fat without acting as an enzyme, pancreatic and intestinal enzymes continue digestion, and the small intestine is the principal absorption site. Vitamins, minerals, protein and energy have different roles; deficiency, absorption failure and excess are separate nutritional problems.

Answer: C. Explanation: Diffusion moves particles down a concentration gradient, osmosis is net water movement across a selectively permeable membrane, and active transport uses cellular energy to move substances against a gradient. Tonicity describes the effect of an external solution on cell water balance, so swelling, plasmolysis and normal volume require the membrane and concentration context. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q12. Which option avoids the standard UPSC close-option trap about Membrane transport and water balance?

A. Ventilation moves air, alveolar gas exchange moves oxygen and carbon dioxide across thin surfaces, blood transports gases, and cellular respiration releases usable energy from nutrients. Haemoglobin binds oxygen while carbon monoxide can disrupt transport through stronger binding; breathing, gas exchange and ATP-producing respiration are linked but not synonymous. B. Digestion converts food into absorbable molecules: salivary amylase begins starch digestion, pepsin works in acidic stomach conditions, bile emulsifies fat without acting as an enzyme, pancreatic and intestinal enzymes continue digestion, and the small intestine is the principal absorption site. Vitamins, minerals, protein and energy have different roles; deficiency, absorption failure and excess are separate nutritional problems. C. Blood contains plasma, red cells carrying haemoglobin, immune white cells and platelets involved in clotting; the four-chambered human heart supports pulmonary and systemic circuits. Arteries carry blood away from the heart and veins return it, so oxygenation is not the defining rule; ABO and Rh compatibility, gas transport and clotting must be treated as different functions. D. Diffusion moves particles down a concentration gradient, osmosis is net water movement across a selectively permeable membrane, and active transport uses cellular energy to move substances against a gradient. Tonicity describes the effect of an external solution on cell water balance, so swelling, plasmolysis and normal volume require the membrane and concentration context.

Answer: D. Explanation: Diffusion moves particles down a concentration gradient, osmosis is net water movement across a selectively permeable membrane, and active transport uses cellular energy to move substances against a gradient. Tonicity describes the effect of an external solution on cell water balance, so swelling, plasmolysis and normal volume require the membrane and concentration context. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q13. Which statement correctly identifies Mitosis meiosis and variation?

A. Mitosis is one division producing two genetically similar daughter cells for growth and repair, while meiosis consists of two divisions producing four haploid products for sexual reproduction; crossing over and independent assortment generate variation. Neither process should be described as gene editing, and chromosome number must be tracked relative to the starting cell. B. DNA replication copies hereditary material, transcription produces RNA from a DNA template, and translation uses messenger RNA, transfer RNA and ribosomes to build protein from codons. RNA also has regulatory roles: microRNA can bind complementary messenger RNA and reduce protein production or promote RNA degradation, so DNA storage, RNA regulation and protein function are distinct stages. C. A gene is a functional hereditary unit on DNA; alleles are alternative forms, segregation separates allele pairs into gametes and independent assortment applies to appropriately unlinked genes. Mutations may be point, frameshift or chromosomal changes and supply variation without guaranteeing benefit; human XX-XY sex determination and X-linked recessive inheritance explain why haemophilia or red-green colour blindness can show sex-biased patterns. D. Digestion converts food into absorbable molecules: salivary amylase begins starch digestion, pepsin works in acidic stomach conditions, bile emulsifies fat without acting as an enzyme, pancreatic and intestinal enzymes continue digestion, and the small intestine is the principal absorption site. Vitamins, minerals, protein and energy have different roles; deficiency, absorption failure and excess are separate nutritional problems.

Answer: A. Explanation: Mitosis is one division producing two genetically similar daughter cells for growth and repair, while meiosis consists of two divisions producing four haploid products for sexual reproduction; crossing over and independent assortment generate variation. Neither process should be described as gene editing, and chromosome number must be tracked relative to the starting cell. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q14. Which option preserves the technical boundary of Mitosis meiosis and variation?

A. Blood contains plasma, red cells carrying haemoglobin, immune white cells and platelets involved in clotting; the four-chambered human heart supports pulmonary and systemic circuits. Arteries carry blood away from the heart and veins return it, so oxygenation is not the defining rule; ABO and Rh compatibility, gas transport and clotting must be treated as different functions. B. Mitosis is one division producing two genetically similar daughter cells for growth and repair, while meiosis consists of two divisions producing four haploid products for sexual reproduction; crossing over and independent assortment generate variation. Neither process should be described as gene editing, and chromosome number must be tracked relative to the starting cell. C. A gene is a functional hereditary unit on DNA; alleles are alternative forms, segregation separates allele pairs into gametes and independent assortment applies to appropriately unlinked genes. Mutations may be point, frameshift or chromosomal changes and supply variation without guaranteeing benefit; human XX-XY sex determination and X-linked recessive inheritance explain why haemophilia or red-green colour blindness can show sex-biased patterns. D. Digestion converts food into absorbable molecules: salivary amylase begins starch digestion, pepsin works in acidic stomach conditions, bile emulsifies fat without acting as an enzyme, pancreatic and intestinal enzymes continue digestion, and the small intestine is the principal absorption site. Vitamins, minerals, protein and energy have different roles; deficiency, absorption failure and excess are separate nutritional problems.

Answer: B. Explanation: Mitosis is one division producing two genetically similar daughter cells for growth and repair, while meiosis consists of two divisions producing four haploid products for sexual reproduction; crossing over and independent assortment generate variation. Neither process should be described as gene editing, and chromosome number must be tracked relative to the starting cell. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q15. Which statement uses Mitosis meiosis and variation without changing its institution, unit or status?

A. Ventilation moves air, alveolar gas exchange moves oxygen and carbon dioxide across thin surfaces, blood transports gases, and cellular respiration releases usable energy from nutrients. Haemoglobin binds oxygen while carbon monoxide can disrupt transport through stronger binding; breathing, gas exchange and ATP-producing respiration are linked but not synonymous. B. Digestion converts food into absorbable molecules: salivary amylase begins starch digestion, pepsin works in acidic stomach conditions, bile emulsifies fat without acting as an enzyme, pancreatic and intestinal enzymes continue digestion, and the small intestine is the principal absorption site. Vitamins, minerals, protein and energy have different roles; deficiency, absorption failure and excess are separate nutritional problems. C. Mitosis is one division producing two genetically similar daughter cells for growth and repair, while meiosis consists of two divisions producing four haploid products for sexual reproduction; crossing over and independent assortment generate variation. Neither process should be described as gene editing, and chromosome number must be tracked relative to the starting cell. D. Blood contains plasma, red cells carrying haemoglobin, immune white cells and platelets involved in clotting; the four-chambered human heart supports pulmonary and systemic circuits. Arteries carry blood away from the heart and veins return it, so oxygenation is not the defining rule; ABO and Rh compatibility, gas transport and clotting must be treated as different functions.

Answer: C. Explanation: Mitosis is one division producing two genetically similar daughter cells for growth and repair, while meiosis consists of two divisions producing four haploid products for sexual reproduction; crossing over and independent assortment generate variation. Neither process should be described as gene editing, and chromosome number must be tracked relative to the starting cell. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q16. Which option avoids the standard UPSC close-option trap about Mitosis meiosis and variation?

A. Blood contains plasma, red cells carrying haemoglobin, immune white cells and platelets involved in clotting; the four-chambered human heart supports pulmonary and systemic circuits. Arteries carry blood away from the heart and veins return it, so oxygenation is not the defining rule; ABO and Rh compatibility, gas transport and clotting must be treated as different functions. B. Ventilation moves air, alveolar gas exchange moves oxygen and carbon dioxide across thin surfaces, blood transports gases, and cellular respiration releases usable energy from nutrients. Haemoglobin binds oxygen while carbon monoxide can disrupt transport through stronger binding; breathing, gas exchange and ATP-producing respiration are linked but not synonymous. C. The nephron filters blood at the glomerulus and then selectively reabsorbs and secretes substances along its tubule to regulate water, electrolytes, acid-base balance and wastes. Dialysis can replace selected filtration functions when kidneys fail but does not reproduce every endocrine and regulatory role of healthy kidneys. D. Mitosis is one division producing two genetically similar daughter cells for growth and repair, while meiosis consists of two divisions producing four haploid products for sexual reproduction; crossing over and independent assortment generate variation. Neither process should be described as gene editing, and chromosome number must be tracked relative to the starting cell.

Answer: D. Explanation: Mitosis is one division producing two genetically similar daughter cells for growth and repair, while meiosis consists of two divisions producing four haploid products for sexual reproduction; crossing over and independent assortment generate variation. Neither process should be described as gene editing, and chromosome number must be tracked relative to the starting cell. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q17. Which statement correctly identifies DNA RNA gene expression and microRNA?

A. DNA replication copies hereditary material, transcription produces RNA from a DNA template, and translation uses messenger RNA, transfer RNA and ribosomes to build protein from codons. RNA also has regulatory roles: microRNA can bind complementary messenger RNA and reduce protein production or promote RNA degradation, so DNA storage, RNA regulation and protein function are distinct stages. B. Blood contains plasma, red cells carrying haemoglobin, immune white cells and platelets involved in clotting; the four-chambered human heart supports pulmonary and systemic circuits. Arteries carry blood away from the heart and veins return it, so oxygenation is not the defining rule; ABO and Rh compatibility, gas transport and clotting must be treated as different functions. C. A gene is a functional hereditary unit on DNA; alleles are alternative forms, segregation separates allele pairs into gametes and

independent assortment applies to appropriately unlinked genes. Mutations may be point, frameshift or chromosomal changes and supply variation without guaranteeing benefit; human XX-XY sex determination and X-linked recessive inheritance explain why haemophilia or red-green colour blindness can show sex-biased patterns. D. Digestion converts food into absorbable molecules: salivary amylase begins starch digestion, pepsin works in acidic stomach conditions, bile emulsifies fat without acting as an enzyme, pancreatic and intestinal enzymes continue digestion, and the small intestine is the principal absorption site. Vitamins, minerals, protein and energy have different roles; deficiency, absorption failure and excess are separate nutritional problems.

Answer: A. Explanation: DNA replication copies hereditary material, transcription produces RNA from a DNA template, and translation uses messenger RNA, transfer RNA and ribosomes to build protein from codons. RNA also has regulatory roles: microRNA can bind complementary messenger RNA and reduce protein production or promote RNA degradation, so DNA storage, RNA regulation and protein function are distinct stages. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q18. Which option preserves the technical boundary of DNA RNA gene expression and microRNA?

A. Ventilation moves air, alveolar gas exchange moves oxygen and carbon dioxide across thin surfaces, blood transports gases, and cellular respiration releases usable energy from nutrients. Haemoglobin binds oxygen while carbon monoxide can disrupt transport through stronger binding; breathing, gas exchange and ATP-producing respiration are linked but not synonymous. B. DNA replication copies hereditary material, transcription produces RNA from a DNA template, and translation uses messenger RNA, transfer RNA and ribosomes to build protein from codons. RNA also has regulatory roles: microRNA can bind complementary messenger RNA and reduce protein production or promote RNA degradation, so DNA storage, RNA regulation and protein function are distinct stages. C. Digestion converts food into absorbable molecules: salivary amylase begins starch digestion, pepsin works in acidic stomach conditions, bile emulsifies fat without acting as an enzyme, pancreatic and intestinal enzymes continue digestion, and the small intestine is the principal absorption site. Vitamins, minerals, protein and energy have different roles; deficiency, absorption failure and excess are separate nutritional problems. D. Blood contains plasma, red cells carrying haemoglobin, immune white cells and platelets involved in clotting; the four-chambered human heart supports pulmonary and systemic circuits. Arteries carry blood away from the heart and veins return it, so oxygenation is not the defining rule; ABO and Rh compatibility, gas transport and clotting must be treated as different functions.

Answer: B. Explanation: DNA replication copies hereditary material, transcription produces RNA from a DNA template, and translation uses messenger RNA, transfer RNA and ribosomes to build protein from codons. RNA also has regulatory roles: microRNA can bind complementary messenger RNA and reduce protein production or promote RNA degradation, so DNA storage, RNA regulation and protein function are distinct stages. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q19. Which statement uses DNA RNA gene expression and microRNA without changing its institution, unit or status?

A. Blood contains plasma, red cells carrying haemoglobin, immune white cells and platelets involved in clotting; the four-chambered human heart supports pulmonary and systemic circuits. Arteries carry blood away from the heart and veins return it, so oxygenation is not the defining rule; ABO and Rh compatibility, gas transport and clotting must be treated as different functions. B. The nephron filters blood at the glomerulus and then selectively reabsorbs and secretes substances along its tubule to regulate water, electrolytes, acid-base balance and wastes. Dialysis can replace selected filtration functions when kidneys fail but does not reproduce every endocrine and regulatory role of healthy kidneys. C. DNA replication copies hereditary material, transcription produces RNA from a DNA template, and translation uses messenger RNA, transfer RNA and ribosomes to build protein from codons. RNA also has regulatory roles: microRNA can bind complementary messenger RNA and reduce protein production or promote RNA degradation, so DNA storage, RNA regulation and protein function are distinct stages. D. Ventilation moves air, alveolar gas exchange moves oxygen and carbon dioxide across thin surfaces, blood transports gases, and cellular respiration releases usable energy from nutrients. Haemoglobin binds oxygen while carbon monoxide can disrupt transport through stronger binding; breathing, gas exchange and ATP-producing respiration are linked but not synonymous.

Answer: C. Explanation: DNA replication copies hereditary material, transcription produces RNA from a DNA template, and translation uses messenger RNA, transfer RNA and ribosomes to build protein from codons. RNA also has regulatory roles: microRNA can bind complementary messenger RNA and reduce protein production or promote RNA degradation, so DNA storage, RNA regulation and protein function are distinct stages. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q20. Which option avoids the standard UPSC close-option trap about DNA RNA gene expression and microRNA?

A. Neurons transmit rapid electrical signals and communicate chemically at synapses, reflex arcs permit fast automatic responses, and the brain and spinal cord coordinate wider activity. Endocrine glands release hormones into blood for target-dependent effects that are often slower and longer-lasting; neural and hormonal control interact through feedback rather than operating as isolated systems. B. Ventilation moves air, alveolar gas exchange moves oxygen and carbon dioxide across thin surfaces, blood transports gases, and cellular respiration releases usable energy from nutrients. Haemoglobin binds oxygen while carbon monoxide can disrupt transport through stronger binding; breathing, gas exchange and ATP-producing respiration are linked but not synonymous. C. The nephron filters blood at the glomerulus and then selectively reabsorbs and secretes substances along its tubule to regulate water, electrolytes, acid-base balance and wastes. Dialysis can replace selected filtration functions when kidneys fail but does not reproduce every endocrine and regulatory role of healthy kidneys. D. DNA replication copies hereditary material, transcription produces RNA from a DNA template, and translation uses messenger RNA, transfer RNA and ribosomes to build protein from codons. RNA also has regulatory roles: microRNA can bind complementary messenger RNA and reduce protein production or promote RNA degradation, so DNA storage, RNA regulation and protein function are distinct stages.

Answer: D. Explanation: DNA replication copies hereditary material, transcription produces RNA from a DNA template, and translation uses messenger RNA, transfer RNA and ribosomes to build protein from codons. RNA also has regulatory roles: microRNA can bind complementary messenger RNA and reduce protein production or promote RNA degradation, so DNA storage, RNA regulation and protein function are distinct stages. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q21. Which statement correctly identifies Inheritance mutation and sex linkage?

A. A gene is a functional hereditary unit on DNA; alleles are alternative forms, segregation separates allele pairs into gametes and independent assortment applies to appropriately unlinked genes. Mutations may be point, frameshift or chromosomal changes and supply variation without guaranteeing benefit; human XX-XY sex determination and X-linked recessive inheritance explain why haemophilia or red-green colour blindness can show sex-biased patterns. B. Blood contains plasma, red cells carrying haemoglobin, immune white cells and platelets involved in clotting; the four-chambered human heart supports pulmonary and systemic circuits. Arteries carry blood away from the heart and veins return it, so oxygenation is not the defining rule; ABO and Rh compatibility, gas transport and clotting must be treated as different functions. C. Ventilation moves air, alveolar gas exchange moves oxygen and carbon dioxide across thin surfaces, blood transports gases, and cellular respiration releases usable energy from nutrients. Haemoglobin binds oxygen while carbon monoxide can disrupt transport through stronger binding; breathing, gas exchange and ATP-producing respiration are linked but not synonymous. D. Digestion converts food into absorbable molecules: salivary amylase begins starch digestion, pepsin works in acidic stomach conditions, bile emulsifies fat without acting as an enzyme, pancreatic and intestinal enzymes continue digestion, and the small intestine is the principal absorption site. Vitamins, minerals, protein and energy have different roles; deficiency, absorption failure and excess are separate nutritional problems.

Answer: A. Explanation: A gene is a functional hereditary unit on DNA; alleles are alternative forms, segregation separates allele pairs into gametes and independent assortment applies to appropriately unlinked genes. Mutations may be point, frameshift or chromosomal changes and supply variation without guaranteeing benefit; human XX-XY sex determination and X-linked recessive inheritance explain why haemophilia or red-green colour blindness can show sex-biased patterns. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q22. Which option preserves the technical boundary of Inheritance mutation and sex linkage?

A. Blood contains plasma, red cells carrying haemoglobin, immune white cells and platelets involved in clotting; the four-chambered human heart supports pulmonary and systemic circuits. Arteries carry blood away from the heart and veins return it, so oxygenation is not the defining rule; ABO and Rh compatibility, gas transport and clotting must be treated as different functions. B. A gene is a functional hereditary unit on DNA; alleles are alternative forms, segregation separates allele pairs into gametes and independent assortment applies to appropriately unlinked genes. Mutations may be point, frameshift or chromosomal changes and supply variation without guaranteeing benefit; human XX-XY sex determination and X-linked recessive inheritance explain why haemophilia or red-green colour blindness can show sex-biased patterns. C. Ventilation moves air, alveolar gas exchange moves oxygen and carbon dioxide across thin surfaces, blood transports gases, and cellular respiration releases usable energy from nutrients. Haemoglobin binds oxygen while carbon monoxide can disrupt transport through stronger binding; breathing, gas exchange and ATP-producing respiration are linked but not synonymous. D. The nephron filters blood at the glomerulus and then selectively reabsorbs and secretes substances along its tubule to regulate water, electrolytes, acid-base balance and wastes. Dialysis can replace selected filtration functions when kidneys fail but does not reproduce every endocrine and regulatory role of healthy kidneys.

Answer: B. Explanation: A gene is a functional hereditary unit on DNA; alleles are alternative forms, segregation separates allele pairs into gametes and independent assortment applies to appropriately unlinked genes. Mutations may be point, frameshift or chromosomal changes and supply variation without guaranteeing benefit; human XX-XY sex determination and X-linked recessive inheritance explain why haemophilia or red-green colour blindness can show sex-biased patterns. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q23. Which statement uses Inheritance mutation and sex linkage without changing its institution, unit or status?

A. Neurons transmit rapid electrical signals and communicate chemically at synapses, reflex arcs permit fast automatic responses, and the brain and spinal cord coordinate wider activity. Endocrine glands release hormones into blood for target-dependent effects that are often slower and longer-lasting; neural and hormonal control interact through feedback rather than operating as isolated systems. B. The nephron filters blood at the glomerulus and then selectively reabsorbs and secretes substances along its tubule to regulate water, electrolytes, acid-base balance and wastes. Dialysis can replace selected filtration functions when kidneys fail but does not reproduce every endocrine and regulatory role of healthy kidneys. C. A gene is a functional hereditary unit on DNA; alleles are alternative forms, segregation separates allele pairs into gametes and independent assortment applies to appropriately unlinked genes. Mutations may be point, frameshift or chromosomal changes and supply variation without guaranteeing benefit; human XX-XY sex determination and X-linked recessive inheritance explain why haemophilia or red-green colour blindness can show sex-biased patterns. D. Ventilation moves air, alveolar gas exchange moves oxygen and carbon dioxide across thin surfaces, blood transports gases, and cellular respiration releases usable energy from nutrients. Haemoglobin binds oxygen while carbon monoxide can disrupt transport through stronger binding; breathing, gas exchange and ATP-producing respiration are linked but not synonymous.

Answer: C. Explanation: A gene is a functional hereditary unit on DNA; alleles are alternative forms, segregation separates allele pairs into gametes and independent assortment applies to appropriately unlinked genes. Mutations may be point, frameshift or chromosomal changes and supply variation without guaranteeing benefit; human XX-XY sex determination and X-linked recessive inheritance explain why haemophilia or red-green colour blindness can show sex-biased patterns. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q24. Which option avoids the standard UPSC close-option trap about Inheritance mutation and sex linkage?

A. The nephron filters blood at the glomerulus and then selectively reabsorbs and secretes substances along its tubule to regulate water, electrolytes, acid-base balance and wastes. Dialysis can replace selected filtration functions when kidneys fail but does not reproduce every endocrine and regulatory role of healthy kidneys. B. Neurons transmit rapid electrical signals and communicate chemically at synapses, reflex arcs permit fast automatic responses, and the brain and spinal cord coordinate wider activity. Endocrine glands release hormones into blood for target-dependent effects that are often slower and longer-lasting; neural and hormonal control interact through feedback rather than operating as isolated systems. C. Insulin lowers elevated blood glucose by promoting uptake and storage while glucagon supports its rise, thyroid hormones influence metabolism, parathyroid hormone helps regulate calcium, adrenal hormones mediate stress responses and pituitary signals coordinate several glands. Homeostasis is dynamic regulation around a range, not an unchanging value; hormone deficiency, resistance and gland failure are different mechanisms. D. A gene is a functional hereditary unit on DNA; alleles are alternative forms, segregation separates allele pairs into gametes and independent assortment applies to appropriately unlinked genes. Mutations may be point, frameshift or chromosomal changes and supply variation without guaranteeing benefit; human XX-XY sex determination and X-linked recessive inheritance explain why haemophilia or red-green colour blindness can show sex-biased patterns.

Answer: D. Explanation: A gene is a functional hereditary unit on DNA; alleles are alternative forms, segregation separates allele pairs into gametes and independent assortment applies to appropriately unlinked genes. Mutations may be point, frameshift or chromosomal changes and supply variation without guaranteeing benefit; human XX-XY sex determination and X-linked recessive inheritance explain why haemophilia or red-green colour blindness can show sex-biased patterns. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q25. Which statement correctly identifies Digestion absorption and nutrition?

A. Digestion converts food into absorbable molecules: salivary amylase begins starch digestion, pepsin works in acidic stomach conditions, bile emulsifies fat without acting as an enzyme, pancreatic and intestinal enzymes continue digestion, and the small intestine is the principal absorption site. Vitamins, minerals, protein and energy have different roles; deficiency, absorption failure and excess are separate nutritional problems. B. Blood contains plasma, red cells carrying haemoglobin, immune white cells and platelets involved in clotting; the four-chambered human heart supports pulmonary and systemic circuits. Arteries carry blood away from the heart and veins return it, so oxygenation is not the defining rule; ABO and Rh compatibility, gas transport and clotting must be treated as different functions. C. The nephron filters blood at the glomerulus and then selectively reabsorbs and secretes substances along its tubule to regulate water, electrolytes, acid-base balance and wastes. Dialysis can replace selected filtration functions when kidneys fail but does not reproduce every endocrine and regulatory role of healthy kidneys. D. Ventilation moves air, alveolar gas exchange moves oxygen and carbon dioxide across thin surfaces, blood transports gases, and cellular respiration releases usable energy from nutrients. Haemoglobin binds oxygen while carbon monoxide can disrupt transport through stronger binding; breathing, gas exchange and ATP-producing respiration are linked but not synonymous.

Answer: A. Explanation: Digestion converts food into absorbable molecules: salivary amylase begins starch digestion, pepsin works in acidic stomach conditions, bile emulsifies fat without acting as an enzyme, pancreatic and intestinal enzymes continue digestion, and the small intestine is the principal absorption site. Vitamins, minerals, protein and energy have different roles; deficiency, absorption failure and excess are separate nutritional problems. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q26. Which option preserves the technical boundary of Digestion absorption and nutrition?

A. The nephron filters blood at the glomerulus and then selectively reabsorbs and secretes substances along its tubule to regulate water, electrolytes, acid-base balance and wastes. Dialysis can replace selected filtration functions when kidneys fail but does not reproduce every endocrine and regulatory role of healthy kidneys. B. Digestion converts food into absorbable molecules: salivary amylase begins starch digestion, pepsin works in acidic stomach conditions, bile emulsifies fat without acting as an enzyme, pancreatic and intestinal enzymes continue digestion, and the small intestine is the principal absorption site. Vitamins, minerals, protein and energy have different roles; deficiency, absorption failure and excess are separate nutritional problems. C. Ventilation moves air, alveolar gas exchange moves oxygen and carbon dioxide across thin surfaces, blood transports gases, and cellular respiration releases usable energy from nutrients. Haemoglobin binds oxygen while carbon monoxide can disrupt transport through stronger binding; breathing, gas exchange and ATP-producing respiration are linked but not synonymous. D. Neurons transmit rapid electrical signals and communicate chemically at synapses, reflex arcs permit fast automatic responses, and the brain and spinal cord coordinate wider activity. Endocrine glands release hormones into blood for target-dependent effects that are often slower and longer-lasting; neural and hormonal control interact through feedback rather than operating as isolated systems.

Answer: B. Explanation: Digestion converts food into absorbable molecules: salivary amylase begins starch digestion, pepsin works in acidic stomach conditions, bile emulsifies fat without acting as an enzyme, pancreatic and intestinal enzymes continue digestion, and the small intestine is the principal absorption site. Vitamins, minerals, protein and energy have different roles; deficiency, absorption failure and excess are separate nutritional problems. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q27. Which statement uses Digestion absorption and nutrition without changing its institution, unit or status?

A. Insulin lowers elevated blood glucose by promoting uptake and storage while glucagon supports its rise, thyroid hormones influence metabolism, parathyroid hormone helps regulate calcium, adrenal hormones mediate stress responses and pituitary signals coordinate several glands. Homeostasis is dynamic regulation around a range, not an unchanging value; hormone deficiency, resistance and gland failure are different mechanisms. B. The nephron filters blood at the glomerulus and then selectively reabsorbs and secretes substances along its tubule to regulate water, electrolytes, acid-base balance and wastes. Dialysis can replace selected filtration functions when kidneys fail but does not reproduce every endocrine and regulatory role of healthy kidneys. C. Digestion converts food into absorbable molecules: salivary amylase begins starch digestion, pepsin works in acidic stomach conditions, bile emulsifies fat without acting as an enzyme, pancreatic and intestinal enzymes continue digestion, and the small intestine is the principal absorption site. Vitamins, minerals, protein and energy have different roles; deficiency, absorption failure and excess are separate nutritional problems. D. Neurons transmit rapid electrical signals and communicate chemically at synapses, reflex arcs permit fast automatic responses, and the brain and spinal cord coordinate wider activity. Endocrine glands release hormones into blood for target-dependent effects that are often slower and longer-lasting; neural and hormonal control interact through feedback rather than operating as isolated systems.

Answer: C. Explanation: Digestion converts food into absorbable molecules: salivary amylase begins starch digestion, pepsin works in acidic stomach conditions, bile emulsifies fat without acting as an enzyme, pancreatic and intestinal enzymes continue digestion, and the small intestine is the principal absorption site. Vitamins, minerals, protein and energy have different roles; deficiency, absorption failure and excess are separate nutritional problems. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q28. Which option avoids the standard UPSC close-option trap about Digestion absorption and nutrition?

A. Neurons transmit rapid electrical signals and communicate chemically at synapses, reflex arcs permit fast automatic responses, and the brain and spinal cord coordinate wider activity. Endocrine glands release hormones into blood for target-dependent effects that are often slower and longer-lasting; neural and hormonal control interact through feedback rather than operating as isolated systems. B. Insulin lowers elevated blood glucose by promoting uptake and storage while glucagon supports its rise, thyroid hormones influence metabolism, parathyroid hormone helps regulate calcium, adrenal hormones mediate stress responses and pituitary signals coordinate several glands. Homeostasis is dynamic regulation around a range, not an unchanging value; hormone deficiency, resistance and gland failure are different mechanisms. C. Innate immunity includes barriers, phagocytes, natural killer cells, complement, interferons, inflammation and fever; adaptive immunity uses antigen-specific B and T lymphocytes and can form memory. B-cell lineages produce antibodies, T cells perform coordinating, cytotoxic or regulatory roles, and peripheral immune tolerance including regulatory T cells limits attack on self; immunity is not antibodies alone. D. Digestion converts food into absorbable molecules: salivary amylase begins starch digestion, pepsin works in acidic stomach conditions, bile emulsifies fat without acting as an enzyme, pancreatic and intestinal enzymes continue digestion, and the small intestine is the principal absorption site. Vitamins, minerals, protein and energy have different roles; deficiency, absorption failure and excess are separate nutritional problems.

Answer: D. Explanation: Digestion converts food into absorbable molecules: salivary amylase begins starch digestion, pepsin works in acidic stomach conditions, bile emulsifies fat without acting as an enzyme, pancreatic and intestinal enzymes continue digestion, and the small intestine is the principal absorption site. Vitamins, minerals, protein and energy have different roles; deficiency, absorption failure and excess are separate nutritional problems. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q29. Which statement correctly identifies Blood heart and double circulation?

A. Blood contains plasma, red cells carrying haemoglobin, immune white cells and platelets involved in clotting; the four-chambered human heart supports pulmonary and systemic circuits. Arteries carry blood away from the heart and veins return it, so oxygenation is not the defining rule; ABO and Rh compatibility, gas transport and clotting must be treated as different functions. B. Neurons transmit rapid electrical signals and communicate chemically at synapses, reflex arcs permit fast automatic responses, and the brain and spinal cord coordinate wider activity. Endocrine glands release hormones into blood for target-dependent effects that are often slower and longer-lasting; neural and hormonal control interact through feedback rather than operating as isolated systems. C. Ventilation moves air, alveolar gas exchange moves oxygen and carbon dioxide across thin surfaces, blood transports gases, and cellular respiration releases usable energy from nutrients. Haemoglobin binds oxygen while carbon monoxide can disrupt transport through stronger binding; breathing, gas exchange and ATP-producing respiration are linked but not synonymous. D. The nephron filters blood at the glomerulus and then selectively reabsorbs and secretes substances along its tubule to regulate water, electrolytes, acid-base balance and wastes. Dialysis can replace selected filtration functions when kidneys fail but does not reproduce every endocrine and regulatory role of healthy kidneys.

Answer: A. Explanation: Blood contains plasma, red cells carrying haemoglobin, immune white cells and platelets involved in clotting; the four-chambered human heart supports pulmonary and systemic circuits. Arteries carry blood away from the heart and veins return it, so oxygenation is not the defining rule; ABO and Rh compatibility, gas transport and clotting must be treated as different functions. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q30. Which option preserves the technical boundary of Blood heart and double circulation?

A. Neurons transmit rapid electrical signals and communicate chemically at synapses, reflex arcs permit fast automatic responses, and the brain and spinal cord coordinate wider activity. Endocrine glands release hormones into blood for target-dependent effects that are often slower and longer-lasting; neural and hormonal control interact through feedback rather than operating as isolated systems. B. Blood contains plasma, red cells carrying haemoglobin, immune white cells and platelets involved in clotting; the four-chambered human heart supports pulmonary and systemic circuits. Arteries carry blood away from the heart and veins return it, so oxygenation is not the defining rule; ABO and Rh compatibility, gas transport and clotting must be treated as different functions. C. The nephron filters blood at the glomerulus and then selectively reabsorbs and secretes substances along its tubule to regulate water, electrolytes, acid-base balance and wastes. Dialysis can replace selected filtration functions when kidneys fail but does not reproduce every endocrine and regulatory role of healthy kidneys. D. Insulin lowers elevated blood glucose by promoting uptake and storage while glucagon supports its rise, thyroid hormones influence metabolism, parathyroid hormone helps regulate calcium, adrenal hormones mediate stress responses and pituitary signals coordinate several glands. Homeostasis is dynamic regulation around a range, not an unchanging value; hormone deficiency, resistance and gland failure are different mechanisms.

Answer: B. Explanation: Blood contains plasma, red cells carrying haemoglobin, immune white cells and platelets involved in clotting; the four-chambered human heart supports pulmonary and systemic circuits. Arteries carry blood away from the heart and veins return it, so oxygenation is not the defining rule; ABO and Rh compatibility, gas transport and clotting must be treated as different functions. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q31. Which statement uses Blood heart and double circulation without changing its institution, unit or status?

A. Innate immunity includes barriers, phagocytes, natural killer cells, complement, interferons, inflammation and fever; adaptive immunity uses antigen-specific B and T lymphocytes and can form memory. B-cell lineages produce antibodies, T cells perform coordinating, cytotoxic or regulatory roles, and peripheral immune tolerance including regulatory T cells limits attack on self; immunity is not antibodies alone. B. Neurons transmit rapid electrical signals and communicate chemically at synapses, reflex arcs permit fast automatic responses, and the brain and spinal cord coordinate wider activity. Endocrine glands release hormones into blood for target-dependent effects that are often slower and longer-lasting; neural and hormonal control interact through feedback rather than operating as isolated systems. C. Blood contains plasma, red cells carrying haemoglobin, immune white cells and platelets involved in clotting; the four-chambered human heart supports pulmonary and systemic circuits. Arteries carry blood away from the heart and veins return it, so oxygenation is not the defining rule; ABO and Rh compatibility, gas transport and clotting must be treated as different functions. D. Insulin lowers elevated blood glucose by promoting uptake and storage while glucagon supports its rise, thyroid hormones influence metabolism, parathyroid hormone helps regulate calcium, adrenal hormones mediate stress responses and pituitary signals coordinate several glands. Homeostasis is dynamic regulation around a range, not an unchanging value; hormone deficiency, resistance and gland failure are different mechanisms.

Answer: C. Explanation: Blood contains plasma, red cells carrying haemoglobin, immune white cells and platelets involved in clotting; the four-chambered human heart supports pulmonary and systemic circuits. Arteries carry blood away from the heart and veins return it, so oxygenation is not the defining rule; ABO and Rh compatibility, gas transport and clotting must be treated as different functions. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q32. Which option avoids the standard UPSC close-option trap about Blood heart and double circulation?

A. Insulin lowers elevated blood glucose by promoting uptake and storage while glucagon supports its rise, thyroid hormones influence metabolism, parathyroid hormone helps regulate calcium, adrenal hormones mediate stress responses and pituitary signals coordinate several glands. Homeostasis is dynamic regulation around a range, not an unchanging value; hormone deficiency, resistance and gland failure are different mechanisms. B. Innate immunity includes barriers, phagocytes, natural killer cells, complement, interferons, inflammation and fever; adaptive immunity uses antigen-specific B and T lymphocytes and can form memory. B-cell lineages produce antibodies, T cells perform coordinating, cytotoxic or regulatory roles, and peripheral immune tolerance including regulatory T cells limits attack on self; immunity is not antibodies alone. C. Bacteria are cellular organisms, viruses are acellular and depend on host-cell machinery, fungi are eukaryotes, protozoa are single-celled eukaryotes and helminths are multicellular parasites. Transmission may be respiratory, faeco-oral, blood-borne, sexual, vector-borne, fomite-linked or zoonotic; a vector such as Aedes, Anopheles, Culex or sandfly carries a pathogen but is not the disease itself. D. Blood contains plasma, red cells carrying haemoglobin, immune white cells and platelets involved in clotting; the four-chambered human heart supports pulmonary and systemic circuits. Arteries carry blood away from the heart and veins return it, so oxygenation is not the defining rule; ABO and Rh compatibility, gas transport and clotting must be treated as different functions.

Answer: D. Explanation: Blood contains plasma, red cells carrying haemoglobin, immune white cells and platelets involved in clotting; the four-chambered human heart supports pulmonary and systemic circuits. Arteries carry blood away from the heart and veins return it, so oxygenation is not the defining rule; ABO and Rh compatibility, gas transport and clotting must be treated as different functions. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q33. Which statement correctly identifies Respiration gas exchange and cellular energy?

A. Ventilation moves air, alveolar gas exchange moves oxygen and carbon dioxide across thin surfaces, blood transports gases, and cellular respiration releases usable energy from nutrients. Haemoglobin binds oxygen while carbon monoxide can disrupt transport through stronger binding; breathing, gas exchange and ATP-producing respiration are linked but not synonymous. B. Neurons transmit rapid electrical signals and communicate chemically at synapses, reflex arcs permit fast automatic responses, and the brain and spinal cord coordinate wider activity. Endocrine glands release hormones into blood for target-dependent effects that are often slower and longer-lasting; neural and hormonal control interact through feedback rather than operating as isolated systems. C. The nephron filters blood at the glomerulus and then selectively reabsorbs and secretes substances along its tubule to regulate water, electrolytes, acid-base balance and wastes. Dialysis can replace selected filtration functions when kidneys fail but does not reproduce every endocrine and regulatory role of healthy kidneys. D. Insulin lowers elevated blood glucose by promoting uptake and storage while glucagon supports its rise, thyroid hormones influence metabolism, parathyroid hormone helps regulate calcium, adrenal hormones mediate stress responses and pituitary signals coordinate several glands. Homeostasis is dynamic regulation around a range, not an unchanging value; hormone deficiency, resistance and gland failure are different mechanisms.

Answer: A. Explanation: Ventilation moves air, alveolar gas exchange moves oxygen and carbon dioxide across thin surfaces, blood transports gases, and cellular respiration releases usable energy from nutrients. Haemoglobin binds oxygen while carbon monoxide can disrupt transport through stronger binding; breathing, gas exchange and ATP-producing respiration are linked but not synonymous. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q34. Which option preserves the technical boundary of Respiration gas exchange and cellular energy?

A. Insulin lowers elevated blood glucose by promoting uptake and storage while glucagon supports its rise, thyroid hormones influence metabolism, parathyroid hormone helps regulate calcium, adrenal hormones mediate stress responses and pituitary signals coordinate several glands. Homeostasis is dynamic regulation around a range, not an unchanging value; hormone deficiency, resistance and gland failure are different mechanisms. B. Ventilation moves air, alveolar gas exchange moves oxygen and carbon dioxide across thin surfaces, blood transports gases, and cellular respiration releases usable energy from nutrients. Haemoglobin binds oxygen while carbon monoxide can disrupt transport through stronger binding; breathing, gas exchange and ATP-producing respiration are linked but not synonymous. C. Innate immunity includes barriers, phagocytes, natural killer cells, complement, interferons, inflammation and fever; adaptive immunity uses antigen-specific B and T lymphocytes and can form memory. B-cell lineages produce antibodies, T cells perform coordinating, cytotoxic or regulatory roles, and peripheral immune tolerance including regulatory T cells limits attack on self; immunity is not antibodies alone. D. Neurons transmit rapid electrical signals and communicate chemically at synapses, reflex arcs permit fast automatic responses, and the brain and spinal cord coordinate wider activity. Endocrine glands release hormones into blood for target-dependent effects that are often slower and longer-lasting; neural and hormonal control interact through feedback rather than operating as isolated systems.

Answer: B. Explanation: Ventilation moves air, alveolar gas exchange moves oxygen and carbon dioxide across thin surfaces, blood transports gases, and cellular respiration releases usable energy from nutrients. Haemoglobin binds oxygen while carbon monoxide can disrupt transport through stronger binding; breathing, gas exchange and ATP-producing respiration are linked but not synonymous. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q35. Which statement uses Respiration gas exchange and cellular energy without changing its institution, unit or status?

A. Bacteria are cellular organisms, viruses are acellular and depend on host-cell machinery, fungi are eukaryotes, protozoa are single-celled eukaryotes and helminths are multicellular parasites. Transmission may be respiratory, faeco-oral, blood-borne, sexual, vector-borne, fomite-linked or zoonotic; a vector such as Aedes, Anopheles, Culex or sandfly carries a pathogen but is not the disease itself. B. Insulin lowers elevated blood glucose by promoting uptake and storage while glucagon supports its rise, thyroid hormones influence metabolism, parathyroid hormone helps regulate calcium, adrenal hormones mediate stress responses and pituitary signals coordinate several glands. Homeostasis is dynamic regulation around a range, not an unchanging value; hormone deficiency, resistance and gland failure are different mechanisms. C. Ventilation moves air, alveolar gas exchange moves oxygen and carbon dioxide across thin surfaces, blood transports gases, and cellular respiration releases usable energy from nutrients. Haemoglobin binds oxygen while carbon monoxide can disrupt transport through stronger binding; breathing, gas exchange and ATP-producing respiration are linked but not synonymous. D. Innate immunity includes barriers, phagocytes, natural killer cells, complement, interferons, inflammation and fever; adaptive immunity uses antigen-specific B and T lymphocytes and can form memory. B-cell lineages produce antibodies, T cells perform coordinating, cytotoxic or regulatory roles, and peripheral immune tolerance including regulatory T cells limits attack on self; immunity is not antibodies alone.

Answer: C. Explanation: Ventilation moves air, alveolar gas exchange moves oxygen and carbon dioxide across thin surfaces, blood transports gases, and cellular respiration releases usable energy from nutrients. Haemoglobin binds oxygen while carbon monoxide can disrupt transport through stronger binding; breathing, gas exchange and ATP-producing respiration are linked but not synonymous. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q36. Which option avoids the standard UPSC close-option trap about Respiration gas exchange and cellular energy?

A. Bacteria are cellular organisms, viruses are acellular and depend on host-cell machinery, fungi are eukaryotes, protozoa are single-celled eukaryotes and helminths are multicellular parasites. Transmission may be respiratory, faeco-oral, blood-borne, sexual, vector-borne, fomite-linked or zoonotic; a vector such as Aedes, Anopheles, Culex or sandfly carries a pathogen but is not the disease itself. B. Innate immunity includes barriers, phagocytes, natural killer cells, complement, interferons, inflammation and fever; adaptive immunity uses antigen-specific B and T lymphocytes and can form memory. B-cell lineages produce antibodies, T cells perform coordinating, cytotoxic or regulatory roles, and peripheral immune tolerance including regulatory T cells limits attack on self; immunity is not antibodies alone. C. Antibiotics target susceptible bacteria rather than viruses, while antivirals interfere with virus-specific replication stages. Antimicrobial resistance is selection and spread of resistant microorganisms under pressures including misuse, incomplete treatment, agricultural use and environmental residues; the microbe or population becomes resistant, not the patient's body. D. Ventilation moves air, alveolar gas exchange moves oxygen and carbon dioxide across thin surfaces, blood transports gases, and cellular respiration releases usable energy from nutrients. Haemoglobin binds oxygen while carbon monoxide can disrupt transport through stronger binding; breathing, gas exchange and ATP-producing respiration are linked but not synonymous.

Answer: D. Explanation: Ventilation moves air, alveolar gas exchange moves oxygen and carbon dioxide across thin surfaces, blood transports gases, and cellular respiration releases usable energy from nutrients. Haemoglobin binds oxygen while carbon monoxide can disrupt transport through stronger binding; breathing, gas exchange and ATP-producing respiration are linked but not synonymous. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q37. Which statement correctly identifies Kidney nephron and homeostasis?

A. The nephron filters blood at the glomerulus and then selectively reabsorbs and secretes substances along its tubule to regulate water, electrolytes, acid-base balance and wastes. Dialysis can replace selected filtration functions when kidneys fail but does not reproduce every endocrine and regulatory role of healthy kidneys. B. Neurons transmit rapid electrical signals and communicate chemically at synapses, reflex arcs permit fast automatic responses, and the brain and spinal cord coordinate wider activity. Endocrine glands release hormones into blood for target-dependent effects that are often slower and longer-lasting; neural and hormonal control interact through feedback rather than operating as isolated systems. C. Insulin lowers elevated blood glucose by promoting uptake and storage while glucagon supports its rise, thyroid hormones influence metabolism, parathyroid hormone helps regulate calcium, adrenal hormones mediate stress responses and pituitary signals coordinate several glands. Homeostasis is dynamic regulation around a range, not an unchanging value; hormone deficiency, resistance and gland failure are different mechanisms. D. Innate immunity includes barriers, phagocytes, natural killer cells, complement, interferons, inflammation and fever; adaptive immunity uses antigen-specific B and T lymphocytes and can form memory. B-cell lineages produce antibodies, T cells perform coordinating, cytotoxic or regulatory roles, and peripheral immune tolerance including regulatory T cells limits attack on self; immunity is not antibodies alone.

Answer: A. Explanation: The nephron filters blood at the glomerulus and then selectively reabsorbs and secretes substances along its tubule to regulate water, electrolytes, acid-base balance and wastes. Dialysis can replace selected filtration functions when kidneys fail but does not reproduce every endocrine and regulatory role of healthy kidneys. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q38. Which option preserves the technical boundary of Kidney nephron and homeostasis?

A. Innate immunity includes barriers, phagocytes, natural killer cells, complement, interferons, inflammation and fever; adaptive immunity uses antigen-specific B and T lymphocytes and can form memory. B-cell lineages produce antibodies, T cells perform coordinating, cytotoxic or regulatory roles, and peripheral immune tolerance including regulatory T cells limits attack on self; immunity is not antibodies alone. B. The nephron filters blood at the glomerulus and then selectively reabsorbs and secretes substances along its tubule to regulate water, electrolytes, acid-base balance and wastes. Dialysis can replace selected filtration functions when kidneys fail but does not reproduce every endocrine and regulatory role of healthy kidneys. C. Insulin lowers elevated blood glucose by promoting uptake and storage while glucagon supports its rise, thyroid hormones influence metabolism, parathyroid hormone helps regulate

calcium, adrenal hormones mediate stress responses and pituitary signals coordinate several glands. Homeostasis is dynamic regulation around a range, not an unchanging value; hormone deficiency, resistance and gland failure are different mechanisms. D. Bacteria are cellular organisms, viruses are acellular and depend on host-cell machinery, fungi are eukaryotes, protozoa are single-celled eukaryotes and helminths are multicellular parasites. Transmission may be respiratory, faeco-oral, blood-borne, sexual, vector-borne, fomite-linked or zoonotic; a vector such as Aedes, Anopheles, Culex or sandfly carries a pathogen but is not the disease itself.

Answer: B. Explanation: The nephron filters blood at the glomerulus and then selectively reabsorbs and secretes substances along its tubule to regulate water, electrolytes, acid-base balance and wastes. Dialysis can replace selected filtration functions when kidneys fail but does not reproduce every endocrine and regulatory role of healthy kidneys. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q39. Which statement uses Kidney nephron and homeostasis without changing its institution, unit or status?

A. Innate immunity includes barriers, phagocytes, natural killer cells, complement, interferons, inflammation and fever; adaptive immunity uses antigen-specific B and T lymphocytes and can form memory. B-cell lineages produce antibodies, T cells perform coordinating, cytotoxic or regulatory roles, and peripheral immune tolerance including regulatory T cells limits attack on self; immunity is not antibodies alone. B. Bacteria are cellular organisms, viruses are acellular and depend on host-cell machinery, fungi are eukaryotes, protozoa are single-celled eukaryotes and helminths are multicellular parasites. Transmission may be respiratory, faeco-oral, blood-borne, sexual, vector-borne, fomite-linked or zoonotic; a vector such as Aedes, Anopheles, Culex or sandfly carries a pathogen but is not the disease itself. C. The nephron filters blood at the glomerulus and then selectively reabsorbs and secretes substances along its tubule to regulate water, electrolytes, acid-base balance and wastes. Dialysis can replace selected filtration functions when kidneys fail but does not reproduce every endocrine and regulatory role of healthy kidneys. D. Antibiotics target susceptible bacteria rather than viruses, while antivirals interfere with virus-specific replication stages. Antimicrobial resistance is selection and spread of resistant microorganisms under pressures including misuse, incomplete treatment, agricultural use and environmental residues; the microbe or population becomes resistant, not the patient's body.

Answer: C. Explanation: The nephron filters blood at the glomerulus and then selectively reabsorbs and secretes substances along its tubule to regulate water, electrolytes, acid-base balance and wastes. Dialysis can replace selected filtration functions when kidneys fail but does not reproduce every endocrine and regulatory role of healthy kidneys. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q40. Which option avoids the standard UPSC close-option trap about Kidney nephron and homeostasis?

A. A biofilm is a surface-associated microbial community embedded in a matrix that can alter persistence and treatment response; the human microbiome contributes to digestion, vitamin production, immune development and pathogen exclusion. Probiotics are live microorganisms intended to confer a benefit in an adequate amount, while thermophiles, psychrophiles and acidophiles are classified by environmental tolerance; none of these labels proves safety or effectiveness in every context. B. Bacteria are cellular organisms, viruses are acellular and depend on host-cell machinery, fungi are eukaryotes, protozoa are single-celled eukaryotes and helminths are multicellular parasites. Transmission may be respiratory, faeco-oral, blood-borne, sexual, vector-borne, fomite-linked or zoonotic; a vector such as Aedes, Anopheles, Culex or sandfly carries a pathogen but is not the disease itself. C. Antibiotics target susceptible bacteria rather than viruses, while antivirals interfere with virus-specific replication stages. Antimicrobial resistance is selection and spread of resistant microorganisms under pressures including misuse, incomplete treatment, agricultural use and environmental residues; the microbe or population becomes resistant, not the patient's body. D. The nephron filters blood at the glomerulus and then selectively reabsorbs and secretes substances along its tubule to regulate water, electrolytes, acid-base balance and wastes. Dialysis can replace selected filtration functions when kidneys fail but does not reproduce every endocrine and regulatory role of healthy kidneys.

Answer: D. Explanation: The nephron filters blood at the glomerulus and then selectively reabsorbs and secretes substances along its tubule to regulate water, electrolytes, acid-base balance and wastes. Dialysis can replace selected filtration functions when kidneys fail but does not reproduce every endocrine and regulatory role of healthy kidneys. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q41. Which statement correctly identifies Nervous endocrine and feedback control?

A. Neurons transmit rapid electrical signals and communicate chemically at synapses, reflex arcs permit fast automatic responses, and the brain and spinal cord coordinate wider activity. Endocrine glands release hormones into blood for target-dependent effects that are often slower and longer-lasting; neural and hormonal control interact through feedback rather than operating as isolated systems. B. Insulin lowers elevated blood glucose by promoting uptake and storage while glucagon supports its rise, thyroid hormones influence metabolism, parathyroid hormone helps regulate calcium, adrenal hormones mediate stress responses and pituitary signals coordinate several glands. Homeostasis is dynamic regulation around a range, not an unchanging value; hormone deficiency, resistance and gland failure are different mechanisms. C. Bacteria are cellular organisms, viruses are acellular and depend on host-cell machinery, fungi are eukaryotes, protozoa are single-celled eukaryotes and helminths are multicellular parasites. Transmission may be respiratory, faeco-oral, blood-borne, sexual, vector-borne, fomite-linked or zoonotic; a vector such as Aedes, Anopheles, Culex or sandfly carries a pathogen but is not the disease itself. D. Innate immunity includes barriers, phagocytes, natural killer cells, complement, interferons, inflammation and fever; adaptive immunity uses antigen-specific B and T lymphocytes and can form memory. B-cell lineages produce antibodies, T cells perform coordinating, cytotoxic or regulatory roles, and peripheral immune tolerance including regulatory T cells limits attack on self; immunity is not antibodies alone.

Answer: A. Explanation: Neurons transmit rapid electrical signals and communicate chemically at synapses, reflex arcs permit fast automatic responses, and the brain and spinal cord coordinate wider activity. Endocrine glands release hormones into blood for target-dependent effects that are often slower and longer-lasting; neural and hormonal control interact through feedback rather than operating as isolated systems. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q42. Which option preserves the technical boundary of Nervous endocrine and feedback control?

A. Innate immunity includes barriers, phagocytes, natural killer cells, complement, interferons, inflammation and fever; adaptive immunity uses antigen-specific B and T lymphocytes and can form memory. B-cell lineages produce antibodies, T cells perform coordinating, cytotoxic or regulatory roles, and peripheral immune tolerance including regulatory T cells limits attack on self; immunity is not antibodies alone. B. Neurons transmit rapid electrical signals and communicate chemically at synapses, reflex arcs permit fast automatic responses, and the brain and spinal cord coordinate wider activity. Endocrine glands release hormones into blood for target-dependent effects that are often slower and longer-lasting; neural and hormonal control interact through feedback rather than operating as isolated systems. C. Bacteria are cellular organisms, viruses are acellular and depend on host-cell machinery, fungi are eukaryotes, protozoa are single-celled eukaryotes and helminths are multicellular parasites. Transmission may be respiratory, faeco-oral, blood-borne, sexual, vector-borne, fomite-linked or zoonotic; a vector such as Aedes, Anopheles, Culex or sandfly carries a pathogen but is not the disease itself. D. Antibiotics target susceptible bacteria rather than viruses, while antivirals interfere with virus-specific replication stages. Antimicrobial resistance is selection and spread of resistant microorganisms under pressures including misuse, incomplete treatment, agricultural use and environmental residues; the microbe or population becomes resistant, not the patient's body.

Answer: B. Explanation: Neurons transmit rapid electrical signals and communicate chemically at synapses, reflex arcs permit fast automatic responses, and the brain and spinal cord coordinate wider activity. Endocrine glands release hormones into blood for target-dependent effects that are often slower and longer-lasting; neural and hormonal control interact through feedback rather than operating as isolated systems. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q43. Which statement uses Nervous endocrine and feedback control without changing its institution, unit or status?

A. A biofilm is a surface-associated microbial community embedded in a matrix that can alter persistence and treatment response; the human microbiome contributes to digestion, vitamin production, immune development and pathogen exclusion. Probiotics are live microorganisms intended to confer a benefit in an adequate amount, while thermophiles, psychrophiles and acidophiles are classified by environmental tolerance; none of these labels proves safety or effectiveness in every context. B. Bacteria are cellular organisms, viruses are acellular and depend on host-cell machinery, fungi are eukaryotes, protozoa are single-celled eukaryotes and helminths are multicellular parasites. Transmission may be respiratory, faeco-oral, blood-borne, sexual, vector-borne, fomite-linked or zoonotic; a vector such as Aedes, Anopheles, Culex or sandfly carries a pathogen but is not the disease itself. C. Neurons transmit rapid electrical signals and communicate chemically at synapses, reflex arcs permit fast automatic responses, and the brain and spinal cord coordinate wider activity. Endocrine glands release hormones into blood for target-dependent effects that are often slower and longer-lasting; neural and hormonal control interact through feedback rather than operating as isolated systems. D. Antibiotics target susceptible bacteria rather than viruses, while antivirals interfere with virus-specific replication stages. Antimicrobial resistance is selection and spread of resistant microorganisms under pressures including misuse, incomplete treatment, agricultural use and environmental residues; the microbe or population becomes resistant, not the patient's body.

Answer: C. Explanation: Neurons transmit rapid electrical signals and communicate chemically at synapses, reflex arcs permit fast automatic responses, and the brain and spinal cord coordinate wider activity. Endocrine glands release hormones into blood for target-dependent effects that are often slower and longer-lasting; neural and hormonal control interact through feedback rather than operating as isolated systems. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q44. Which option avoids the standard UPSC close-option trap about Nervous endocrine and feedback control?

A. Antibiotics target susceptible bacteria rather than viruses, while antivirals interfere with virus-specific replication stages. Antimicrobial resistance is selection and spread of resistant microorganisms under pressures including misuse, incomplete treatment, agricultural use and environmental residues; the microbe or population becomes resistant, not the patient's body. B. A biofilm is a surface-associated microbial community embedded in a matrix that can alter persistence and treatment response; the human microbiome contributes to digestion, vitamin production, immune development and pathogen exclusion. Probiotics are live microorganisms intended to confer a benefit in an adequate amount, while thermophiles, psychrophiles and acidophiles are classified by environmental tolerance; none of these labels proves safety or effectiveness in every context. C. Photosynthesis stores light energy in chemical form while plants also respire continuously; xylem conducts water and minerals and phloem translocates organic food. Stomata regulate gas exchange and water loss, transpiration contributes to upward water movement, and auxin, gibberellin, cytokinin, abscisic acid and ethylene have distinct growth or stress roles; C3, C4 and CAM are different carbon-fixation strategies. D. Neurons transmit rapid electrical signals and communicate chemically at synapses, reflex arcs permit fast automatic responses, and the brain and spinal cord coordinate wider activity. Endocrine glands release hormones into blood for target-dependent effects that are often slower and longer-lasting; neural and hormonal control interact through feedback rather than operating as isolated systems.

Answer: D. Explanation: Neurons transmit rapid electrical signals and communicate chemically at synapses, reflex arcs permit fast automatic responses, and the brain and spinal cord coordinate wider activity. Endocrine glands release hormones into blood for target-dependent effects that are often slower and longer-lasting; neural and hormonal control interact through feedback rather than operating as isolated systems. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q45. Which statement correctly identifies Hormones metabolism and internal stability?

A. Insulin lowers elevated blood glucose by promoting uptake and storage while glucagon supports its rise, thyroid hormones influence metabolism, parathyroid hormone helps regulate calcium, adrenal hormones mediate stress responses and pituitary signals coordinate several glands. Homeostasis is dynamic regulation around a range, not an unchanging value; hormone deficiency, resistance and gland failure are different mechanisms. B. Innate immunity includes barriers, phagocytes, natural killer cells, complement, interferons, inflammation and fever; adaptive immunity uses antigen-specific B and T lymphocytes and can form memory. B-cell lineages produce antibodies, T cells perform coordinating, cytotoxic or regulatory roles, and peripheral immune tolerance including regulatory T cells limits attack on self; immunity is not antibodies alone. C. Antibiotics target susceptible bacteria rather than viruses, while antivirals interfere with virus-specific replication stages. Antimicrobial resistance is selection and spread of resistant microorganisms under pressures including misuse, incomplete treatment, agricultural use and environmental residues; the microbe or population becomes resistant, not the patient's body. D. Bacteria are cellular organisms, viruses are acellular and depend on host-cell machinery, fungi are eukaryotes, protozoa are single-celled eukaryotes and helminths are multicellular parasites. Transmission may be respiratory, faeco-oral, blood-borne, sexual, vector-borne, fomite-linked or zoonotic; a vector such as Aedes, Anopheles, Culex or sandfly carries a pathogen but is not the disease itself.

Answer: A. Explanation: Insulin lowers elevated blood glucose by promoting uptake and storage while glucagon supports its rise, thyroid hormones influence metabolism, parathyroid hormone helps regulate calcium, adrenal hormones mediate stress responses and pituitary signals coordinate several glands. Homeostasis is dynamic regulation around a range, not an unchanging value; hormone deficiency, resistance and gland failure are different mechanisms. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q46. Which option preserves the technical boundary of Hormones metabolism and internal stability?

A. A biofilm is a surface-associated microbial community embedded in a matrix that can alter persistence and treatment response; the human microbiome contributes to digestion, vitamin production, immune development and pathogen exclusion. Probiotics are live microorganisms intended to confer a benefit in an adequate amount, while thermophiles, psychrophiles and acidophiles are classified by environmental tolerance; none of these labels proves safety or effectiveness in every context. B. Insulin lowers elevated blood glucose by promoting uptake and storage while glucagon supports its rise, thyroid hormones influence metabolism, parathyroid hormone helps regulate calcium, adrenal hormones mediate stress responses and pituitary signals coordinate several glands. Homeostasis is dynamic regulation around a range, not an unchanging value; hormone deficiency, resistance and gland failure are different mechanisms. C. Bacteria are cellular organisms, viruses are acellular and depend on host-cell machinery, fungi are eukaryotes, protozoa are single-celled eukaryotes and helminths are multicellular parasites. Transmission may be respiratory, faeco-oral, blood-borne, sexual, vector-borne, fomite-linked or zoonotic; a vector such as Aedes, Anopheles, Culex or sandfly carries a pathogen but is not the disease itself. D. Antibiotics target susceptible bacteria rather than viruses, while antivirals interfere with virus-specific replication stages. Antimicrobial resistance is selection and spread of resistant microorganisms under pressures including misuse, incomplete treatment, agricultural use and environmental residues; the microbe or population becomes resistant, not the patient's body.

Answer: B. Explanation: Insulin lowers elevated blood glucose by promoting uptake and storage while glucagon supports its rise, thyroid hormones influence metabolism, parathyroid hormone helps regulate calcium, adrenal hormones mediate stress responses and pituitary signals coordinate several glands. Homeostasis is dynamic regulation around a range, not an unchanging value; hormone deficiency, resistance and gland failure are different mechanisms. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q47. Which statement uses Hormones metabolism and internal stability without changing its institution, unit or status?

A. Antibiotics target susceptible bacteria rather than viruses, while antivirals interfere with virus-specific replication stages. Antimicrobial resistance is selection and spread of resistant microorganisms under pressures including misuse, incomplete treatment, agricultural use and environmental residues; the microbe or population becomes resistant, not the patient's body. B. A biofilm is a surface-associated microbial community embedded in a matrix that can alter persistence and treatment response; the human microbiome contributes to digestion, vitamin production, immune development and pathogen exclusion. Probiotics are live microorganisms intended to confer a benefit in an adequate amount, while thermophiles, psychrophiles and acidophiles are classified by environmental tolerance; none of these labels proves safety or effectiveness in every context. C. Insulin lowers elevated blood glucose by promoting uptake and storage while glucagon supports its rise, thyroid hormones influence metabolism, parathyroid hormone helps regulate calcium, adrenal hormones mediate stress responses and pituitary signals coordinate several glands. Homeostasis is dynamic regulation around a range, not an unchanging value; hormone deficiency, resistance and gland failure are different mechanisms. D. Photosynthesis stores light energy in chemical form while plants also respire continuously; xylem conducts water and minerals and phloem translocates organic food. Stomata regulate gas exchange and water loss, transpiration contributes to upward water movement, and auxin, gibberellin, cytokinin, abscisic acid and ethylene have distinct growth or stress roles; C3, C4 and CAM are different carbon-fixation strategies.

Answer: C. Explanation: Insulin lowers elevated blood glucose by promoting uptake and storage while glucagon supports its rise, thyroid hormones influence metabolism, parathyroid hormone helps regulate calcium, adrenal hormones mediate stress responses and pituitary signals coordinate several glands. Homeostasis is dynamic regulation around a range, not an unchanging value; hormone deficiency, resistance and gland failure are different mechanisms. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q48. Which option avoids the standard UPSC close-option trap about Hormones metabolism and internal stability?

A. A biofilm is a surface-associated microbial community embedded in a matrix that can alter persistence and treatment response; the human microbiome contributes to digestion, vitamin production, immune development and pathogen exclusion. Probiotics are live microorganisms intended to confer a benefit in an adequate amount, while thermophiles, psychrophiles and acidophiles are classified by environmental tolerance; none of these labels proves safety or effectiveness in every context. B. Photosynthesis stores light energy in chemical form while plants also respire continuously; xylem conducts water and minerals and phloem translocates organic food. Stomata regulate gas exchange and water loss, transpiration contributes to upward water movement, and auxin, gibberellin, cytokinin, abscisic acid and ethylene have distinct growth or stress roles; C3, C4 and CAM are different carbon-fixation strategies. C. Groundnut, horse-gram and soybean are routed pea-family examples; cicada, frog hopper and pond skater are insects; poisonous species occur in multiple animal groups, so a broad group name does not prove toxicity. Cassava is a woody shrub, ginger a rhizomatous herb, Malabar spinach a climbing vine, mint a herb and papaya a large herb; mushrooms can have species-specific edible, medicinal, psychoactive, insect-pathogenic or bioluminescent properties. D. Insulin lowers elevated blood glucose by promoting uptake and storage while glucagon supports its rise, thyroid hormones influence metabolism, parathyroid hormone helps regulate calcium, adrenal hormones mediate stress responses and pituitary signals coordinate several glands. Homeostasis is dynamic regulation around a range, not an unchanging value; hormone deficiency, resistance and gland failure are different mechanisms.

Answer: D. Explanation: Insulin lowers elevated blood glucose by promoting uptake and storage while glucagon supports its rise, thyroid hormones influence metabolism, parathyroid hormone helps regulate calcium, adrenal hormones mediate stress responses and pituitary signals coordinate several glands. Homeostasis is dynamic regulation around a range, not an unchanging value; hormone deficiency, resistance and gland failure are different mechanisms. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q49. Which statement correctly identifies Innate adaptive and tolerance architecture?

A. Innate immunity includes barriers, phagocytes, natural killer cells, complement, interferons, inflammation and fever; adaptive immunity uses antigen-specific B and T lymphocytes and can form memory. B-cell lineages produce antibodies, T cells perform coordinating, cytotoxic or regulatory roles, and peripheral immune tolerance including regulatory T cells limits attack on self; immunity is not antibodies alone. B. Antibiotics target susceptible bacteria rather than viruses, while antivirals interfere with virus-specific replication stages. Antimicrobial resistance is selection and spread of resistant microorganisms under pressures including misuse, incomplete treatment, agricultural use and environmental residues; the microbe or population becomes resistant, not the patient's body. C. A biofilm is a surface-associated microbial community embedded in a matrix that can alter persistence and treatment response; the human microbiome contributes to digestion, vitamin production, immune development and pathogen exclusion. Probiotics are live microorganisms intended to confer a benefit in an adequate amount, while thermophiles, psychrophiles and acidophiles are classified by environmental tolerance; none of these labels proves safety or effectiveness in every context. D. Bacteria are cellular organisms, viruses are acellular and depend on host-cell machinery, fungi are eukaryotes, protozoa are single-celled eukaryotes and helminths are multicellular parasites. Transmission may be respiratory, faeco-oral, blood-borne, sexual, vector-borne, fomite-linked or zoonotic; a vector such as Aedes, Anopheles, Culex or sandfly carries a pathogen but is not the disease itself.

Answer: A. Explanation: Innate immunity includes barriers, phagocytes, natural killer cells, complement, interferons, inflammation and fever; adaptive immunity uses antigen-specific B and T lymphocytes and can form memory. B-cell lineages produce antibodies, T cells perform coordinating, cytotoxic or regulatory roles, and peripheral immune tolerance including regulatory T cells limits attack on self; immunity is not antibodies alone. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q50. Which option preserves the technical boundary of Innate adaptive and tolerance architecture?

A. Photosynthesis stores light energy in chemical form while plants also respire continuously; xylem conducts water and minerals and phloem translocates organic food. Stomata regulate gas exchange and water loss, transpiration contributes to upward water movement, and auxin, gibberellin, cytokinin, abscisic acid and ethylene have distinct growth or stress roles; C3, C4 and CAM are different carbon-fixation strategies. B. Innate immunity includes barriers, phagocytes, natural killer cells, complement, interferons, inflammation and fever; adaptive immunity uses antigen-specific B and T lymphocytes and can form memory. B-cell lineages produce antibodies, T cells perform coordinating, cytotoxic or regulatory roles, and peripheral immune tolerance including regulatory T cells limits attack on self; immunity is not antibodies alone. C. Antibiotics target susceptible bacteria rather than viruses, while antivirals interfere with virus-specific replication stages. Antimicrobial resistance is selection and spread of resistant microorganisms under pressures including misuse, incomplete treatment, agricultural use and environmental residues; the microbe or population becomes resistant, not the patient's body. D. A biofilm is a surface-associated microbial community embedded in a matrix that can alter persistence and treatment response; the human microbiome contributes to digestion, vitamin production, immune development and pathogen exclusion. Probiotics are live microorganisms intended to confer a benefit in an adequate amount, while thermophiles, psychrophiles and acidophiles are classified by environmental tolerance; none of these labels proves safety or effectiveness in every context.

Answer: B. Explanation: Innate immunity includes barriers, phagocytes, natural killer cells, complement, interferons, inflammation and fever; adaptive immunity uses antigen-specific B and T lymphocytes and can form memory. B-cell lineages produce antibodies, T cells perform coordinating, cytotoxic or regulatory roles, and peripheral immune tolerance including regulatory T cells limits attack on self; immunity is not antibodies alone. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q51. Which statement uses Innate adaptive and tolerance architecture without changing its institution, unit or status?

A. Groundnut, horse-gram and soybean are routed pea-family examples; cicada, froghopper and pond skater are insects; poisonous species occur in multiple animal groups, so a broad group name does not prove toxicity. Cassava is a woody shrub, ginger a rhizomatous herb, Malabar spinach a climbing vine, mint a herb and papaya a large herb; mushrooms can have species-specific edible, medicinal, psychoactive, insect-pathogenic or bioluminescent properties. B. Photosynthesis stores light energy in chemical form while plants also respire continuously; xylem conducts water and minerals and phloem translocates organic food. Stomata regulate gas exchange and water loss, transpiration contributes to upward water movement, and auxin, gibberellin, cytokinin, abscisic acid and ethylene have distinct growth or stress roles; C3, C4 and CAM are different carbon-fixation strategies. C. Innate immunity includes barriers, phagocytes, natural killer cells, complement, interferons, inflammation and fever; adaptive immunity uses antigen-specific B and T lymphocytes and can form memory. B-cell lineages produce antibodies, T cells perform coordinating, cytotoxic or regulatory roles, and peripheral immune tolerance including regulatory T cells limits attack on self; immunity is not antibodies alone. D. A biofilm is a surface-associated microbial community embedded in a matrix that can alter persistence and treatment response; the human microbiome contributes to digestion, vitamin production, immune development and pathogen exclusion. Probiotics are live microorganisms intended to confer a benefit in an adequate amount, while thermophiles, psychrophiles and acidophiles are classified by environmental tolerance; none of these labels proves safety or effectiveness in every context.

Answer: C. Explanation: Innate immunity includes barriers, phagocytes, natural killer cells, complement, interferons, inflammation and fever; adaptive immunity uses antigen-specific B and T lymphocytes and can form memory. B-cell lineages produce antibodies, T cells perform coordinating, cytotoxic or regulatory roles, and peripheral immune tolerance including regulatory T cells limits attack on self; immunity is not antibodies alone. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q52. Which option avoids the standard UPSC close-option trap about Innate adaptive and tolerance architecture?

A. Photosynthesis stores light energy in chemical form while plants also respire continuously; xylem conducts water and minerals and phloem translocates organic food. Stomata regulate gas exchange and water loss, transpiration contributes to upward water movement, and auxin, gibberellin, cytokinin, abscisic acid and ethylene have distinct growth or stress roles; C3, C4 and CAM are different carbon-fixation strategies. B. Groundnut, horse-gram and soybean are routed pea-family examples; cicada, froghopper and pond skater are insects; poisonous species occur in multiple animal groups, so a broad group name does not prove toxicity. Cassava is a woody shrub, ginger a rhizomatous herb, Malabar spinach a climbing vine, mint a herb and papaya a large herb; mushrooms can have species-specific edible, medicinal, psychoactive, insect-pathogenic or bioluminescent properties. C. Cellulose is a structural plant polysaccharide; decomposer fungi and bacteria secrete cellulases that hydrolyse it into smaller sugars, after which microbial metabolism returns carbon through soil and food-web processes. Moisture, temperature, oxygen, lignin and community composition constrain the rate, and natural decomposition is not proof of an industrially viable biofuel process. D. Innate immunity includes barriers, phagocytes, natural killer cells, complement, interferons, inflammation and fever; adaptive immunity uses antigen-specific B and T lymphocytes and can form memory. B-cell lineages produce antibodies, T cells perform coordinating, cytotoxic or regulatory roles, and peripheral immune tolerance including regulatory T cells limits attack on self; immunity is not antibodies alone.

Answer: D. Explanation: Innate immunity includes barriers, phagocytes, natural killer cells, complement, interferons, inflammation and fever; adaptive immunity uses antigen-specific B and T lymphocytes and can form memory. B-cell lineages produce antibodies, T cells perform coordinating, cytotoxic or regulatory roles, and peripheral immune tolerance including regulatory T cells limits attack on self; immunity is not antibodies alone. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q53. Which statement correctly identifies Pathogen transmission vector and zoonosis map?

A. Bacteria are cellular organisms, viruses are acellular and depend on host-cell machinery, fungi are eukaryotes, protozoa are single-celled eukaryotes and helminths are multicellular parasites. Transmission may be respiratory, faeco-oral, blood-borne, sexual, vector-borne, fomite-linked or zoonotic; a vector such as Aedes, Anopheles, Culex or sandfly carries a pathogen but is not the disease itself. B. Photosynthesis stores light energy in chemical form while plants also respire continuously; xylem conducts water and minerals and phloem translocates organic food. Stomata regulate gas exchange and water loss, transpiration contributes to upward water movement, and auxin, gibberellin, cytokinin, abscisic acid and ethylene have distinct growth or stress roles; C3, C4 and CAM are different carbon-fixation strategies. C. Antibiotics target susceptible bacteria rather than viruses, while antivirals interfere with virus-specific replication stages. Antimicrobial resistance is selection and spread of resistant microorganisms under pressures including misuse, incomplete treatment, agricultural use and environmental residues; the microbe or population becomes resistant, not the patient's body. D. A biofilm is a surface-associated microbial community embedded in a matrix that can alter persistence and treatment response; the human microbiome contributes to digestion, vitamin production, immune development and pathogen exclusion. Probiotics are live microorganisms intended to confer a benefit in an adequate amount, while thermophiles, psychrophiles and acidophiles are classified by environmental tolerance; none of these labels proves safety or effectiveness in every context.

Answer: A. Explanation: Bacteria are cellular organisms, viruses are acellular and depend on host-cell machinery, fungi are eukaryotes, protozoa are single-celled eukaryotes and helminths are multicellular parasites. Transmission may be respiratory, faeco-oral, blood-borne, sexual, vector-borne, fomite-linked or zoonotic; a vector such as Aedes, Anopheles, Culex or sandfly carries a pathogen but is not the disease itself. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q54. Which option preserves the technical boundary of Pathogen transmission vector and zoonosis map?

A. Groundnut, horse-gram and soybean are routed pea-family examples; cicada, frog hopper and pond skater are insects; poisonous species occur in multiple animal groups, so a broad group name does not prove toxicity. Cassava is a woody shrub, ginger a rhizomatous herb, Malabar spinach a climbing vine, mint a herb and papaya a large herb; mushrooms can have species-specific edible, medicinal, psychoactive, insect-pathogenic or bioluminescent properties. B. Bacteria are cellular organisms, viruses are acellular and depend on host-cell machinery, fungi are eukaryotes, protozoa are single-celled eukaryotes and helminths are multicellular parasites. Transmission may be respiratory, faeco-oral, blood-borne, sexual, vector-borne, fomite-linked or zoonotic; a vector such as Aedes, Anopheles, Culex or sandfly carries a pathogen but is not the disease itself. C. Photosynthesis stores light energy in chemical form while plants also respire continuously; xylem conducts water and minerals and phloem translocates organic food. Stomata regulate gas exchange and water loss, transpiration contributes to upward water movement, and auxin, gibberellin, cytokinin, abscisic acid and ethylene have distinct growth or stress roles; C3, C4 and CAM are different carbon-fixation strategies. D. A biofilm is a surface-associated microbial community embedded in a matrix that can alter persistence and treatment response; the human microbiome contributes to digestion, vitamin production, immune development and pathogen exclusion. Probiotics are live microorganisms intended to confer a benefit in an adequate amount, while thermophiles, psychrophiles and acidophiles are classified by environmental tolerance; none of these labels proves safety or effectiveness in every context.

Answer: B. Explanation: Bacteria are cellular organisms, viruses are acellular and depend on host-cell machinery, fungi are eukaryotes, protozoa are single-celled eukaryotes and helminths are multicellular parasites. Transmission may be respiratory, faeco-oral, blood-borne, sexual, vector-borne, fomite-linked or zoonotic; a vector such as Aedes, Anopheles, Culex or sandfly carries a pathogen but is not the disease itself. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q55. Which statement uses Pathogen transmission vector and zoonosis map without changing its institution, unit or status?

A. Cellulose is a structural plant polysaccharide; decomposer fungi and bacteria secrete cellulases that hydrolyse it into smaller sugars, after which microbial metabolism returns carbon through soil and food-web processes. Moisture, temperature, oxygen, lignin and community composition constrain the rate, and natural decomposition is not proof of an industrially viable biofuel process. B. Photosynthesis stores light energy in chemical form while plants also respire continuously; xylem conducts water and minerals and phloem translocates organic food. Stomata regulate gas exchange and water loss, transpiration contributes to upward water movement, and auxin, gibberellin, cytokinin, abscisic acid and ethylene have distinct growth or stress roles; C3, C4 and CAM are different carbon-fixation strategies. C. Bacteria are cellular organisms, viruses are acellular and depend on host-cell machinery, fungi are eukaryotes, protozoa are single-celled eukaryotes and helminths are multicellular parasites. Transmission may be respiratory, faeco-oral, blood-borne, sexual, vector-borne, fomite-linked or zoonotic; a vector such as Aedes, Anopheles, Culex or sandfly carries a pathogen but is not the disease itself. D. Groundnut, horse-gram and soybean are routed pea-family examples; cicada, frog hopper and pond skater are insects; poisonous species occur in multiple animal groups, so a broad group name does not prove toxicity. Cassava is a woody shrub, ginger a rhizomatous herb, Malabar spinach a climbing vine, mint a herb and papaya a large herb; mushrooms can have species-specific edible, medicinal, psychoactive, insect-pathogenic or bioluminescent properties.

Answer: C. Explanation: Bacteria are cellular organisms, viruses are acellular and depend on host-cell machinery, fungi are eukaryotes, protozoa are single-celled eukaryotes and helminths are multicellular parasites. Transmission may be respiratory, faeco-oral, blood-borne, sexual, vector-borne, fomite-linked or zoonotic; a vector such as Aedes, Anopheles, Culex or sandfly carries a pathogen but is not the disease itself. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q56. Which option avoids the standard UPSC close-option trap about Pathogen transmission vector and zoonosis map?

A. Cellulose is a structural plant polysaccharide; decomposer fungi and bacteria secrete cellulases that hydrolyse it into smaller sugars, after which microbial metabolism returns carbon through soil and food-web processes. Moisture, temperature, oxygen, lignin and community composition constrain the rate, and natural decomposition is not proof of an industrially viable biofuel process. B. ICMR performs and supports biomedical research, NCDC and the IDSP or IHIP network support surveillance and outbreak response under MoHFW, State systems deliver programmes, and CDSCO regulates drugs and vaccines; DBT, CSIR-CCMB and CSIR-IGIB occupy different research roles. A biological mechanism, association, surveillance signal, WHO validation, elimination as a public-health problem, regulatory approval and eradication are separate evidence rungs. C. Groundnut, horse-gram and soybean are routed pea-family examples; cicada, frog hopper and pond skater are insects; poisonous species occur in multiple animal groups, so a broad group name does not prove toxicity. Cassava is a woody shrub, ginger a rhizomatous herb, Malabar spinach a climbing vine, mint a herb and papaya a large herb; mushrooms can have species-specific edible, medicinal, psychoactive, insect-pathogenic or bioluminescent properties. D. Bacteria are cellular organisms, viruses are acellular and depend on host-cell machinery, fungi are eukaryotes, protozoa are single-celled eukaryotes and helminths are multicellular parasites. Transmission may be respiratory, faeco-oral, blood-borne, sexual, vector-borne, fomite-linked or zoonotic; a vector such as Aedes, Anopheles, Culex or sandfly carries a pathogen but is not the disease itself.

Answer: D. Explanation: Bacteria are cellular organisms, viruses are acellular and depend on host-cell machinery, fungi are eukaryotes, protozoa are single-celled eukaryotes and helminths are multicellular parasites. Transmission may be respiratory, faeco-oral, blood-borne, sexual, vector-borne, fomite-linked or zoonotic; a vector such as Aedes, Anopheles, Culex or sandfly carries a pathogen but is not the disease itself. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q57. Which statement correctly identifies Antibiotic antiviral and AMR boundary?

A. Antibiotics target susceptible bacteria rather than viruses, while antivirals interfere with virus-specific replication stages. Antimicrobial resistance is selection and spread of resistant microorganisms under pressures including misuse, incomplete treatment, agricultural use and environmental residues; the microbe or population becomes resistant, not the patient's body. B. A biofilm is a surface-associated microbial community embedded in a matrix that can alter persistence and treatment response; the human microbiome contributes to digestion, vitamin production, immune development and pathogen exclusion. Probiotics are live microorganisms intended to confer a benefit in an adequate amount, while thermophiles, psychrophiles and acidophiles are classified by environmental tolerance; none of these labels proves safety or effectiveness in every context. C. Groundnut, horse-gram and soybean are routed pea-family examples; cicada, frog hopper and pond skater are insects; poisonous species occur in multiple animal groups, so a broad group name does not prove toxicity. Cassava is a woody shrub, ginger a rhizomatous herb, Malabar spinach a climbing vine, mint a herb and papaya a large herb; mushrooms can have species-specific edible, medicinal, psychoactive, insect-pathogenic or bioluminescent properties. D. Photosynthesis stores light energy in chemical form while plants also respire continuously; xylem conducts water and minerals and phloem translocates organic food. Stomata regulate gas exchange and water loss, transpiration contributes to upward water movement, and auxin, gibberellin, cytokinin, abscisic acid and ethylene have distinct growth or stress roles; C3, C4 and CAM are different carbon-fixation strategies.

Answer: A. Explanation: Antibiotics target susceptible bacteria rather than viruses, while antivirals interfere with virus-specific replication stages. Antimicrobial resistance is selection and spread of resistant microorganisms under pressures including misuse, incomplete treatment, agricultural use and environmental residues; the microbe or population becomes resistant, not the patient's body. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q58. Which option preserves the technical boundary of Antibiotic antiviral and AMR boundary?

A. Cellulose is a structural plant polysaccharide; decomposer fungi and bacteria secrete cellulases that hydrolyse it into smaller sugars, after which microbial metabolism returns carbon through soil and food-web processes. Moisture, temperature, oxygen, lignin and community composition constrain the rate, and natural decomposition is not proof of an industrially viable biofuel process. B. Antibiotics target susceptible bacteria rather than viruses, while antivirals interfere with virus-specific replication stages. Antimicrobial resistance is selection and spread of resistant microorganisms under pressures including misuse, incomplete treatment, agricultural use and environmental residues; the microbe or population becomes resistant, not the patient's body. C. Groundnut, horse-gram and soybean are routed pea-family examples; cicada, frog hopper and pond skater are insects; poisonous species occur in multiple animal groups, so a broad group name does not prove toxicity. Cassava is a woody shrub, ginger a rhizomatous herb, Malabar spinach a climbing vine, mint a herb and papaya a large herb; mushrooms can have species-specific edible, medicinal, psychoactive, insect-pathogenic or bioluminescent properties. D. Photosynthesis stores light energy in chemical form while plants also respire continuously; xylem conducts water and minerals and phloem translocates organic food. Stomata regulate gas exchange and water loss, transpiration contributes to upward water movement, and auxin, gibberellin, cytokinin, abscisic acid and ethylene have distinct growth or stress roles; C3, C4 and CAM are different carbon-fixation strategies.

Answer: B. Explanation: Antibiotics target susceptible bacteria rather than viruses, while antivirals interfere with virus-specific replication stages. Antimicrobial resistance is selection and spread of resistant microorganisms under pressures including misuse, incomplete treatment, agricultural use and environmental residues; the microbe or population becomes resistant, not the patient's body. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q59. Which statement uses Antibiotic antiviral and AMR boundary without changing its institution, unit or status?

A. Groundnut, horse-gram and soybean are routed pea-family examples; cicada, froghopper and pond skater are insects; poisonous species occur in multiple animal groups, so a broad group name does not prove toxicity. Cassava is a woody shrub, ginger a rhizomatous herb, Malabar spinach a climbing vine, mint a herb and papaya a large herb; mushrooms can have species-specific edible, medicinal, psychoactive, insect-pathogenic or bioluminescent properties. B. ICMR performs and supports biomedical research, NCDC and the IDSP or IHIP network support surveillance and outbreak response under MoHFW, State systems deliver programmes, and CDSCO regulates drugs and vaccines; DBT, CSIR-CCMB and CSIR-IGIB occupy different research roles. A biological mechanism, association, surveillance signal, WHO validation, elimination as a public-health problem, regulatory approval and eradication are separate evidence rungs. C. Antibiotics target susceptible bacteria rather than viruses, while antivirals interfere with virus-specific replication stages. Antimicrobial resistance is selection and spread of resistant microorganisms under pressures including misuse, incomplete treatment, agricultural use and environmental residues; the microbe or population becomes resistant, not the patient's body. D. Cellulose is a structural plant polysaccharide; decomposer fungi and bacteria secrete cellulases that hydrolyse it into smaller sugars, after which microbial metabolism returns carbon through soil and food-web processes. Moisture, temperature, oxygen, lignin and community composition constrain the rate, and natural decomposition is not proof of an industrially viable biofuel process.

Answer: C. Explanation: Antibiotics target susceptible bacteria rather than viruses, while antivirals interfere with virus-specific replication stages. Antimicrobial resistance is selection and spread of resistant microorganisms under pressures including misuse, incomplete treatment, agricultural use and environmental residues; the microbe or population becomes resistant, not the patient's body. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q60. Which option avoids the standard UPSC close-option trap about Antibiotic antiviral and AMR boundary?

A. Cellulose is a structural plant polysaccharide; decomposer fungi and bacteria secrete cellulases that hydrolyse it into smaller sugars, after which microbial metabolism returns carbon through soil and food-web processes. Moisture, temperature, oxygen, lignin and community composition constrain the rate, and natural decomposition is not proof of an industrially viable biofuel process. B. ICMR performs and supports biomedical research, NCDC and the IDSP or IHIP network support surveillance and outbreak response under MoHFW, State systems deliver programmes, and CDSCO regulates drugs and vaccines; DBT, CSIR-CCMB and CSIR-IGIB occupy different research roles. A biological mechanism, association, surveillance signal, WHO validation, elimination as a public-health problem, regulatory approval and eradication are separate evidence rungs. C. A cell is the basic structural and functional unit of life; prokaryotes lack a membrane-bound nucleus and membrane-bound organelles, whereas eukaryotes possess them. The nucleus stores most genetic material, ribosomes synthesize protein, mitochondria support aerobic ATP generation, the endoplasmic reticulum and Golgi system process and move products, lysosomes contain hydrolytic enzymes, and plant cells additionally possess a cellulose wall, chloroplasts and a large vacuole. D. Antibiotics target susceptible bacteria rather than viruses, while antivirals interfere with virus-specific replication stages. Antimicrobial resistance is selection and spread of resistant microorganisms under pressures including misuse, incomplete treatment, agricultural use and environmental residues; the microbe or population becomes resistant, not the patient's body.

Answer: D. Explanation: Antibiotics target susceptible bacteria rather than viruses, while antivirals interfere with virus-specific replication stages. Antimicrobial resistance is selection and spread of resistant microorganisms under pressures including misuse, incomplete treatment, agricultural use and environmental residues; the microbe or population becomes resistant, not the patient's body. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q61. Which statement correctly identifies Biofilms microbiome probiotics and extremophiles?

A. A biofilm is a surface-associated microbial community embedded in a matrix that can alter persistence and treatment response; the human microbiome contributes to digestion, vitamin production, immune development and pathogen exclusion. Probiotics are live microorganisms intended to confer a benefit in an adequate amount, while thermophiles, psychrophiles and acidophiles are classified by environmental tolerance; none of these labels proves safety or effectiveness in every context. B. Photosynthesis stores light energy in chemical form while plants also respire continuously; xylem conducts water and minerals and phloem translocates organic food. Stomata regulate gas exchange and water loss, transpiration contributes to upward water movement, and auxin, gibberellin, cytokinin, abscisic acid and ethylene have distinct growth or stress roles; C3, C4 and CAM are different carbon-fixation strategies. C. Groundnut, horse-gram and soybean are routed pea-family examples; cicada, frog hopper and pond skater are insects; poisonous species occur in multiple animal groups, so a broad group name does not prove toxicity. Cassava is a woody shrub, ginger a rhizomatous herb, Malabar spinach a climbing vine, mint a herb and papaya a large herb; mushrooms can have species-specific edible, medicinal, psychoactive, insect-pathogenic or bioluminescent properties. D. Cellulose is a structural plant polysaccharide; decomposer fungi and bacteria secrete cellulases that hydrolyse it into smaller sugars, after which microbial metabolism returns carbon through soil and food-web processes. Moisture, temperature, oxygen, lignin and community composition constrain the rate, and natural decomposition is not proof of an industrially viable biofuel process.

Answer: A. Explanation: A biofilm is a surface-associated microbial community embedded in a matrix that can alter persistence and treatment response; the human microbiome contributes to digestion, vitamin production, immune development and pathogen exclusion. Probiotics are live microorganisms intended to confer a benefit in an adequate amount, while thermophiles, psychrophiles and acidophiles are classified by environmental tolerance; none of these labels proves safety or effectiveness in every context. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q62. Which option preserves the technical boundary of Biofilms microbiome probiotics and extremophiles?

A. Cellulose is a structural plant polysaccharide; decomposer fungi and bacteria secrete cellulases that hydrolyse it into smaller sugars, after which microbial metabolism returns carbon through soil and food-web processes. Moisture, temperature, oxygen, lignin and community composition constrain the rate, and natural decomposition is not proof of an industrially viable biofuel process. B. A biofilm is a surface-associated microbial community embedded in a matrix that can alter persistence and treatment response; the human microbiome contributes to digestion, vitamin production, immune development and pathogen exclusion. Probiotics are live microorganisms intended to confer a benefit in an adequate amount, while thermophiles, psychrophiles and acidophiles are classified by environmental tolerance; none of these labels proves safety or effectiveness in every context. C. Groundnut, horse-gram and soybean are routed pea-family examples; cicada, frog hopper and pond skater are insects; poisonous species occur in multiple animal groups, so a broad group name does not prove toxicity. Cassava is a woody shrub, ginger a rhizomatous herb, Malabar spinach a climbing vine, mint a herb and papaya a large herb; mushrooms can have species-specific edible, medicinal, psychoactive, insect-pathogenic or bioluminescent properties. D. ICMR performs and supports biomedical research, NCDC and the IDSP or IHIP network support surveillance and outbreak response under MoHFW, State systems deliver programmes, and CDSCO regulates drugs and vaccines; DBT, CSIR-CCMB and CSIR-IGIB occupy different research roles. A biological mechanism, association, surveillance signal, WHO validation, elimination as a public-health problem, regulatory approval and eradication are separate evidence rungs.

Answer: B. Explanation: A biofilm is a surface-associated microbial community embedded in a matrix that can alter persistence and treatment response; the human microbiome contributes to digestion, vitamin production, immune development and pathogen exclusion. Probiotics are live microorganisms intended to confer a benefit in an adequate amount, while thermophiles, psychrophiles and acidophiles are classified by environmental tolerance; none of these labels proves safety or effectiveness in every context. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q63. Which statement uses Biofilms microbiome probiotics and extremophiles without changing its institution, unit or status?

A. A cell is the basic structural and functional unit of life; prokaryotes lack a membrane-bound nucleus and membrane-bound organelles, whereas eukaryotes possess them. The nucleus stores most genetic material, ribosomes synthesize protein, mitochondria support aerobic ATP generation, the endoplasmic reticulum and Golgi system process and move products, lysosomes contain hydrolytic enzymes, and plant cells additionally possess a cellulose wall, chloroplasts and a large vacuole. B. Cellulose is a structural plant polysaccharide; decomposer fungi and bacteria secrete cellulases that hydrolyse it into smaller sugars, after which microbial metabolism returns carbon through soil and food-web processes. Moisture, temperature, oxygen, lignin and community composition constrain the rate, and natural decomposition is not proof of an industrially viable biofuel process. C. A biofilm is a surface-associated microbial community embedded in a matrix that can alter persistence and treatment response; the human microbiome contributes to digestion, vitamin production, immune development and pathogen exclusion. Probiotics are live microorganisms intended to confer a benefit in an adequate amount, while thermophiles, psychrophiles and acidophiles are classified by environmental tolerance; none of these labels proves safety or effectiveness in every context. D. ICMR performs and supports biomedical research, NCDC and the IDSP or IHIP network support surveillance and outbreak response under MoHFW, State systems deliver programmes, and CDSCO regulates drugs and vaccines; DBT, CSIR-CCMB and CSIR-IGIB occupy different research roles. A biological mechanism, association, surveillance signal, WHO validation, elimination as a public-health problem, regulatory approval and eradication are separate evidence rungs.

Answer: C. Explanation: A biofilm is a surface-associated microbial community embedded in a matrix that can alter persistence and treatment response; the human microbiome contributes to digestion, vitamin production, immune development and pathogen exclusion. Probiotics are live microorganisms intended to confer a benefit in an adequate amount, while thermophiles, psychrophiles and acidophiles are classified by environmental tolerance; none of these labels proves safety or effectiveness in every context. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q64. Which option avoids the standard UPSC close-option trap about Biofilms microbiome probiotics and extremophiles?

A. A cell is the basic structural and functional unit of life; prokaryotes lack a membrane-bound nucleus and membrane-bound organelles, whereas eukaryotes possess them. The nucleus stores most genetic material, ribosomes synthesize protein, mitochondria support aerobic ATP generation, the endoplasmic reticulum and Golgi system process and move products, lysosomes contain hydrolytic enzymes, and plant cells additionally possess a cellulose wall, chloroplasts and a large vacuole. B. Carbohydrates commonly supply or store energy and provide structures such as cellulose, lipids store energy and form membranes, proteins perform structural, transport, receptor and catalytic roles, and nucleic acids store or express information. Enzymes are biological catalysts whose activity depends on substrate fit and conditions such as temperature and pH; a biomolecule class does not establish one universal function. C. ICMR performs and supports biomedical research, NCDC and the IDSP or IHIP network support surveillance and outbreak response under MoHFW, State systems deliver programmes, and CDSCO regulates drugs and vaccines; DBT, CSIR-CCMB and CSIR-IGIB occupy different research roles. A biological mechanism, association, surveillance signal, WHO validation, elimination as a public-health problem, regulatory approval and eradication are separate evidence rungs. D. A biofilm is a surface-associated microbial community embedded in a matrix that can alter persistence and treatment response; the human microbiome contributes to digestion, vitamin production, immune development and pathogen exclusion. Probiotics are live microorganisms intended to confer a benefit in an adequate amount, while thermophiles, psychrophiles and acidophiles are classified by environmental tolerance; none of these labels proves safety or effectiveness in every context.

Answer: D. Explanation: A biofilm is a surface-associated microbial community embedded in a matrix that can alter persistence and treatment response; the human microbiome contributes to digestion, vitamin production, immune development and pathogen exclusion. Probiotics are live microorganisms intended to confer a benefit in an adequate amount, while thermophiles, psychrophiles and acidophiles are classified by environmental tolerance; none of these labels proves safety or effectiveness in every context. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q65. Which statement correctly identifies Plant physiology transport and stress response?

A. Photosynthesis stores light energy in chemical form while plants also respire continuously; xylem conducts water and minerals and phloem translocates organic food. Stomata regulate gas exchange and water loss, transpiration contributes to upward water movement, and auxin, gibberellin, cytokinin, abscisic acid and ethylene have distinct growth or stress roles; C₃, C₄ and CAM are different carbon-fixation strategies. B. Groundnut, horse-gram and soybean are routed pea-family examples; cicada, frog hopper and pond skater are insects; poisonous species occur in multiple animal groups, so a broad group name does not prove toxicity. Cassava is a woody shrub, ginger a rhizomatous herb, Malabar spinach a climbing vine, mint a herb and papaya a large herb; mushrooms can have species-specific edible, medicinal, psychoactive, insect-pathogenic or bioluminescent properties. C. ICMR performs and supports biomedical research, NCDC and the IDSP or IHIP network support surveillance and outbreak response under MoHFW, State systems deliver programmes, and CDSCO regulates drugs and vaccines; DBT, CSIR-CCMB and CSIR-IGIB occupy different research roles. A biological mechanism, association, surveillance signal, WHO validation, elimination as a public-health problem, regulatory approval and eradication are separate evidence rungs. D. Cellulose is a structural plant polysaccharide; decomposer fungi and bacteria secrete cellulases that hydrolyse it into smaller sugars, after which microbial metabolism returns carbon through soil and food-web processes. Moisture, temperature, oxygen, lignin and community composition constrain the rate, and natural decomposition is not proof of an industrially viable biofuel process.

Answer: A. Explanation: Photosynthesis stores light energy in chemical form while plants also respire continuously; xylem conducts water and minerals and phloem translocates organic food. Stomata regulate gas exchange and water loss, transpiration contributes to upward water movement, and auxin, gibberellin, cytokinin, abscisic acid and ethylene have distinct growth or stress roles; C₃, C₄ and CAM are different carbon-fixation strategies. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q66. Which option preserves the technical boundary of Plant physiology transport and stress response?

A. ICMR performs and supports biomedical research, NCDC and the IDSP or IHIP network support surveillance and outbreak response under MoHFW, State systems deliver programmes, and CDSCO regulates drugs and vaccines; DBT, CSIR-CCMB and CSIR-IGIB occupy different research roles. A biological mechanism, association, surveillance signal, WHO validation, elimination as a public-health problem, regulatory approval and eradication are separate evidence rungs. B. Photosynthesis stores light energy in chemical form while plants also respire continuously; xylem conducts water and minerals and phloem translocates organic food. Stomata regulate gas exchange and water loss, transpiration contributes to upward water movement, and auxin, gibberellin, cytokinin, abscisic acid and ethylene have distinct growth or stress roles; C₃, C₄ and CAM are different carbon-fixation strategies. C. A cell is the basic structural and functional unit of life; prokaryotes lack a membrane-bound nucleus and membrane-bound organelles, whereas eukaryotes possess them. The nucleus stores most genetic material, ribosomes synthesize protein, mitochondria support aerobic ATP generation, the endoplasmic reticulum and Golgi system process and move products, lysosomes contain hydrolytic enzymes, and plant cells additionally possess a cellulose wall, chloroplasts and a large vacuole. D. Cellulose is a structural plant polysaccharide; decomposer fungi and bacteria secrete cellulases that hydrolyse it into smaller sugars, after which microbial metabolism returns carbon through soil and food-web processes. Moisture, temperature, oxygen, lignin and community composition constrain the rate, and natural decomposition is not proof of an industrially viable biofuel process.

Answer: B. Explanation: Photosynthesis stores light energy in chemical form while plants also respire continuously; xylem conducts water and minerals and phloem translocates organic food. Stomata regulate gas exchange and water loss, transpiration contributes to upward water movement, and auxin, gibberellin, cytokinin, abscisic acid and ethylene have distinct growth or stress roles; C₃, C₄ and CAM are different carbon-fixation strategies. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q67. Which statement uses Plant physiology transport and stress response without changing its institution, unit or status?

A. Carbohydrates commonly supply or store energy and provide structures such as cellulose, lipids store energy and form membranes, proteins perform structural, transport, receptor and catalytic roles, and nucleic acids store or express information. Enzymes are biological catalysts whose activity depends on substrate fit and conditions such as temperature and pH; a biomolecule class does not establish one universal function. B. A cell is the basic structural and functional unit of life; prokaryotes lack a membrane-bound nucleus and membrane-bound organelles, whereas eukaryotes possess them. The nucleus stores most genetic material, ribosomes synthesize protein, mitochondria support aerobic ATP generation, the endoplasmic reticulum and Golgi system process and move products, lysosomes contain hydrolytic enzymes, and plant cells additionally possess a cellulose wall, chloroplasts and a large vacuole. C. Photosynthesis stores light energy in chemical form while plants also respire continuously; xylem conducts water and minerals and phloem translocates organic food. Stomata regulate gas exchange and water loss, transpiration contributes to upward water movement, and auxin, gibberellin, cytokinin, abscisic acid and ethylene have distinct growth or stress roles; C₃, C₄ and CAM are different carbon-fixation strategies. D. ICMR performs and supports biomedical research, NCDC and the IDSP or IHIP network support surveillance and outbreak response under MoHFW, State systems deliver programmes, and CDSCO regulates drugs and vaccines; DBT, CSIR-CCMB and CSIR-IGIB occupy different research roles. A biological mechanism, association, surveillance signal, WHO validation, elimination as a public-health problem, regulatory approval and eradication are separate evidence rungs.

Answer: C. Explanation: Photosynthesis stores light energy in chemical form while plants also respire continuously; xylem conducts water and minerals and phloem translocates organic food. Stomata regulate gas exchange and water loss, transpiration contributes to upward water movement, and auxin, gibberellin, cytokinin, abscisic acid and ethylene have distinct growth or stress roles; C₃, C₄ and CAM are different carbon-fixation strategies. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q68. Which option avoids the standard UPSC close-option trap about Plant physiology transport and stress response?

A. A cell is the basic structural and functional unit of life; prokaryotes lack a membrane-bound nucleus and membrane-bound organelles, whereas eukaryotes possess them. The nucleus stores most genetic material, ribosomes synthesize protein, mitochondria support aerobic ATP generation, the endoplasmic reticulum and Golgi system process and move products, lysosomes contain hydrolytic enzymes, and plant cells additionally possess a cellulose wall, chloroplasts and a large vacuole. B. Carbohydrates commonly supply or store energy and provide structures such as cellulose, lipids store energy and form membranes, proteins perform structural, transport, receptor and catalytic roles, and nucleic acids store or express information. Enzymes are biological catalysts whose activity depends on substrate fit and conditions such as temperature and pH; a biomolecule class does not establish one universal function. C. Diffusion moves particles down a concentration gradient, osmosis is net water movement across a selectively permeable membrane, and active transport uses cellular energy to move substances against a gradient. Tonicity describes the effect of an external solution on cell water balance, so swelling, plasmolysis and normal volume require the membrane and concentration context. D. Photosynthesis stores light energy in chemical form while plants also respire continuously; xylem conducts water and minerals and phloem translocates organic food. Stomata regulate gas exchange and water loss, transpiration contributes to upward water movement, and auxin, gibberellin, cytokinin, abscisic acid and ethylene have distinct growth or stress roles; C₃, C₄ and CAM are different carbon-fixation strategies.

Answer: D. Explanation: Photosynthesis stores light energy in chemical form while plants also respire continuously; xylem conducts water and minerals and phloem translocates organic food. Stomata regulate gas exchange and water loss, transpiration contributes to upward water movement, and auxin, gibberellin, cytokinin, abscisic acid and ethylene have distinct growth or stress roles; C₃, C₄ and CAM are different carbon-fixation strategies. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q69. Which statement correctly identifies Routed taxonomy and organism-form distinctions?

A. Groundnut, horse-gram and soybean are routed pea-family examples; cicada, frog hopper and pond skater are insects; poisonous species occur in multiple animal groups, so a broad group name does not prove toxicity. Cassava is a woody shrub, ginger a rhizomatous herb, Malabar spinach a climbing vine, mint a herb and papaya a large herb; mushrooms can have species-specific edible, medicinal, psychoactive, insect-pathogenic or bioluminescent properties. B. ICMR performs and supports biomedical research, NCDC and the IDSP or IHIP network support surveillance and outbreak response under MoHFW, State systems deliver programmes, and CDSCO regulates drugs and vaccines; DBT, CSIR-CCMB and CSIR-IGIB occupy different research roles. A biological mechanism, association, surveillance signal, WHO validation, elimination as a public-health problem, regulatory approval and eradication are separate evidence rungs. C. A cell is the basic structural and functional unit of life; prokaryotes lack a membrane-bound nucleus and membrane-bound organelles, whereas eukaryotes possess them. The nucleus stores most genetic material, ribosomes synthesize protein, mitochondria support aerobic ATP generation, the endoplasmic reticulum and Golgi system process and move products, lysosomes contain hydrolytic enzymes, and plant cells additionally possess a cellulose wall, chloroplasts and a large vacuole. D. Cellulose is a structural plant polysaccharide; decomposer fungi and bacteria secrete cellulases that hydrolyse it into smaller sugars, after which microbial metabolism returns carbon through soil and food-web processes. Moisture, temperature, oxygen, lignin and community composition constrain the rate, and natural decomposition is not proof of an industrially viable biofuel process.

Answer: A. Explanation: Groundnut, horse-gram and soybean are routed pea-family examples; cicada, frog hopper and pond skater are insects; poisonous species occur in multiple animal groups, so a broad group name does not prove toxicity. Cassava is a woody shrub, ginger a rhizomatous herb, Malabar spinach a climbing vine, mint a herb and papaya a large herb; mushrooms can have species-specific edible, medicinal, psychoactive, insect-pathogenic or bioluminescent properties. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q70. Which option preserves the technical boundary of Routed taxonomy and organism-form distinctions?

A. ICMR performs and supports biomedical research, NCDC and the IDSP or IHIP network support surveillance and outbreak response under MoHFW, State systems deliver programmes, and CDSCO regulates drugs and vaccines; DBT, CSIR-CCMB and CSIR-IGIB occupy different research roles. A biological mechanism, association, surveillance signal, WHO validation, elimination as a public-health problem, regulatory approval and eradication are separate evidence rungs. B. Groundnut, horse-gram and soybean are routed pea-family examples; cicada, frog hopper and pond skater are insects; poisonous species occur in multiple animal groups, so a broad group name does not prove toxicity. Cassava is a woody shrub, ginger a rhizomatous herb, Malabar spinach a climbing vine, mint a herb and papaya a large herb; mushrooms can have species-specific edible, medicinal, psychoactive, insect-pathogenic or bioluminescent properties. C. A cell is the basic structural and functional unit of life; prokaryotes lack a membrane-bound nucleus and membrane-bound organelles, whereas eukaryotes possess them. The nucleus stores most genetic material, ribosomes synthesize protein, mitochondria support aerobic ATP generation, the endoplasmic reticulum and Golgi system process and move products, lysosomes contain hydrolytic enzymes, and plant cells additionally possess a cellulose wall, chloroplasts and a large vacuole. D. Carbohydrates commonly supply or store energy and provide structures such as cellulose, lipids store energy and form membranes, proteins perform structural, transport, receptor and catalytic roles, and nucleic acids store or express information. Enzymes are biological catalysts whose activity depends on substrate fit and conditions such as temperature and pH; a biomolecule class does not establish one universal function.

Answer: B. Explanation: Groundnut, horse-gram and soybean are routed pea-family examples; cicada, frog hopper and pond skater are insects; poisonous species occur in multiple animal groups, so a broad group name does not prove toxicity. Cassava is a woody shrub, ginger a rhizomatous herb, Malabar spinach a climbing vine, mint a herb and papaya a large herb; mushrooms can have species-specific edible, medicinal, psychoactive, insect-pathogenic or bioluminescent properties. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q71. Which statement uses Routed taxonomy and organism-form distinctions without changing its institution, unit or status?

A. A cell is the basic structural and functional unit of life; prokaryotes lack a membrane-bound nucleus and membrane-bound organelles, whereas eukaryotes possess them. The nucleus stores most genetic material, ribosomes synthesize protein, mitochondria support aerobic ATP generation, the endoplasmic reticulum and Golgi system process and move products, lysosomes contain hydrolytic enzymes, and plant cells additionally possess a cellulose wall, chloroplasts and a large vacuole. B. Diffusion moves particles down a concentration gradient, osmosis is net water movement across a selectively permeable membrane, and active transport uses cellular energy to move substances against a gradient. Tonicity describes the effect of an external solution on cell water balance, so swelling, plasmolysis and normal volume require the membrane and concentration context. C. Groundnut, horse-gram and soybean are routed pea-family examples; cicada, froghopper and pond skater are insects; poisonous species occur in multiple animal groups, so a broad group name does not prove toxicity. Cassava is a woody shrub, ginger a rhizomatous herb, Malabar spinach a climbing vine, mint a herb and papaya a large herb; mushrooms can have species-specific edible, medicinal, psychoactive, insect-pathogenic or bioluminescent properties. D. Carbohydrates commonly supply or store energy and provide structures such as cellulose, lipids store energy and form membranes, proteins perform structural, transport, receptor and catalytic roles, and nucleic acids store or express information. Enzymes are biological catalysts whose activity depends on substrate fit and conditions such as temperature and pH; a biomolecule class does not establish one universal function.

Answer: C. Explanation: Groundnut, horse-gram and soybean are routed pea-family examples; cicada, froghopper and pond skater are insects; poisonous species occur in multiple animal groups, so a broad group name does not prove toxicity. Cassava is a woody shrub, ginger a rhizomatous herb, Malabar spinach a climbing vine, mint a herb and papaya a large herb; mushrooms can have species-specific edible, medicinal, psychoactive, insect-pathogenic or bioluminescent properties. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q72. Which option avoids the standard UPSC close-option trap about Routed taxonomy and organism-form distinctions?

A. Mitosis is one division producing two genetically similar daughter cells for growth and repair, while meiosis consists of two divisions producing four haploid products for sexual reproduction; crossing over and independent assortment generate variation. Neither process should be described as gene editing, and chromosome number must be tracked relative to the starting cell. B. Carbohydrates commonly supply or store energy and provide structures such as cellulose, lipids store energy and form membranes, proteins perform structural, transport, receptor and catalytic roles, and nucleic acids store or express information. Enzymes are biological catalysts whose activity depends on substrate fit and conditions such as temperature and pH; a biomolecule class does not establish one universal function. C. Diffusion moves particles down a concentration gradient, osmosis is net water movement across a selectively permeable membrane, and active transport uses cellular energy to move substances against a gradient. Tonicity describes the effect of an external solution on cell water balance, so swelling, plasmolysis and normal volume require the membrane and concentration context. D. Groundnut, horse-gram and soybean are routed pea-family examples; cicada, froghopper and pond skater are insects; poisonous species occur in multiple animal groups, so a broad group name does not prove toxicity. Cassava is a woody shrub, ginger a rhizomatous herb, Malabar spinach a climbing vine, mint a herb and papaya a large herb; mushrooms can have species-specific edible, medicinal, psychoactive, insect-pathogenic or bioluminescent properties.

Answer: D. Explanation: Groundnut, horse-gram and soybean are routed pea-family examples; cicada, froghopper and pond skater are insects; poisonous species occur in multiple animal groups, so a broad group name does not prove toxicity. Cassava is a woody shrub, ginger a rhizomatous herb, Malabar spinach a climbing vine, mint a herb and papaya a large herb; mushrooms can have species-specific edible, medicinal, psychoactive, insect-pathogenic or bioluminescent properties. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q73. Which statement correctly identifies Cellulose decomposition route?

A. Cellulose is a structural plant polysaccharide; decomposer fungi and bacteria secrete cellulases that hydrolyse it into smaller sugars, after which microbial metabolism returns carbon through soil and food-web processes. Moisture, temperature, oxygen, lignin and community composition constrain the rate, and natural decomposition is not proof of an industrially viable biofuel process. B. Carbohydrates commonly supply or store energy and provide structures such as cellulose, lipids store energy and form membranes, proteins perform structural, transport, receptor and catalytic roles, and nucleic acids store or express information. Enzymes are biological catalysts whose activity depends on substrate fit and conditions such as temperature and pH; a biomolecule class does not establish one universal function. C. ICMR performs and supports biomedical research, NCDC and the IDSP or IHIP network support surveillance and outbreak response under MoHFW, State systems deliver programmes, and CDSCO regulates drugs and vaccines; DBT, CSIR-CCMB and CSIR-IGIB occupy different research roles. A biological mechanism, association, surveillance signal, WHO validation, elimination as a public-health problem, regulatory approval and eradication are separate evidence rungs. D. A cell is the basic structural and functional unit of life; prokaryotes lack a membrane-bound nucleus and membrane-bound organelles, whereas eukaryotes possess them. The nucleus stores most genetic material, ribosomes synthesize protein, mitochondria support aerobic ATP generation, the endoplasmic reticulum and Golgi system process and move products, lysosomes contain hydrolytic enzymes, and plant cells additionally possess a cellulose wall, chloroplasts and a large vacuole.

Answer: A. Explanation: Cellulose is a structural plant polysaccharide; decomposer fungi and bacteria secrete cellulases that hydrolyse it into smaller sugars, after which microbial metabolism returns carbon through soil and food-web processes. Moisture, temperature, oxygen, lignin and community composition constrain the rate, and natural decomposition is not proof of an industrially viable biofuel process. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q74. Which option preserves the technical boundary of Cellulose decomposition route?

A. Carbohydrates commonly supply or store energy and provide structures such as cellulose, lipids store energy and form membranes, proteins perform structural, transport, receptor and catalytic roles, and nucleic acids store or express information. Enzymes are biological catalysts whose activity depends on substrate fit and conditions such as temperature and pH; a biomolecule class does not establish one universal function. B. Cellulose is a structural plant polysaccharide; decomposer fungi and bacteria secrete cellulases that hydrolyse it into smaller sugars, after which microbial metabolism returns carbon through soil and food-web processes. Moisture, temperature, oxygen, lignin and community composition constrain the rate, and natural decomposition is not proof of an industrially viable biofuel process. C. Diffusion moves particles down a concentration gradient, osmosis is net water movement across a selectively permeable membrane, and active transport uses cellular energy to move substances against a gradient. Tonicity describes the effect of an external solution on cell water balance, so swelling, plasmolysis and normal volume require the membrane and concentration context. D. A cell is the basic structural and functional unit of life; prokaryotes lack a membrane-bound nucleus and membrane-bound organelles, whereas eukaryotes possess them. The nucleus stores most genetic material, ribosomes synthesize protein, mitochondria support aerobic ATP generation, the endoplasmic reticulum and Golgi system process and move products, lysosomes contain hydrolytic enzymes, and plant cells additionally possess a cellulose wall, chloroplasts and a large vacuole.

Answer: B. Explanation: Cellulose is a structural plant polysaccharide; decomposer fungi and bacteria secrete cellulases that hydrolyse it into smaller sugars, after which microbial metabolism returns carbon through soil and food-web processes. Moisture, temperature, oxygen, lignin and community composition constrain the rate, and natural decomposition is not proof of an industrially viable biofuel process. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q75. Which statement uses Cellulose decomposition route without changing its institution, unit or status?

A. Carbohydrates commonly supply or store energy and provide structures such as cellulose, lipids store energy and form membranes, proteins perform structural, transport, receptor and catalytic roles, and nucleic acids store or express information. Enzymes are biological catalysts whose activity depends on substrate fit and conditions such as temperature and pH; a biomolecule class does not establish one universal function. B. Mitosis is one division producing two genetically similar daughter cells for growth and repair, while meiosis consists of two divisions producing four haploid products for sexual reproduction; crossing over and independent assortment generate variation. Neither process should be described as gene editing, and chromosome number must be tracked relative to the starting cell. C. Cellulose is a structural plant polysaccharide; decomposer fungi and bacteria secrete cellulases that hydrolyse it into smaller sugars, after which microbial metabolism returns carbon through soil and food-web processes. Moisture, temperature, oxygen, lignin and community composition constrain the rate, and natural decomposition is not proof of an industrially viable biofuel process. D. Diffusion moves particles down a concentration gradient, osmosis is net water movement across a selectively permeable membrane, and active transport uses cellular energy to move substances against a gradient. Tonicity describes the effect of an external solution on cell water balance, so swelling, plasmolysis and normal volume require the membrane and concentration context.

Answer: C. Explanation: Cellulose is a structural plant polysaccharide; decomposer fungi and bacteria secrete cellulases that hydrolyse it into smaller sugars, after which microbial metabolism returns carbon through soil and food-web processes. Moisture, temperature, oxygen, lignin and community composition constrain the rate, and natural decomposition is not proof of an industrially viable biofuel process. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q76. Which option avoids the standard UPSC close-option trap about Cellulose decomposition route?

A. Diffusion moves particles down a concentration gradient, osmosis is net water movement across a selectively permeable membrane, and active transport uses cellular energy to move substances against a gradient. Tonicity describes the effect of an external solution on cell water balance, so swelling, plasmolysis and normal volume require the membrane and concentration context. B. Mitosis is one division producing two genetically similar daughter cells for growth and repair, while meiosis consists of two divisions producing four haploid products for sexual reproduction; crossing over and independent assortment generate variation. Neither process should be described as gene editing, and chromosome number must be tracked relative to the starting cell. C. DNA replication copies hereditary material, transcription produces RNA from a DNA template, and translation uses messenger RNA, transfer RNA and ribosomes to build protein from codons. RNA also has regulatory roles: microRNA can bind complementary messenger RNA and reduce protein production or promote RNA degradation, so DNA storage, RNA regulation and protein function are distinct stages. D. Cellulose is a structural plant polysaccharide; decomposer fungi and bacteria secrete cellulases that hydrolyse it into smaller sugars, after which microbial metabolism returns carbon through soil and food-web processes. Moisture, temperature, oxygen, lignin and community composition constrain the rate, and natural decomposition is not proof of an industrially viable biofuel process.

Answer: D. Explanation: Cellulose is a structural plant polysaccharide; decomposer fungi and bacteria secrete cellulases that hydrolyse it into smaller sugars, after which microbial metabolism returns carbon through soil and food-web processes. Moisture, temperature, oxygen, lignin and community composition constrain the rate, and natural decomposition is not proof of an industrially viable biofuel process. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q77. Which statement correctly identifies Health institution and evidence-status firewall?

A. ICMR performs and supports biomedical research, NCDC and the IDSP or IHIP network support surveillance and outbreak response under MoHFW, State systems deliver programmes, and CDSCO regulates drugs and vaccines; DBT, CSIR-CCMB and CSIR-IGIB occupy different research roles. A biological mechanism, association, surveillance signal, WHO validation, elimination as a public-health problem, regulatory approval and eradication are separate evidence rungs. B. Carbohydrates commonly supply or store energy and provide structures such as cellulose, lipids store energy and form membranes, proteins perform structural, transport, receptor and catalytic roles, and nucleic acids store or express information. Enzymes are biological catalysts whose activity depends on substrate fit and conditions such as temperature and pH; a biomolecule class does not establish one universal function. C. A cell is the basic structural and functional unit of life; prokaryotes lack a membrane-bound nucleus and membrane-bound organelles, whereas eukaryotes possess them. The nucleus stores most genetic material, ribosomes synthesize protein, mitochondria support aerobic ATP generation, the endoplasmic reticulum and Golgi system process and move products, lysosomes contain hydrolytic enzymes, and plant cells additionally possess a cellulose wall, chloroplasts and a large vacuole. D. Diffusion moves particles down a concentration gradient, osmosis is net water movement across a selectively permeable membrane, and active transport uses cellular energy to move substances against a gradient. Tonicity describes the effect of an external solution on cell water balance, so swelling, plasmolysis and normal volume require the membrane and concentration context.

Answer: A. Explanation: ICMR performs and supports biomedical research, NCDC and the IDSP or IHIP network support surveillance and outbreak response under MoHFW, State systems deliver programmes, and CDSCO regulates drugs and vaccines; DBT, CSIR-CCMB and CSIR-IGIB occupy different research roles. A biological mechanism, association, surveillance signal, WHO validation, elimination as a public-health problem, regulatory approval and eradication are separate evidence rungs. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q78. Which option preserves the technical boundary of Health institution and evidence-status firewall?

A. Carbohydrates commonly supply or store energy and provide structures such as cellulose, lipids store energy and form membranes, proteins perform structural, transport, receptor and catalytic roles, and nucleic acids store or express information. Enzymes are biological catalysts whose activity depends on substrate fit and conditions such as temperature and pH; a biomolecule class does not establish one universal function. B. ICMR performs and supports biomedical research, NCDC and the IDSP or IHIP network support surveillance and outbreak response under MoHFW, State systems deliver programmes, and CDSCO regulates drugs and vaccines; DBT, CSIR-CCMB and CSIR-IGIB occupy different research roles. A biological mechanism, association, surveillance signal, WHO validation, elimination as a public-health problem, regulatory approval and eradication are separate evidence rungs. C. Diffusion moves particles down a concentration gradient, osmosis is net water movement across a selectively permeable membrane, and active transport uses cellular energy to move substances against a gradient. Tonicity describes the effect of an external solution on cell water balance, so swelling, plasmolysis and normal volume require the membrane and concentration context. D. Mitosis is one division producing two genetically similar daughter cells for growth and repair, while meiosis consists of two divisions producing four haploid products for sexual reproduction; crossing over and independent assortment generate variation. Neither process should be described as gene editing, and chromosome number must be tracked relative to the starting cell.

Answer: B. Explanation: ICMR performs and supports biomedical research, NCDC and the IDSP or IHIP network support surveillance and outbreak response under MoHFW, State systems deliver programmes, and CDSCO regulates drugs and vaccines; DBT, CSIR-CCMB and CSIR-IGIB occupy different research roles. A biological mechanism, association, surveillance signal, WHO validation, elimination as a public-health problem, regulatory approval and eradication are separate evidence rungs. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q79. Which statement uses Health institution and evidence-status firewall without changing its institution, unit or status?

A. Mitosis is one division producing two genetically similar daughter cells for growth and repair, while meiosis consists of two divisions producing four haploid products for sexual reproduction; crossing over and independent assortment generate variation. Neither process should be described as gene editing, and chromosome number must be tracked relative to the starting cell. B. DNA replication copies hereditary material, transcription produces RNA from a DNA template, and translation uses messenger RNA, transfer RNA and ribosomes to build protein from codons. RNA also has regulatory roles: microRNA can bind complementary messenger RNA and reduce protein production or promote RNA degradation, so DNA storage, RNA regulation and protein function are distinct stages. C. ICMR performs and supports biomedical research, NCDC and the IDSP or IHIP network support surveillance and outbreak response under MoHFW, State systems deliver programmes, and CDSCO regulates drugs and vaccines; DBT, CSIR-CCMB and CSIR-IGIB occupy different research roles. A biological mechanism, association, surveillance signal, WHO validation, elimination as a public-health problem, regulatory approval and eradication are separate evidence rungs. D. Diffusion moves particles down a concentration gradient, osmosis is net water movement across a selectively permeable membrane, and active transport uses cellular energy to move substances against a gradient. Tonicity describes the effect of an external solution on cell water balance, so swelling, plasmolysis and normal volume require the membrane and concentration context.

Answer: C. Explanation: ICMR performs and supports biomedical research, NCDC and the IDSP or IHIP network support surveillance and outbreak response under MoHFW, State systems deliver programmes, and CDSCO regulates drugs and vaccines; DBT, CSIR-CCMB and CSIR-IGIB occupy different research roles. A biological mechanism, association, surveillance signal, WHO validation, elimination as a public-health problem, regulatory approval and eradication are separate evidence rungs. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q80. Which option avoids the standard UPSC close-option trap about Health institution and evidence-status firewall?

A. A gene is a functional hereditary unit on DNA; alleles are alternative forms, segregation separates allele pairs into gametes and independent assortment applies to appropriately unlinked genes. Mutations may be point, frameshift or chromosomal changes and supply variation without guaranteeing benefit; human XX-XY sex determination and X-linked recessive inheritance explain why haemophilia or red-green colour blindness can show sex-biased patterns. B. Mitosis is one division producing two genetically similar daughter cells for growth and repair, while meiosis consists of two divisions producing four haploid products for sexual reproduction; crossing over and independent assortment generate variation. Neither process should be described as gene editing, and chromosome number must be tracked relative to the starting cell. C. DNA replication copies hereditary material, transcription produces RNA from a DNA template, and translation uses messenger RNA, transfer RNA and ribosomes to build protein from codons. RNA also has regulatory roles: microRNA can bind complementary messenger RNA and reduce protein production or promote RNA degradation, so DNA storage, RNA regulation and protein function are distinct stages. D. ICMR performs and supports biomedical research, NCDC and the IDSP or IHIP network support surveillance and outbreak response under MoHFW, State systems deliver programmes, and CDSCO regulates drugs and vaccines; DBT, CSIR-CCMB and CSIR-IGIB occupy different research roles. A biological mechanism, association, surveillance signal, WHO validation, elimination as a public-health problem, regulatory approval and eradication are separate evidence rungs.

Answer: D. Explanation: ICMR performs and supports biomedical research, NCDC and the IDSP or IHIP network support surveillance and outbreak response under MoHFW, State systems deliver programmes, and CDSCO regulates drugs and vaccines; DBT, CSIR-CCMB and CSIR-IGIB occupy different research roles. A biological mechanism, association, surveillance signal, WHO validation, elimination as a public-health problem, regulatory approval and eradication are separate evidence rungs. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Semantic-completeness coverage drills - Topic 23

Use each drill as a 60-second plan before returning to the solved PYQs.

Drill	Prompt	Minimum answer route	Fatal trap
A	A gene is a functional hereditary unit on DNA; alleles are alternative forms, segregation separates allele pairs into gametes and independent assortment applies to appropriately unlinked genes. Mutations may be point, frameshift or chromosomal changes and supply variation without guaranteeing benefit; human XX-XY sex determination and X-linked recessive inheritance explain why haemophilia or red-green colour blindness can show sex-biased patterns.		
B	Mitosis is one division producing two genetically similar daughter cells for growth and repair, while meiosis consists of two divisions producing four haploid products for sexual reproduction; crossing over and independent assortment generate variation. Neither process should be described as gene editing, and chromosome number must be tracked relative to the starting cell.		
C	DNA replication copies hereditary material, transcription produces RNA from a DNA template, and translation uses messenger RNA, transfer RNA and ribosomes to build protein from codons. RNA also has regulatory roles: microRNA can bind complementary messenger RNA and reduce protein production or promote RNA degradation, so DNA storage, RNA regulation and protein function are distinct stages.		
D	ICMR performs and supports biomedical research, NCDC and the IDSP or IHIP network support surveillance and outbreak response under MoHFW, State systems deliver programmes, and CDSCO regulates drugs and vaccines; DBT, CSIR-CCMB and CSIR-IGIB occupy different research roles. A biological mechanism, association, surveillance signal, WHO validation, elimination as a public-health problem, regulatory approval and eradication are separate evidence rungs.		
E	Was Brahmanization one-way cultural replacement?	grant and migration -> status/ritual -> local cult adaptation -> language localization -> unequal reciprocity	homogenization
F	What does multilingual epigraphy prove?	Prakrit/Sanskrit/Tamil/Kannada and bounded Telugu transition -> institutional audience -> literacy limit	language equals ethnicity
G	What can queenly patronage prove?	Lokamahadevi/Virupaksha and Trailokyamahadevi/Mallikarjuna -> elite agency -> no universal gender claim	general equality from two patrons
H	Where does Topic 23 end?	Rashtrakuta takeover c. 753 and continuing Vengi line -> bounded Chola link	all Chalukyas vanish together

Self-check model: every response should contain one named item of evidence, one mechanism, one source or regional limit and one qualified conclusion.

PYQS AND ANSWER PRACTICE

AUDITED TOPIC-SPECIFIC PYQ OWNERSHIP

Audited ledgers route thirteen historical Prelims demands from 2018-2023, six Prelims demands from 2024-2025 and the 2022 GS-III cellulose-decomposition demand to this owner. Three representative cards preserve historical objective, recent objective and Mains routes without reproducing answer keys. Vaccine-platform, monoclonal-antibody and gene-editing demands remain with Topics 15 and 14.

PYQ DEMAND CARD 1 - 2018-2023 Prelims GS-I

Demand: Assess routed distinctions on plant adaptations, cell structure, pathogens, artificial culture, viral genomes, ACE2, biofilms, probiotics, B and T cells, mushrooms and extremophiles.

Status: Representative historical objective card covering thirteen routed demands; official keys are unavailable locally, so no option, answer letter or inferred key is supplied.

Model solution: Cell organisation and organelle boundary: A cell is the basic structural and functional unit of life; prokaryotes lack a membrane-bound nucleus and membrane-bound organelles, whereas eukaryotes possess them. The nucleus stores most genetic material, ribosomes synthesize protein, mitochondria support aerobic ATP generation, the endoplasmic reticulum and Golgi system process and move products, lysosomes contain hydrolytic enzymes, and plant cells additionally possess a cellulose wall, chloroplasts and a large vacuole. **Innate adaptive and tolerance architecture:** Innate immunity includes barriers, phagocytes, natural killer cells, complement, interferons, inflammation and fever; adaptive immunity uses antigen-specific B and T lymphocytes and can form memory. B-cell lineages produce antibodies, T cells perform coordinating, cytotoxic or regulatory roles, and peripheral immune tolerance including regulatory T cells limits attack on self; immunity is not antibodies alone. **Pathogen transmission vector and zoonosis map:** Bacteria are cellular organisms, viruses are acellular and depend on host-cell machinery, fungi are eukaryotes, protozoa are single-celled eukaryotes and helminths are multicellular parasites. Transmission may be respiratory, faeco-oral, blood-borne, sexual, vector-borne, fomite-linked or zoonotic; a vector such as Aedes, Anopheles, Culex or sandfly carries a pathogen but is not the disease itself. **Antibiotic antiviral and AMR boundary:** Antibiotics target susceptible bacteria rather than viruses, while antivirals interfere with virus-specific replication stages. Antimicrobial resistance is selection and spread of resistant microorganisms under pressures including misuse, incomplete treatment, agricultural use and environmental residues; the microbe or population becomes resistant, not the patient's body. **Biofilms microbiome probiotics and extremophiles:** A biofilm is a surface-associated microbial community embedded in a matrix that can alter persistence and treatment response; the human microbiome contributes to digestion, vitamin production, immune development and pathogen exclusion. Probiotics are live microorganisms intended to confer a benefit in an adequate amount, while thermophiles, psychrophiles and acidophiles are classified by environmental tolerance; none of these labels proves safety or effectiveness in every context. **Routed taxonomy and organism-form distinctions:** Groundnut, horse-gram and soybean are routed pea-family examples; cicada, frog hopper and pond skater are insects; poisonous species occur in multiple animal groups, so a broad group name does not prove toxicity. Cassava is a woody shrub, ginger a rhizomatous herb, Malabar spinach a climbing vine, mint a herb and papaya a large herb; mushrooms can have species-specific edible, medicinal, psychoactive, insect-pathogenic or bioluminescent properties. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

Demand decoding: Treat "PYQ DEMAND CARD 1 - 2018-2023 Prelims GS-I" as a mechanism, category, institution, unit, date and status problem.

Detailed examiner-grade model answer:

Introduction and thesis: Cell organisation and organelle boundary: A cell is the basic structural and functional unit of life; prokaryotes lack a membrane-bound nucleus and membrane-bound organelles, whereas eukaryotes possess them. The nucleus stores most genetic material, ribosomes synthesize protein, mitochondria support aerobic ATP generation, the endoplasmic reticulum and Golgi system process and move products, lysosomes contain hydrolytic enzymes, and plant cells additionally possess a cellulose wall, chloroplasts and a large vacuole. **Innate adaptive and tolerance architecture:** Innate immunity includes barriers, phagocytes, natural killer cells, complement, interferons, inflammation and fever; adaptive immunity uses antigen-specific B and T lymphocytes and can form memory. B-cell lineages produce antibodies, T cells perform coordinating, cytotoxic or regulatory roles, and

peripheral immune tolerance including regulatory T cells limits attack on self; immunity is not antibodies alone.

Pathogen transmission vector and zoonosis map: Bacteria are cellular organisms, viruses are acellular and depend on host-cell machinery, fungi are eukaryotes, protozoa are single-celled eukaryotes and helminths are multicellular parasites. Transmission may be respiratory, faeco-oral, blood-borne, sexual, vector-borne, fomite-linked or zoonotic; a vector such as Aedes, Anopheles, Culex or sandfly carries a pathogen but is not the disease itself.

Antibiotic antiviral and AMR boundary: Antibiotics target susceptible bacteria rather than viruses, while antivirals interfere with virus-specific replication stages. Antimicrobial resistance is selection and spread of resistant microorganisms under pressures including misuse, incomplete treatment, agricultural use and environmental residues; the microbe or population becomes resistant, not the patient's body. **Biofilms microbiome probiotics and extremophiles:** A biofilm is a surface-associated microbial community embedded in a matrix that can alter persistence and treatment response; the human microbiome contributes to digestion, vitamin production, immune development and pathogen exclusion. Probiotics are live microorganisms intended to confer a benefit in an adequate amount, while thermophiles, psychrophiles and acidophiles are classified by environmental tolerance; none of these labels proves safety or effectiveness in every context. **Routed taxonomy and organism-form distinctions:** Groundnut, horse-gram and soybean are routed pea-family examples; cicada, frog hopper and pond skater are insects; poisonous species occur in multiple animal groups, so a broad group name does not prove toxicity. Cassava is a woody shrub, ginger a rhizomatous herb, Malabar spinach a climbing vine, mint a herb and papaya a large herb; mushrooms can have species-specific edible, medicinal, psychoactive, insect-pathogenic or bioluminescent properties. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

Analytical body:

1. **Claim and named evidence:** Demand: Assess routed distinctions on plant adaptations, cell structure, pathogens, artificial culture, viral genomes, ACE2, biofilms, probiotics, B and T cells, mushrooms and extremophiles. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
2. **Claim and named evidence:** Status: Representative historical objective card covering thirteen routed demands; official keys are unavailable locally, so no option, answer letter or inferred key is supplied. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

Counter-position / limit: A launch, test, target, budget, capacity, approval, contract, dataset, benchmark or prototype cannot alone establish deployment, induction, commercial operation, safety, inclusion or observed outcome; test mechanism, status, scale and evidence.

Qualified conclusion: Cell organisation and organelle boundary: A cell is the basic structural and functional unit of life; prokaryotes lack a membrane-bound nucleus and membrane-bound organelles, whereas eukaryotes possess them. The nucleus stores most genetic material, ribosomes synthesize protein, mitochondria support aerobic ATP generation, the endoplasmic reticulum and Golgi system process and move products, lysosomes contain hydrolytic enzymes, and plant cells additionally possess a cellulose wall, chloroplasts and a large vacuole. **Innate adaptive and tolerance architecture:** Innate immunity includes barriers, phagocytes, natural killer cells, complement, interferons, inflammation and fever; adaptive immunity uses antigen-specific B and T lymphocytes and can form memory. B-cell lineages produce antibodies, T cells perform coordinating, cytotoxic or regulatory roles, and peripheral immune tolerance including regulatory T cells limits attack on self; immunity is not antibodies alone. **Pathogen transmission vector and zoonosis map:** Bacteria are cellular organisms, viruses are acellular and depend on host-cell machinery, fungi are eukaryotes, protozoa are single-celled eukaryotes and helminths are multicellular parasites. Transmission may be respiratory, faeco-oral, blood-borne, sexual, vector-borne, fomite-linked or zoonotic; a vector such as Aedes, Anopheles, Culex or sandfly carries a pathogen but is not the disease itself. **Antibiotic antiviral and AMR boundary:** Antibiotics target susceptible bacteria rather than viruses, while antivirals interfere with virus-specific replication stages. Antimicrobial resistance is selection and spread of resistant microorganisms under pressures including misuse, incomplete treatment, agricultural use and environmental residues; the microbe or population becomes resistant, not the patient's body. **Biofilms microbiome probiotics and extremophiles:** A biofilm is a surface-associated microbial community embedded in a matrix that can alter persistence and treatment response; the

human microbiome contributes to digestion, vitamin production, immune development and pathogen exclusion. Probiotics are live microorganisms intended to confer a benefit in an adequate amount, while thermophiles, psychrophiles and acidophiles are classified by environmental tolerance; none of these labels proves safety or effectiveness in every context. **Routed taxonomy and organism-form distinctions:** Groundnut, horse-gram and soybean are routed pea-family examples; cicada, frog hopper and pond skater are insects; poisonous species occur in multiple animal groups, so a broad group name does not prove toxicity. Cassava is a woody shrub, ginger a rhizomatous herb, Malabar spinach a climbing vine, mint a herb and papaya a large herb; mushrooms can have species-specific edible, medicinal, psychoactive, insect-pathogenic or bioluminescent properties. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

Executable exam-length answer / compression plan: Identify the object and actor; trace the technical mechanism; fix the measurement and platform; test each statement against source date, status rung and closest exception.

Why this earns marks: It prevents organisation, platform, process, application, regulatory role, target and observed outcome from being conflated.

How to improve this answer: For "PYQ DEMAND CARD 1 - 2018-2023 Prelims GS-I", explain why the nearest distractor fails on mechanism, platform, unit, institution, legal status, maturity or date.

PYQ DEMAND CARD 2 - 2024-2025 Prelims GS-I

Demand: Assess routed organism-family, insect, poisonous-species, nitric-oxide, plant-form and virus-host distinctions.

Status: Representative recent objective card covering six routed demands; official Set-A keys exist locally but no option, answer letter or objective key is reproduced.

Model solution: Hormones metabolism and internal stability: Insulin lowers elevated blood glucose by promoting uptake and storage while glucagon supports its rise, thyroid hormones influence metabolism, parathyroid hormone helps regulate calcium, adrenal hormones mediate stress responses and pituitary signals coordinate several glands. Homeostasis is dynamic regulation around a range, not an unchanging value; hormone deficiency, resistance and gland failure are different mechanisms. **Pathogen transmission vector and zoonosis map:** Bacteria are cellular organisms, viruses are acellular and depend on host-cell machinery, fungi are eukaryotes, protozoa are single-celled eukaryotes and helminths are multicellular parasites. Transmission may be respiratory, faeco-oral, blood-borne, sexual, vector-borne, fomite-linked or zoonotic; a vector such as Aedes, Anopheles, Culex or sandfly carries a pathogen but is not the disease itself. **Routed taxonomy and organism-form distinctions:** Groundnut, horse-gram and soybean are routed pea-family examples; cicada, frog hopper and pond skater are insects; poisonous species occur in multiple animal groups, so a broad group name does not prove toxicity. Cassava is a woody shrub, ginger a rhizomatous herb, Malabar spinach a climbing vine, mint a herb and papaya a large herb; mushrooms can have species-specific edible, medicinal, psychoactive, insect-pathogenic or bioluminescent properties. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

Demand decoding: Treat "PYQ DEMAND CARD 2 - 2024-2025 Prelims GS-I" as a mechanism, category, institution, unit, date and status problem.

Detailed examiner-grade model answer:

Introduction and thesis: Hormones metabolism and internal stability: Insulin lowers elevated blood glucose by promoting uptake and storage while glucagon supports its rise, thyroid hormones influence metabolism, parathyroid hormone helps regulate calcium, adrenal hormones mediate stress responses and pituitary signals coordinate several glands. Homeostasis is dynamic regulation around a range, not an unchanging value; hormone deficiency, resistance and gland failure are different mechanisms. **Pathogen transmission vector and zoonosis map:** Bacteria are cellular organisms, viruses are acellular and depend on host-cell machinery, fungi are eukaryotes, protozoa are single-celled eukaryotes and helminths are multicellular parasites. Transmission may be respiratory, faeco-oral, blood-borne, sexual, vector-borne, fomite-linked or zoonotic; a vector such as Aedes, Anopheles, Culex or sandfly carries a pathogen but is not the disease itself. **Routed taxonomy and organism-form distinctions:** Groundnut, horse-gram and soybean are routed pea-family examples; cicada, frog hopper and pond skater are insects; poisonous species occur in multiple animal groups, so a broad group name does not prove toxicity. Cassava is a woody shrub, ginger a rhizomatous herb, Malabar spinach a climbing vine, mint a herb and papaya a large herb; mushrooms can have species-specific edible, medicinal, psychoactive, insect-pathogenic or bioluminescent properties. The solution

therefore answers the routed demand without inventing an official model answer or an objective answer key.

Analytical body:

1. **Claim and named evidence:** Demand: Assess routed organism-family, insect, poisonous-species, nitric-oxide, plant-form and virus-host distinctions. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
2. **Claim and named evidence:** Status: Representative recent objective card covering six routed demands; official Set-A keys exist locally but no option, answer letter or objective key is reproduced. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

Counter-position / limit: A launch, test, target, budget, capacity, approval, contract, dataset, benchmark or prototype cannot alone establish deployment, induction, commercial operation, safety, inclusion or observed outcome; test mechanism, status, scale and evidence.

Qualified conclusion: Hormones metabolism and internal stability: Insulin lowers elevated blood glucose by promoting uptake and storage while glucagon supports its rise, thyroid hormones influence metabolism, parathyroid hormone helps regulate calcium, adrenal hormones mediate stress responses and pituitary signals coordinate several glands. Homeostasis is dynamic regulation around a range, not an unchanging value; hormone deficiency, resistance and gland failure are different mechanisms. **Pathogen transmission vector and zoonosis map:** Bacteria are cellular organisms, viruses are acellular and depend on host-cell machinery, fungi are eukaryotes, protozoa are single-celled eukaryotes and helminths are multicellular parasites. Transmission may be respiratory, faeco-oral, blood-borne, sexual, vector-borne, fomite-linked or zoonotic; a vector such as Aedes, Anopheles, Culex or sandfly carries a pathogen but is not the disease itself. **Routed taxonomy and organism-form distinctions:** Groundnut, horse-gram and soybean are routed pea-family examples; cicada, frog hopper and pond skater are insects; poisonous species occur in multiple animal groups, so a broad group name does not prove toxicity. Cassava is a woody shrub, ginger a rhizomatous herb, Malabar spinach a climbing vine, mint a herb and papaya a large herb; mushrooms can have species-specific edible, medicinal, psychoactive, insect-pathogenic or bioluminescent properties. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

Executable exam-length answer / compression plan: Identify the object and actor; trace the technical mechanism; fix the measurement and platform; test each statement against source date, status rung and closest exception.

Why this earns marks: It prevents organisation, platform, process, application, regulatory role, target and observed outcome from being conflated.

How to improve this answer: For "PYQ DEMAND CARD 2 - 2024-2025 Prelims GS-I", explain why the nearest distractor fails on mechanism, platform, unit, institution, legal status, maturity or date.

PYQ DEMAND CARD 3 - 2022 GS-III

Demand: Discuss the natural processes through which cellulose undergoes decomposition on the Earth's surface.

Status: Verified routed Mains demand at 10 marks and 150 words; the response supplies a mechanism-first answer route and qualification, not an official model answer.

Model solution: Biomolecules and enzyme specificity: Carbohydrates commonly supply or store energy and provide structures such as cellulose, lipids store energy and form membranes, proteins perform structural, transport, receptor and catalytic roles, and nucleic acids store or express information. Enzymes are biological catalysts whose activity depends on substrate fit and conditions such as temperature and pH; a biomolecule class does not establish one universal function. **Biofilms microbiome probiotics and extremophiles:** A biofilm is a surface-associated microbial community embedded in a matrix that can alter persistence and treatment response; the human microbiome contributes to digestion, vitamin production, immune development and pathogen exclusion. Probiotics are live microorganisms intended to confer a benefit in an adequate amount, while thermophiles, psychrophiles and acidophiles are classified by environmental tolerance; none of these labels proves safety or effectiveness in every context. **Cellulose decomposition route:** Cellulose is a structural plant polysaccharide; decomposer fungi and bacteria secrete cellulases that hydrolyse it into smaller sugars, after which microbial metabolism returns carbon

through soil and food-web processes. Moisture, temperature, oxygen, lignin and community composition constrain the rate, and natural decomposition is not proof of an industrially viable biofuel process. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

Demand decoding: The directive **answer** requires a direct position on "PYQ DEMAND CARD 3 - 2022 GS-III", each clause, the exact technical mechanism, named Indian institution, dated status, application, risk and qualification.

Detailed examiner-grade model answer:

Introduction and thesis: Biomolecules and enzyme specificity: Carbohydrates commonly supply or store energy and provide structures such as cellulose, lipids store energy and form membranes, proteins perform structural, transport, receptor and catalytic roles, and nucleic acids store or express information. Enzymes are biological catalysts whose activity depends on substrate fit and conditions such as temperature and pH; a biomolecule class does not establish one universal function. **Biofilms microbiome probiotics and extremophiles:** A biofilm is a surface-associated microbial community embedded in a matrix that can alter persistence and treatment response; the human microbiome contributes to digestion, vitamin production, immune development and pathogen exclusion. Probiotics are live microorganisms intended to confer a benefit in an adequate amount, while thermophiles, psychrophiles and acidophiles are classified by environmental tolerance; none of these labels proves safety or effectiveness in every context. **Cellulose decomposition route:** Cellulose is a structural plant polysaccharide; decomposer fungi and bacteria secrete cellulases that hydrolyse it into smaller sugars, after which microbial metabolism returns carbon through soil and food-web processes. Moisture, temperature, oxygen, lignin and community composition constrain the rate, and natural decomposition is not proof of an industrially viable biofuel process. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

Analytical body:

1. **Claim and named evidence:** Demand: Discuss the natural processes through which cellulose undergoes decomposition on the Earth's surface. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
2. **Claim and named evidence:** Status: Verified routed Mains demand at 10 marks and 150 words; the response supplies a mechanism-first answer route and qualification, not an official model answer. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

Counter-position / limit: A launch, test, target, budget, capacity, approval, contract, dataset, benchmark or prototype cannot alone establish deployment, induction, commercial operation, safety, inclusion or observed outcome; test mechanism, status, scale and evidence.

Qualified conclusion: Biomolecules and enzyme specificity: Carbohydrates commonly supply or store energy and provide structures such as cellulose, lipids store energy and form membranes, proteins perform structural, transport, receptor and catalytic roles, and nucleic acids store or express information. Enzymes are biological catalysts whose activity depends on substrate fit and conditions such as temperature and pH; a biomolecule class does not establish one universal function. **Biofilms microbiome probiotics and extremophiles:** A biofilm is a surface-associated microbial community embedded in a matrix that can alter persistence and treatment response; the human microbiome contributes to digestion, vitamin production, immune development and pathogen exclusion. Probiotics are live microorganisms intended to confer a benefit in an adequate amount, while thermophiles, psychrophiles and acidophiles are classified by environmental tolerance; none of these labels proves safety or effectiveness in every context. **Cellulose decomposition route:** Cellulose is a structural plant polysaccharide; decomposer fungi and bacteria secrete cellulases that hydrolyse it into smaller sugars, after which microbial metabolism returns carbon through soil and food-web processes. Moisture, temperature, oxygen, lignin and community composition constrain the rate, and natural decomposition is not proof of an industrially viable biofuel process. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

Executable exam-length answer / compression plan: For 15 marks, spend one-sixth of the time decoding the directive and drawing definition -> mechanism -> system/institution -> application/status -> risk/outcome; write four to seven claim -> named evidence -> analysis -> qualification points; reserve the final minute for unit, date, actor,

platform, approval/test/deployment rung and residual-risk checks.

Why this earns marks: The answer obeys the directive, explains science rather than listing schemes, uses India-centric evidence and preserves technical, institutional, regulatory and maturity distinctions.

How to improve this answer: For "PYQ DEMAND CARD 3 - 2022 GS-III", replace the weakest catalogue point with one exact mechanism, named mission/institution/platform, dated status, measurable boundary, implementation constraint and answer-specific qualification.

ORIGINAL MAINS 1 - 10 MARKS

Question: Distinguish cell types, organelles, biomolecules and membrane-transport mechanisms. Answer in about 150 words.

Model thesis: Claim: Cell organisation and organelle boundary. **Named evidence/example:** A cell is the basic structural and functional unit of life; prokaryotes lack a membrane-bound nucleus and membrane-bound organelles, whereas eukaryotes possess them. The nucleus stores most genetic material, ribosomes synthesize protein, mitochondria support aerobic ATP generation, the endoplasmic reticulum and Golgi system process and move products, lysosomes contain hydrolytic enzymes, and plant cells additionally possess a cellulose wall, chloroplasts and a large vacuole. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Biomolecules and enzyme specificity. **Named evidence/example:** Carbohydrates commonly supply or store energy and provide structures such as cellulose, lipids store energy and form membranes, proteins perform structural, transport, receptor and catalytic roles, and nucleic acids store or express information. Enzymes are biological catalysts whose activity depends on substrate fit and conditions such as temperature and pH; a biomolecule class does not establish one universal function. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Membrane transport and water balance. **Named evidence/example:** Diffusion moves particles down a concentration gradient, osmosis is net water movement across a selectively permeable membrane, and active transport uses cellular energy to move substances against a gradient. Tonicity describes the effect of an external solution on cell water balance, so swelling, plasmolysis and normal volume require the membrane and concentration context. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Claim -> named evidence -> analysis -> qualification:

- A cell is the basic structural and functional unit of life; prokaryotes lack a membrane-bound nucleus and membrane-bound organelles, whereas eukaryotes possess them. The nucleus stores most genetic material, ribosomes synthesize protein, mitochondria support aerobic ATP generation, the endoplasmic reticulum and Golgi system process and move products, lysosomes contain hydrolytic enzymes, and plant cells additionally possess a cellulose wall, chloroplasts and a large vacuole.
- Carbohydrates commonly supply or store energy and provide structures such as cellulose, lipids store energy and form membranes, proteins perform structural, transport, receptor and catalytic roles, and nucleic acids store or express information. Enzymes are biological catalysts whose activity depends on substrate fit and conditions such as temperature and pH; a biomolecule class does not establish one universal function.
- Diffusion moves particles down a concentration gradient, osmosis is net water movement across a selectively permeable membrane, and active transport uses cellular energy to move substances against a gradient. Tonicity describes the effect of an external solution on cell water balance, so swelling, plasmolysis and normal volume require the membrane and concentration context.

Qualified conclusion: Claim: Cell organisation and organelle boundary. **Named evidence/example:** A cell is the basic structural and functional unit of life; prokaryotes lack a membrane-bound nucleus and membrane-bound organelles, whereas eukaryotes possess them. The nucleus stores most genetic material, ribosomes synthesize protein, mitochondria support aerobic ATP generation, the endoplasmic reticulum and Golgi system process and move products, lysosomes contain hydrolytic enzymes, and plant cells additionally possess a cellulose wall,

chloroplasts and a large vacuole. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Biomolecules and enzyme specificity. **Named evidence/example:** Carbohydrates commonly supply or store energy and provide structures such as cellulose, lipids store energy and form membranes, proteins perform structural, transport, receptor and catalytic roles, and nucleic acids store or express information. Enzymes are biological catalysts whose activity depends on substrate fit and conditions such as temperature and pH; a biomolecule class does not establish one universal function. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Membrane transport and water balance. **Named evidence/example:** Diffusion moves particles down a concentration gradient, osmosis is net water movement across a selectively permeable membrane, and active transport uses cellular energy to move substances against a gradient. Tonicity describes the effect of an external solution on cell water balance, so swelling, plasmolysis and normal volume require the membrane and concentration context. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Demand decoding: The directive **answer** requires a direct position on "Distinguish cell types, organelles, biomolecules and membrane-transport mechanisms. Answer in...", each clause, the exact technical mechanism, named Indian institution, dated status, application, risk and qualification.

Detailed examiner-grade model answer:

Introduction and thesis: Claim: Cell organisation and organelle boundary. **Named evidence/example:** A cell is the basic structural and functional unit of life; prokaryotes lack a membrane-bound nucleus and membrane-bound organelles, whereas eukaryotes possess them. The nucleus stores most genetic material, ribosomes synthesize protein, mitochondria support aerobic ATP generation, the endoplasmic reticulum and Golgi system process and move products, lysosomes contain hydrolytic enzymes, and plant cells additionally possess a cellulose wall, chloroplasts and a large vacuole. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Biomolecules and enzyme specificity. **Named evidence/example:** Carbohydrates commonly supply or store energy and provide structures such as cellulose, lipids store energy and form membranes, proteins perform structural, transport, receptor and catalytic roles, and nucleic acids store or express information. Enzymes are biological catalysts whose activity depends on substrate fit and conditions such as temperature and pH; a biomolecule class does not establish one universal function. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Membrane transport and water balance. **Named evidence/example:** Diffusion moves particles down a concentration gradient, osmosis is net water movement across a selectively permeable membrane, and active transport uses cellular energy to move substances against a gradient. Tonicity describes the effect of an external solution on cell water balance, so swelling, plasmolysis and normal volume require the membrane and concentration context. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Analytical body:

1. **Claim and named evidence:** A cell is the basic structural and functional unit of life; prokaryotes lack a membrane-bound nucleus and membrane-bound organelles, whereas eukaryotes possess them. The nucleus stores most genetic material, ribosomes synthesize protein, mitochondria support aerobic ATP generation, the endoplasmic reticulum and Golgi system process and move products, lysosomes contain hydrolytic enzymes, and plant cells additionally possess a cellulose wall, chloroplasts and a large vacuole. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

2. **Claim and named evidence:** Carbohydrates commonly supply or store energy and provide structures such as cellulose, lipids store energy and form membranes, proteins perform structural, transport, receptor and catalytic roles, and nucleic acids store or express information. Enzymes are biological catalysts whose activity depends on substrate fit and conditions such as temperature and pH; a biomolecule class does not establish one universal function. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

3. **Claim and named evidence:** Diffusion moves particles down a concentration gradient, osmosis is net water movement across a selectively permeable membrane, and active transport uses cellular energy to move substances against a gradient. Tonicity describes the effect of an external solution on cell water balance, so swelling, plasmolysis and normal volume require the membrane and concentration context. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

Counter-position / limit: A launch, test, target, budget, capacity, approval, contract, dataset, benchmark or prototype cannot alone establish deployment, induction, commercial operation, safety, inclusion or observed outcome; test mechanism, status, scale and evidence.

Qualified conclusion: Claim: Cell organisation and organelle boundary. **Named evidence/example:** A cell is the basic structural and functional unit of life; prokaryotes lack a membrane-bound nucleus and membrane-bound organelles, whereas eukaryotes possess them. The nucleus stores most genetic material, ribosomes synthesize protein, mitochondria support aerobic ATP generation, the endoplasmic reticulum and Golgi system process and move products, lysosomes contain hydrolytic enzymes, and plant cells additionally possess a cellulose wall, chloroplasts and a large vacuole. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Biomolecules and enzyme specificity. **Named evidence/example:** Carbohydrates commonly supply or store energy and provide structures such as cellulose, lipids store energy and form membranes, proteins perform structural, transport, receptor and catalytic roles, and nucleic acids store or express information. Enzymes are biological catalysts whose activity depends on substrate fit and conditions such as temperature and pH; a biomolecule class does not establish one universal function. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Membrane transport and water balance. **Named evidence/example:** Diffusion moves particles down a concentration gradient, osmosis is net water movement across a selectively permeable membrane, and active transport uses cellular energy to move substances against a gradient. Tonicity describes the effect of an external solution on cell water balance, so swelling, plasmolysis and normal volume require the membrane and concentration context. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Executable exam-length answer / compression plan: For 10 marks, spend one-sixth of the time decoding the directive and drawing definition -> mechanism -> system/institution -> application/status -> risk/outcome; write four to seven claim -> named evidence -> analysis -> qualification points; reserve the final minute for unit, date, actor, platform, approval/test/deployment rung and residual-risk checks.

Why this earns marks: The answer obeys the directive, explains science rather than listing schemes, uses India-centric evidence and preserves technical, institutional, regulatory and maturity distinctions.

How to improve this answer: For "Distinguish cell types, organelles, biomolecules and membrane-transport mechanisms. Answer in...", replace the weakest catalogue point with one exact mechanism, named mission/institution/platform, dated status, measurable boundary, implementation constraint and answer-specific qualification.

ORIGINAL MAINS 2 - 10 MARKS

Question: Explain how cell division, gene expression, inheritance and mutation generate biological continuity and variation. Answer in about 150 words.

Model thesis: Claim: Mitosis meiosis and variation. **Named evidence/example:** Mitosis is one division producing two genetically similar daughter cells for growth and repair, while meiosis consists of two divisions producing four haploid products for sexual reproduction; crossing over and independent assortment generate variation. Neither process should be described as gene editing, and chromosome number must be tracked relative to the starting cell. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** DNA RNA gene expression and microRNA. **Named evidence/example:** DNA replication copies hereditary material, transcription produces RNA from a DNA template, and translation uses messenger RNA, transfer RNA and ribosomes to build protein from codons. RNA also has regulatory roles: microRNA can bind complementary messenger RNA and reduce protein production or promote RNA degradation, so DNA storage, RNA regulation and protein function are distinct stages. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Inheritance mutation and sex linkage. **Named evidence/example:** A gene is a functional hereditary unit on DNA; alleles are alternative forms, segregation separates allele pairs into gametes and independent assortment applies to appropriately unlinked genes. Mutations may be point, frameshift or chromosomal changes and supply variation without guaranteeing benefit; human XX-XY sex determination and X-linked recessive inheritance explain why haemophilia or red-green colour blindness can show sex-biased patterns. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Claim -> named evidence -> analysis -> qualification:

- Mitosis is one division producing two genetically similar daughter cells for growth and repair, while meiosis consists of two divisions producing four haploid products for sexual reproduction; crossing over and independent assortment generate variation. Neither process should be described as gene editing, and chromosome number must be tracked relative to the starting cell.
- DNA replication copies hereditary material, transcription produces RNA from a DNA template, and translation uses messenger RNA, transfer RNA and ribosomes to build protein from codons. RNA also has regulatory roles: microRNA can bind complementary messenger RNA and reduce protein production or promote RNA degradation, so DNA storage, RNA regulation and protein function are distinct stages.
- A gene is a functional hereditary unit on DNA; alleles are alternative forms, segregation separates allele pairs into gametes and independent assortment applies to appropriately unlinked genes. Mutations may be point, frameshift or chromosomal changes and supply variation without guaranteeing benefit; human XX-XY sex determination and X-linked recessive inheritance explain why haemophilia or red-green colour blindness can show sex-biased patterns.

Qualified conclusion: Claim: Mitosis meiosis and variation. **Named evidence/example:** Mitosis is one division producing two genetically similar daughter cells for growth and repair, while meiosis consists of two divisions producing four haploid products for sexual reproduction; crossing over and independent assortment generate variation. Neither process should be described as gene editing, and chromosome number must be tracked relative to the starting cell. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** DNA RNA gene expression and microRNA. **Named evidence/example:** DNA replication copies hereditary material, transcription produces RNA from a DNA template, and translation uses messenger RNA, transfer RNA and ribosomes to build protein from codons. RNA also has regulatory roles: microRNA can bind complementary messenger RNA and reduce protein production or promote RNA degradation, so DNA storage, RNA regulation and protein function are distinct stages. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution,

system category, legal character and development-to-deployment status boundary. **Claim:** Inheritance mutation and sex linkage. **Named evidence/example:** A gene is a functional hereditary unit on DNA; alleles are alternative forms, segregation separates allele pairs into gametes and independent assortment applies to appropriately unlinked genes. Mutations may be point, frameshift or chromosomal changes and supply variation without guaranteeing benefit; human XX-XY sex determination and X-linked recessive inheritance explain why haemophilia or red-green colour blindness can show sex-biased patterns. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Demand decoding: The directive **explain** requires a direct position on "Explain how cell division, gene expression, inheritance and mutation generate biological...", each clause, the exact technical mechanism, named Indian institution, dated status, application, risk and qualification.

Detailed examiner-grade model answer:

Introduction and thesis: **Claim:** Mitosis meiosis and variation. **Named evidence/example:** Mitosis is one division producing two genetically similar daughter cells for growth and repair, while meiosis consists of two divisions producing four haploid products for sexual reproduction; crossing over and independent assortment generate variation. Neither process should be described as gene editing, and chromosome number must be tracked relative to the starting cell. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** DNA RNA gene expression and microRNA. **Named evidence/example:** DNA replication copies hereditary material, transcription produces RNA from a DNA template, and translation uses messenger RNA, transfer RNA and ribosomes to build protein from codons. RNA also has regulatory roles: microRNA can bind complementary messenger RNA and reduce protein production or promote RNA degradation, so DNA storage, RNA regulation and protein function are distinct stages. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Inheritance mutation and sex linkage. **Named evidence/example:** A gene is a functional hereditary unit on DNA; alleles are alternative forms, segregation separates allele pairs into gametes and independent assortment applies to appropriately unlinked genes. Mutations may be point, frameshift or chromosomal changes and supply variation without guaranteeing benefit; human XX-XY sex determination and X-linked recessive inheritance explain why haemophilia or red-green colour blindness can show sex-biased patterns. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Analytical body:

1. **Claim and named evidence:** Mitosis is one division producing two genetically similar daughter cells for growth and repair, while meiosis consists of two divisions producing four haploid products for sexual reproduction; crossing over and independent assortment generate variation. Neither process should be described as gene editing, and chromosome number must be tracked relative to the starting cell. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
2. **Claim and named evidence:** DNA replication copies hereditary material, transcription produces RNA from a DNA template, and translation uses messenger RNA, transfer RNA and ribosomes to build protein from codons. RNA also has regulatory roles: microRNA can bind complementary messenger RNA and reduce protein production or promote RNA degradation, so DNA storage, RNA regulation and protein function are distinct stages. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
3. **Claim and named evidence:** A gene is a functional hereditary unit on DNA; alleles are alternative forms, segregation separates allele pairs into gametes and independent assortment applies to appropriately unlinked genes. Mutations may be point, frameshift or chromosomal changes and supply variation without guaranteeing

benefit; human XX-XY sex determination and X-linked recessive inheritance explain why haemophilia or red-green colour blindness can show sex-biased patterns. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

Counter-position / limit: A launch, test, target, budget, capacity, approval, contract, dataset, benchmark or prototype cannot alone establish deployment, induction, commercial operation, safety, inclusion or observed outcome; test mechanism, status, scale and evidence.

Qualified conclusion: Claim: Mitosis meiosis and variation. **Named evidence/example:** Mitosis is one division producing two genetically similar daughter cells for growth and repair, while meiosis consists of two divisions producing four haploid products for sexual reproduction; crossing over and independent assortment generate variation. Neither process should be described as gene editing, and chromosome number must be tracked relative to the starting cell. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** DNA RNA gene expression and microRNA. **Named evidence/example:** DNA replication copies hereditary material, transcription produces RNA from a DNA template, and translation uses messenger RNA, transfer RNA and ribosomes to build protein from codons. RNA also has regulatory roles: microRNA can bind complementary messenger RNA and reduce protein production or promote RNA degradation, so DNA storage, RNA regulation and protein function are distinct stages. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Inheritance mutation and sex linkage. **Named evidence/example:** A gene is a functional hereditary unit on DNA; alleles are alternative forms, segregation separates allele pairs into gametes and independent assortment applies to appropriately unlinked genes. Mutations may be point, frameshift or chromosomal changes and supply variation without guaranteeing benefit; human XX-XY sex determination and X-linked recessive inheritance explain why haemophilia or red-green colour blindness can show sex-biased patterns. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Executable exam-length answer / compression plan: For 10 marks, spend one-sixth of the time decoding the directive and drawing definition -> mechanism -> system/institution -> application/status -> risk/outcome; write four to seven claim -> named evidence -> analysis -> qualification points; reserve the final minute for unit, date, actor, platform, approval/test/deployment rung and residual-risk checks.

Why this earns marks: The answer obeys the directive, explains science rather than listing schemes, uses India-centric evidence and preserves technical, institutional, regulatory and maturity distinctions.

How to improve this answer: For "Explain how cell division, gene expression, inheritance and mutation generate biological...", replace the weakest catalogue point with one exact mechanism, named mission/institution/platform, dated status, measurable boundary, implementation constraint and answer-specific qualification.

ORIGINAL MAINS 3 - 15 MARKS

Question: Analyse digestion, circulation, respiration and excretion as an integrated physiological system. Answer in about 250 words.

Model thesis: Claim: Digestion absorption and nutrition. **Named evidence/example:** Digestion converts food into absorbable molecules: salivary amylase begins starch digestion, pepsin works in acidic stomach conditions, bile emulsifies fat without acting as an enzyme, pancreatic and intestinal enzymes continue digestion, and the small intestine is the principal absorption site. Vitamins, minerals, protein and energy have different roles; deficiency, absorption failure and excess are separate nutritional problems. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Blood heart and double circulation. **Named evidence/example:** Blood contains plasma, red cells carrying haemoglobin, immune white cells and platelets involved in clotting; the four-chambered human heart supports pulmonary and systemic circuits. Arteries carry blood away from the heart and veins return it, so oxygenation

is not the defining rule; ABO and Rh compatibility, gas transport and clotting must be treated as different functions.

Analysis: This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Respiration gas exchange and cellular energy. **Named evidence/example:** Ventilation moves air, alveolar gas exchange moves oxygen and carbon dioxide across thin surfaces, blood transports gases, and cellular respiration releases usable energy from nutrients. Haemoglobin binds oxygen while carbon monoxide can disrupt transport through stronger binding; breathing, gas exchange and ATP-producing respiration are linked but not synonymous. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication.

Qualification: The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Kidney nephron and homeostasis. **Named evidence/example:** The nephron filters blood at the glomerulus and then selectively reabsorbs and secretes substances along its tubule to regulate water, electrolytes, acid-base balance and wastes. Dialysis can replace selected filtration functions when kidneys fail but does not reproduce every endocrine and regulatory role of healthy kidneys. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Claim -> named evidence -> analysis -> qualification:

- Digestion converts food into absorbable molecules: salivary amylase begins starch digestion, pepsin works in acidic stomach conditions, bile emulsifies fat without acting as an enzyme, pancreatic and intestinal enzymes continue digestion, and the small intestine is the principal absorption site. Vitamins, minerals, protein and energy have different roles; deficiency, absorption failure and excess are separate nutritional problems.
- Blood contains plasma, red cells carrying haemoglobin, immune white cells and platelets involved in clotting; the four-chambered human heart supports pulmonary and systemic circuits. Arteries carry blood away from the heart and veins return it, so oxygenation is not the defining rule; ABO and Rh compatibility, gas transport and clotting must be treated as different functions.
- Ventilation moves air, alveolar gas exchange moves oxygen and carbon dioxide across thin surfaces, blood transports gases, and cellular respiration releases usable energy from nutrients. Haemoglobin binds oxygen while carbon monoxide can disrupt transport through stronger binding; breathing, gas exchange and ATP-producing respiration are linked but not synonymous.
- The nephron filters blood at the glomerulus and then selectively reabsorbs and secretes substances along its tubule to regulate water, electrolytes, acid-base balance and wastes. Dialysis can replace selected filtration functions when kidneys fail but does not reproduce every endocrine and regulatory role of healthy kidneys.

Qualified conclusion: Claim: Digestion absorption and nutrition. **Named evidence/example:** Digestion converts food into absorbable molecules: salivary amylase begins starch digestion, pepsin works in acidic stomach conditions, bile emulsifies fat without acting as an enzyme, pancreatic and intestinal enzymes continue digestion, and the small intestine is the principal absorption site. Vitamins, minerals, protein and energy have different roles; deficiency, absorption failure and excess are separate nutritional problems. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Blood heart and double circulation. **Named evidence/example:** Blood contains plasma, red cells carrying haemoglobin, immune white cells and platelets involved in clotting; the four-chambered human heart supports pulmonary and systemic circuits. Arteries carry blood away from the heart and veins return it, so oxygenation is not the defining rule; ABO and Rh compatibility, gas transport and clotting must be treated as different functions.

Analysis: This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Respiration gas exchange and cellular energy. **Named evidence/example:** Ventilation moves air, alveolar gas exchange moves oxygen and carbon dioxide across thin surfaces, blood transports gases, and cellular respiration releases usable energy from nutrients. Haemoglobin binds oxygen while carbon monoxide can disrupt transport through stronger binding; breathing, gas exchange and ATP-producing respiration are linked but not synonymous. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication.

Qualification: The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Kidney nephron and homeostasis. **Named evidence/example:** The nephron filters blood at the glomerulus and then selectively reabsorbs and secretes substances along its tubule to regulate water, electrolytes, acid-base balance and wastes. Dialysis can replace selected filtration functions when kidneys fail but does not reproduce every endocrine and regulatory role of healthy kidneys. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Demand decoding: The directive **analyse** requires a direct position on "Analyse digestion, circulation, respiration and excretion as an integrated physiological...", each clause, the exact technical mechanism, named Indian institution, dated status, application, risk and qualification.

Detailed examiner-grade model answer:

Introduction and thesis: **Claim:** Digestion absorption and nutrition. **Named evidence/example:** Digestion converts food into absorbable molecules: salivary amylase begins starch digestion, pepsin works in acidic stomach conditions, bile emulsifies fat without acting as an enzyme, pancreatic and intestinal enzymes continue digestion, and the small intestine is the principal absorption site. Vitamins, minerals, protein and energy have different roles; deficiency, absorption failure and excess are separate nutritional problems. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Blood heart and double circulation. **Named evidence/example:** Blood contains plasma, red cells carrying haemoglobin, immune white cells and platelets involved in clotting; the four-chambered human heart supports pulmonary and systemic circuits. Arteries carry blood away from the heart and veins return it, so oxygenation is not the defining rule; ABO and Rh compatibility, gas transport and clotting must be treated as different functions. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Respiration gas exchange and cellular energy. **Named evidence/example:** Ventilation moves air, alveolar gas exchange moves oxygen and carbon dioxide across thin surfaces, blood transports gases, and cellular respiration releases usable energy from nutrients. Haemoglobin binds oxygen while carbon monoxide can disrupt transport through stronger binding; breathing, gas exchange and ATP-producing respiration are linked but not synonymous. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Kidney nephron and homeostasis. **Named evidence/example:** The nephron filters blood at the glomerulus and then selectively reabsorbs and secretes substances along its tubule to regulate water, electrolytes, acid-base balance and wastes. Dialysis can replace selected filtration functions when kidneys fail but does not reproduce every endocrine and regulatory role of healthy kidneys. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Analytical body:

1. **Claim and named evidence:** Digestion converts food into absorbable molecules: salivary amylase begins starch digestion, pepsin works in acidic stomach conditions, bile emulsifies fat without acting as an enzyme, pancreatic and intestinal enzymes continue digestion, and the small intestine is the principal absorption site. Vitamins, minerals, protein and energy have different roles; deficiency, absorption failure and excess are separate nutritional problems. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
2. **Claim and named evidence:** Blood contains plasma, red cells carrying haemoglobin, immune white cells and platelets involved in clotting; the four-chambered human heart supports pulmonary and systemic circuits. Arteries carry blood away from the heart and veins return it, so oxygenation is not the defining rule; ABO and Rh compatibility, gas transport and clotting must be treated as different functions. **Analysis:** Connect mechanism ->

named system/institution -> application or implementation pathway -> public or strategic consequence.

Qualification: State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

3. **Claim and named evidence:** Ventilation moves air, alveolar gas exchange moves oxygen and carbon dioxide across thin surfaces, blood transports gases, and cellular respiration releases usable energy from nutrients. Haemoglobin binds oxygen while carbon monoxide can disrupt transport through stronger binding; breathing, gas exchange and ATP-producing respiration are linked but not synonymous. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence.

Qualification: State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

4. **Claim and named evidence:** The nephron filters blood at the glomerulus and then selectively reabsorbs and secretes substances along its tubule to regulate water, electrolytes, acid-base balance and wastes. Dialysis can replace selected filtration functions when kidneys fail but does not reproduce every endocrine and regulatory role of healthy kidneys. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

Counter-position / limit: A launch, test, target, budget, capacity, approval, contract, dataset, benchmark or prototype cannot alone establish deployment, induction, commercial operation, safety, inclusion or observed outcome; test mechanism, status, scale and evidence.

Qualified conclusion: Claim: Digestion absorption and nutrition. **Named evidence/example:** Digestion converts food into absorbable molecules: salivary amylase begins starch digestion, pepsin works in acidic stomach conditions, bile emulsifies fat without acting as an enzyme, pancreatic and intestinal enzymes continue digestion, and the small intestine is the principal absorption site. Vitamins, minerals, protein and energy have different roles; deficiency, absorption failure and excess are separate nutritional problems. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Blood heart and double circulation. **Named evidence/example:** Blood contains plasma, red cells carrying haemoglobin, immune white cells and platelets involved in clotting; the four-chambered human heart supports pulmonary and systemic circuits. Arteries carry blood away from the heart and veins return it, so oxygenation is not the defining rule; ABO and Rh compatibility, gas transport and clotting must be treated as different functions.

Analysis: This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Respiration gas exchange and cellular energy. **Named evidence/example:** Ventilation moves air, alveolar gas exchange moves oxygen and carbon dioxide across thin surfaces, blood transports gases, and cellular respiration releases usable energy from nutrients. Haemoglobin binds oxygen while carbon monoxide can disrupt transport through stronger binding; breathing, gas exchange and ATP-producing respiration are linked but not synonymous. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication.

Qualification: The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Kidney nephron and homeostasis. **Named evidence/example:** The nephron filters blood at the glomerulus and then selectively reabsorbs and secretes substances along its tubule to regulate water, electrolytes, acid-base balance and wastes. Dialysis can replace selected filtration functions when kidneys fail but does not reproduce every endocrine and regulatory role of healthy kidneys. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Executable exam-length answer / compression plan: For 15 marks, spend one-sixth of the time decoding the directive and drawing definition -> mechanism -> system/institution -> application/status -> risk/outcome; write four to seven claim -> named evidence -> analysis -> qualification points; reserve the final minute for unit, date, actor, platform, approval/test/deployment rung and residual-risk checks.

Why this earns marks: The answer obeys the directive, explains science rather than listing schemes, uses India-centric evidence and preserves technical, institutional, regulatory and maturity distinctions.

How to improve this answer: For "Analyse digestion, circulation, respiration and excretion as an integrated physiological...", replace the weakest catalogue point with one exact mechanism, named mission/institution/platform, dated status, measurable boundary, implementation constraint and answer-specific qualification.

ORIGINAL MAINS 4 - 15 MARKS

Question: Explain how nervous, endocrine and immune systems maintain regulated internal function. Answer in about 250 words.

Model thesis: Claim: Nervous endocrine and feedback control. **Named evidence/example:** Neurons transmit rapid electrical signals and communicate chemically at synapses, reflex arcs permit fast automatic responses, and the brain and spinal cord coordinate wider activity. Endocrine glands release hormones into blood for target-dependent effects that are often slower and longer-lasting; neural and hormonal control interact through feedback rather than operating as isolated systems. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Hormones metabolism and internal stability. **Named evidence/example:** Insulin lowers elevated blood glucose by promoting uptake and storage while glucagon supports its rise, thyroid hormones influence metabolism, parathyroid hormone helps regulate calcium, adrenal hormones mediate stress responses and pituitary signals coordinate several glands. Homeostasis is dynamic regulation around a range, not an unchanging value; hormone deficiency, resistance and gland failure are different mechanisms. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Innate adaptive and tolerance architecture. **Named evidence/example:** Innate immunity includes barriers, phagocytes, natural killer cells, complement, interferons, inflammation and fever; adaptive immunity uses antigen-specific B and T lymphocytes and can form memory. B-cell lineages produce antibodies, T cells perform coordinating, cytotoxic or regulatory roles, and peripheral immune tolerance including regulatory T cells limits attack on self; immunity is not antibodies alone. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Claim -> named evidence -> analysis -> qualification:

- Neurons transmit rapid electrical signals and communicate chemically at synapses, reflex arcs permit fast automatic responses, and the brain and spinal cord coordinate wider activity. Endocrine glands release hormones into blood for target-dependent effects that are often slower and longer-lasting; neural and hormonal control interact through feedback rather than operating as isolated systems.
- Insulin lowers elevated blood glucose by promoting uptake and storage while glucagon supports its rise, thyroid hormones influence metabolism, parathyroid hormone helps regulate calcium, adrenal hormones mediate stress responses and pituitary signals coordinate several glands. Homeostasis is dynamic regulation around a range, not an unchanging value; hormone deficiency, resistance and gland failure are different mechanisms.
- Innate immunity includes barriers, phagocytes, natural killer cells, complement, interferons, inflammation and fever; adaptive immunity uses antigen-specific B and T lymphocytes and can form memory. B-cell lineages produce antibodies, T cells perform coordinating, cytotoxic or regulatory roles, and peripheral immune tolerance including regulatory T cells limits attack on self; immunity is not antibodies alone.

Qualified conclusion: Claim: Nervous endocrine and feedback control. **Named evidence/example:** Neurons transmit rapid electrical signals and communicate chemically at synapses, reflex arcs permit fast automatic responses, and the brain and spinal cord coordinate wider activity. Endocrine glands release hormones into blood for target-dependent effects that are often slower and longer-lasting; neural and hormonal control interact through feedback rather than operating as isolated systems. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Hormones metabolism and internal stability. **Named evidence/example:** Insulin lowers elevated blood glucose by promoting uptake and storage while glucagon supports its rise, thyroid hormones influence metabolism, parathyroid hormone helps regulate calcium, adrenal hormones mediate stress responses and pituitary signals coordinate several glands. Homeostasis is dynamic regulation around a range, not an unchanging value;

hormone deficiency, resistance and gland failure are different mechanisms. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Innate adaptive and tolerance architecture. **Named evidence/example:** Innate immunity includes barriers, phagocytes, natural killer cells, complement, interferons, inflammation and fever; adaptive immunity uses antigen-specific B and T lymphocytes and can form memory. B-cell lineages produce antibodies, T cells perform coordinating, cytotoxic or regulatory roles, and peripheral immune tolerance including regulatory T cells limits attack on self; immunity is not antibodies alone. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Demand decoding: The directive **explain** requires a direct position on "Explain how nervous, endocrine and immune systems maintain regulated internal function....", each clause, the exact technical mechanism, named Indian institution, dated status, application, risk and qualification.

Detailed examiner-grade model answer:

Introduction and thesis: Claim: Nervous endocrine and feedback control. **Named evidence/example:** Neurons transmit rapid electrical signals and communicate chemically at synapses, reflex arcs permit fast automatic responses, and the brain and spinal cord coordinate wider activity. Endocrine glands release hormones into blood for target-dependent effects that are often slower and longer-lasting; neural and hormonal control interact through feedback rather than operating as isolated systems. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Hormones metabolism and internal stability. **Named evidence/example:** Insulin lowers elevated blood glucose by promoting uptake and storage while glucagon supports its rise, thyroid hormones influence metabolism, parathyroid hormone helps regulate calcium, adrenal hormones mediate stress responses and pituitary signals coordinate several glands. Homeostasis is dynamic regulation around a range, not an unchanging value; hormone deficiency, resistance and gland failure are different mechanisms. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Innate adaptive and tolerance architecture. **Named evidence/example:** Innate immunity includes barriers, phagocytes, natural killer cells, complement, interferons, inflammation and fever; adaptive immunity uses antigen-specific B and T lymphocytes and can form memory. B-cell lineages produce antibodies, T cells perform coordinating, cytotoxic or regulatory roles, and peripheral immune tolerance including regulatory T cells limits attack on self; immunity is not antibodies alone. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Analytical body:

1. **Claim and named evidence:** Neurons transmit rapid electrical signals and communicate chemically at synapses, reflex arcs permit fast automatic responses, and the brain and spinal cord coordinate wider activity. Endocrine glands release hormones into blood for target-dependent effects that are often slower and longer-lasting; neural and hormonal control interact through feedback rather than operating as isolated systems. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
2. **Claim and named evidence:** Insulin lowers elevated blood glucose by promoting uptake and storage while glucagon supports its rise, thyroid hormones influence metabolism, parathyroid hormone helps regulate calcium, adrenal hormones mediate stress responses and pituitary signals coordinate several glands. Homeostasis is dynamic regulation around a range, not an unchanging value; hormone deficiency, resistance and gland failure are different mechanisms. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit,

source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

3. **Claim and named evidence:** Innate immunity includes barriers, phagocytes, natural killer cells, complement, interferons, inflammation and fever; adaptive immunity uses antigen-specific B and T lymphocytes and can form memory. B-cell lineages produce antibodies, T cells perform coordinating, cytotoxic or regulatory roles, and peripheral immune tolerance including regulatory T cells limits attack on self; immunity is not antibodies alone.

Analysis: Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

Counter-position / limit: A launch, test, target, budget, capacity, approval, contract, dataset, benchmark or prototype cannot alone establish deployment, induction, commercial operation, safety, inclusion or observed outcome; test mechanism, status, scale and evidence.

Qualified conclusion: Claim: Nervous endocrine and feedback control. **Named evidence/example:** Neurons transmit rapid electrical signals and communicate chemically at synapses, reflex arcs permit fast automatic responses, and the brain and spinal cord coordinate wider activity. Endocrine glands release hormones into blood for target-dependent effects that are often slower and longer-lasting; neural and hormonal control interact through feedback rather than operating as isolated systems. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Hormones metabolism and internal stability. **Named evidence/example:** Insulin lowers elevated blood glucose by promoting uptake and storage while glucagon supports its rise, thyroid hormones influence metabolism, parathyroid hormone helps regulate calcium, adrenal hormones mediate stress responses and pituitary signals coordinate several glands. Homeostasis is dynamic regulation around a range, not an unchanging value; hormone deficiency, resistance and gland failure are different mechanisms. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Innate adaptive and tolerance architecture. **Named evidence/example:** Innate immunity includes barriers, phagocytes, natural killer cells, complement, interferons, inflammation and fever; adaptive immunity uses antigen-specific B and T lymphocytes and can form memory. B-cell lineages produce antibodies, T cells perform coordinating, cytotoxic or regulatory roles, and peripheral immune tolerance including regulatory T cells limits attack on self; immunity is not antibodies alone. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Executable exam-length answer / compression plan: For 15 marks, spend one-sixth of the time decoding the directive and drawing definition -> mechanism -> system/institution -> application/status -> risk/outcome; write four to seven claim -> named evidence -> analysis -> qualification points; reserve the final minute for unit, date, actor, platform, approval/test/deployment rung and residual-risk checks.

Why this earns marks: The answer obeys the directive, explains science rather than listing schemes, uses India-centric evidence and preserves technical, institutional, regulatory and maturity distinctions.

How to improve this answer: For "Explain how nervous, endocrine and immune systems maintain regulated internal function....", replace the weakest catalogue point with one exact mechanism, named mission/institution/platform, dated status, measurable boundary, implementation constraint and answer-specific qualification.

ORIGINAL MAINS 5 - 20 MARKS

Question: Examine disease transmission, antimicrobial resistance, biofilms and microbiome interactions through a One Health and public-health lens. Answer in about 300 words.

Model thesis: **Claim:** Pathogen transmission vector and zoonosis map. **Named evidence/example:** Bacteria are cellular organisms, viruses are acellular and depend on host-cell machinery, fungi are eukaryotes, protozoa are single-celled eukaryotes and helminths are multicellular parasites. Transmission may be respiratory, faeco-oral, blood-borne, sexual, vector-borne, fomite-linked or zoonotic; a vector such as Aedes, Anopheles, Culex or sandfly carries a pathogen but is not the disease itself. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Antibiotic antiviral and AMR boundary. **Named evidence/example:** Antibiotics target susceptible bacteria rather than viruses, while antivirals interfere with virus-specific replication stages. Antimicrobial resistance is selection and spread of resistant microorganisms under pressures including misuse, incomplete treatment, agricultural use and environmental residues; the microbe or population becomes resistant, not the patient's body. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Biofilms microbiome probiotics and extremophiles. **Named evidence/example:** A biofilm is a surface-associated microbial community embedded in a matrix that can alter persistence and treatment response; the human microbiome contributes to digestion, vitamin production, immune development and pathogen exclusion. Probiotics are live microorganisms intended to confer a benefit in an adequate amount, while thermophiles, psychrophiles and acidophiles are classified by environmental tolerance; none of these labels proves safety or effectiveness in every context. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Health institution and evidence-status firewall. **Named evidence/example:** ICMR performs and supports biomedical research, NCDC and the IDSP or IHIP network support surveillance and outbreak response under MoHFW, State systems deliver programmes, and CDSCO regulates drugs and vaccines; DBT, CSIR-CCMB and CSIR-IGIB occupy different research roles. A biological mechanism, association, surveillance signal, WHO validation, elimination as a public-health problem, regulatory approval and eradication are separate evidence rungs. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Claim -> named evidence -> analysis -> qualification:

- Bacteria are cellular organisms, viruses are acellular and depend on host-cell machinery, fungi are eukaryotes, protozoa are single-celled eukaryotes and helminths are multicellular parasites. Transmission may be respiratory, faeco-oral, blood-borne, sexual, vector-borne, fomite-linked or zoonotic; a vector such as Aedes, Anopheles, Culex or sandfly carries a pathogen but is not the disease itself.
- Antibiotics target susceptible bacteria rather than viruses, while antivirals interfere with virus-specific replication stages. Antimicrobial resistance is selection and spread of resistant microorganisms under pressures including misuse, incomplete treatment, agricultural use and environmental residues; the microbe or population becomes resistant, not the patient's body.
- A biofilm is a surface-associated microbial community embedded in a matrix that can alter persistence and treatment response; the human microbiome contributes to digestion, vitamin production, immune development and pathogen exclusion. Probiotics are live microorganisms intended to confer a benefit in an adequate amount, while thermophiles, psychrophiles and acidophiles are classified by environmental tolerance; none of these labels proves safety or effectiveness in every context.
- ICMR performs and supports biomedical research, NCDC and the IDSP or IHIP network support surveillance and outbreak response under MoHFW, State systems deliver programmes, and CDSCO regulates drugs and vaccines; DBT, CSIR-CCMB and CSIR-IGIB occupy different research roles. A biological mechanism, association, surveillance signal, WHO validation, elimination as a public-health problem, regulatory approval and eradication are separate evidence rungs.

Qualified conclusion: Claim: Pathogen transmission vector and zoonosis map. **Named evidence/example:** Bacteria are cellular organisms, viruses are acellular and depend on host-cell machinery, fungi are eukaryotes, protozoa are single-celled eukaryotes and helminths are multicellular parasites. Transmission may be respiratory, faeco-oral, blood-borne, sexual, vector-borne, fomite-linked or zoonotic; a vector such as Aedes, Anopheles, Culex or sandfly carries a pathogen but is not the disease itself. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Antibiotic antiviral and AMR boundary. **Named evidence/example:** Antibiotics target susceptible bacteria rather than viruses, while antivirals interfere with virus-specific replication stages. Antimicrobial resistance is selection and spread of resistant microorganisms under pressures including misuse, incomplete treatment, agricultural use and environmental residues; the microbe or population becomes resistant, not the patient's body. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Biofilms microbiome probiotics and extremophiles. **Named evidence/example:** A biofilm is a surface-associated microbial community embedded in a matrix that can alter persistence and treatment response; the human microbiome contributes to digestion, vitamin production, immune development and pathogen exclusion. Probiotics are live microorganisms intended to confer a benefit in an adequate amount, while thermophiles, psychrophiles and acidophiles are classified by environmental tolerance; none of these labels proves safety or effectiveness in every context. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Health institution and evidence-status firewall. **Named evidence/example:** ICMR performs and supports biomedical research, NCDC and the IDSP or IHIP network support surveillance and outbreak response under MoHFW, State systems deliver programmes, and CDSCO regulates drugs and vaccines; DBT, CSIR-CCMB and CSIR-IGIB occupy different research roles. A biological mechanism, association, surveillance signal, WHO validation, elimination as a public-health problem, regulatory approval and eradication are separate evidence rungs. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Demand decoding: The directive **examine** requires a direct position on "Examine disease transmission, antimicrobial resistance, biofilms and microbiome interactions...", each clause, the exact technical mechanism, named Indian institution, dated status, application, risk and qualification.

Detailed examiner-grade model answer:

Introduction and thesis: Claim: Pathogen transmission vector and zoonosis map. **Named evidence/example:** Bacteria are cellular organisms, viruses are acellular and depend on host-cell machinery, fungi are eukaryotes, protozoa are single-celled eukaryotes and helminths are multicellular parasites. Transmission may be respiratory, faeco-oral, blood-borne, sexual, vector-borne, fomite-linked or zoonotic; a vector such as Aedes, Anopheles, Culex or sandfly carries a pathogen but is not the disease itself. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Antibiotic antiviral and AMR boundary. **Named evidence/example:** Antibiotics target susceptible bacteria rather than viruses, while antivirals interfere with virus-specific replication stages. Antimicrobial resistance is selection and spread of resistant microorganisms under pressures including misuse, incomplete treatment, agricultural use and environmental residues; the microbe or population becomes resistant, not the patient's body. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Biofilms microbiome probiotics and extremophiles. **Named evidence/example:** A biofilm is a surface-associated microbial community embedded in a matrix that can alter persistence and treatment response; the human microbiome contributes to digestion, vitamin production, immune development and pathogen exclusion. Probiotics are live microorganisms intended to confer a benefit in an adequate amount, while thermophiles, psychrophiles and acidophiles are classified by environmental tolerance; none of these labels proves safety or effectiveness in every context. **Analysis:** This fixes the scientific

process, system boundary, institution, application and verified capability rung before drawing the policy implication.

Qualification: The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Health institution and evidence-status firewall. **Named**

evidence/example: ICMR performs and supports biomedical research, NCDC and the IDSP or IHIP network support surveillance and outbreak response under MoHFW, State systems deliver programmes, and CDSCO regulates drugs and vaccines; DBT, CSIR-CCMB and CSIR-IGIB occupy different research roles. A biological mechanism, association, surveillance signal, WHO validation, elimination as a public-health problem, regulatory approval and eradication are separate evidence rungs. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Analytical body:

1. **Claim and named evidence:** Bacteria are cellular organisms, viruses are acellular and depend on host-cell machinery, fungi are eukaryotes, protozoa are single-celled eukaryotes and helminths are multicellular parasites. Transmission may be respiratory, faeco-oral, blood-borne, sexual, vector-borne, fomite-linked or zoonotic; a vector such as Aedes, Anopheles, Culex or sandfly carries a pathogen but is not the disease itself. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
2. **Claim and named evidence:** Antibiotics target susceptible bacteria rather than viruses, while antivirals interfere with virus-specific replication stages. Antimicrobial resistance is selection and spread of resistant microorganisms under pressures including misuse, incomplete treatment, agricultural use and environmental residues; the microbe or population becomes resistant, not the patient's body. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
3. **Claim and named evidence:** A biofilm is a surface-associated microbial community embedded in a matrix that can alter persistence and treatment response; the human microbiome contributes to digestion, vitamin production, immune development and pathogen exclusion. Probiotics are live microorganisms intended to confer a benefit in an adequate amount, while thermophiles, psychrophiles and acidophiles are classified by environmental tolerance; none of these labels proves safety or effectiveness in every context. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
4. **Claim and named evidence:** ICMR performs and supports biomedical research, NCDC and the IDSP or IHIP network support surveillance and outbreak response under MoHFW, State systems deliver programmes, and CDSCO regulates drugs and vaccines; DBT, CSIR-CCMB and CSIR-IGIB occupy different research roles. A biological mechanism, association, surveillance signal, WHO validation, elimination as a public-health problem, regulatory approval and eradication are separate evidence rungs. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

Counter-position / limit: A launch, test, target, budget, capacity, approval, contract, dataset, benchmark or prototype cannot alone establish deployment, induction, commercial operation, safety, inclusion or observed outcome; test mechanism, status, scale and evidence.

Qualified conclusion: **Claim:** Pathogen transmission vector and zoonosis map. **Named evidence/example:**

Bacteria are cellular organisms, viruses are acellular and depend on host-cell machinery, fungi are eukaryotes, protozoa are single-celled eukaryotes and helminths are multicellular parasites. Transmission may be respiratory, faeco-oral, blood-borne, sexual, vector-borne, fomite-linked or zoonotic; a vector such as Aedes, Anopheles, Culex or sandfly carries a pathogen but is not the disease itself. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Antibiotic antiviral and AMR boundary. **Named evidence/example:** Antibiotics target susceptible bacteria rather than viruses, while antivirals interfere with virus-specific replication stages. Antimicrobial resistance is selection and spread of resistant microorganisms under pressures including misuse, incomplete treatment, agricultural use and environmental residues; the microbe or population becomes resistant, not the patient's body. **Analysis:** This

fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Biofilms microbiome probiotics and extremophiles. **Named evidence/example:** A biofilm is a surface-associated microbial community embedded in a matrix that can alter persistence and treatment response; the human microbiome contributes to digestion, vitamin production, immune development and pathogen exclusion. Probiotics are live microorganisms intended to confer a benefit in an adequate amount, while thermophiles, psychrophiles and acidophiles are classified by environmental tolerance; none of these labels proves safety or effectiveness in every context. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Health institution and evidence-status firewall. **Named evidence/example:** ICMR performs and supports biomedical research, NCDC and the IDSP or IHIP network support surveillance and outbreak response under MoHFW, State systems deliver programmes, and CDSCO regulates drugs and vaccines; DBT, CSIR-CCMB and CSIR-IGIB occupy different research roles. A biological mechanism, association, surveillance signal, WHO validation, elimination as a public-health problem, regulatory approval and eradication are separate evidence rungs. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Executable exam-length answer / compression plan: For 20 marks, spend one-sixth of the time decoding the directive and drawing definition -> mechanism -> system/institution -> application/status -> risk/outcome; write four to seven claim -> named evidence -> analysis -> qualification points; reserve the final minute for unit, date, actor, platform, approval/test/deployment rung and residual-risk checks.

Why this earns marks: The answer obeys the directive, explains science rather than listing schemes, uses India-centric evidence and preserves technical, institutional, regulatory and maturity distinctions.

How to improve this answer: For "Examine disease transmission, antimicrobial resistance, biofilms and microbiome interactions...", replace the weakest catalogue point with one exact mechanism, named mission/institution/platform, dated status, measurable boundary, implementation constraint and answer-specific qualification.

ORIGINAL MAINS 6 - 20 MARKS

Question: Discuss plant physiology, organism classification and cellulose decomposition while preserving biotechnology and vaccine owner boundaries. Answer in about 300 words.

Model thesis: Claim: Plant physiology transport and stress response. **Named evidence/example:** Photosynthesis stores light energy in chemical form while plants also respire continuously; xylem conducts water and minerals and phloem translocates organic food. Stomata regulate gas exchange and water loss, transpiration contributes to upward water movement, and auxin, gibberellin, cytokinin, abscisic acid and ethylene have distinct growth or stress roles; C3, C4 and CAM are different carbon-fixation strategies. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Routed taxonomy and organism-form distinctions. **Named evidence/example:** Groundnut, horse-gram and soybean are routed pea-family examples; cicada, frog hopper and pond skater are insects; poisonous species occur in multiple animal groups, so a broad group name does not prove toxicity. Cassava is a woody shrub, ginger a rhizomatous herb, Malabar spinach a climbing vine, mint a herb and papaya a large herb; mushrooms can have species-specific edible, medicinal, psychoactive, insect-pathogenic or bioluminescent properties. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Cellulose decomposition route. **Named evidence/example:** Cellulose is a structural plant polysaccharide; decomposer fungi and bacteria secrete cellulases that hydrolyse it into smaller sugars, after which microbial metabolism returns carbon through soil and food-web processes. Moisture, temperature, oxygen, lignin and community composition constrain the rate, and natural decomposition is not proof of an industrially viable biofuel process. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment

status boundary.

Claim -> named evidence -> analysis -> qualification:

- Photosynthesis stores light energy in chemical form while plants also respire continuously; xylem conducts water and minerals and phloem translocates organic food. Stomata regulate gas exchange and water loss, transpiration contributes to upward water movement, and auxin, gibberellin, cytokinin, abscisic acid and ethylene have distinct growth or stress roles; C3, C4 and CAM are different carbon-fixation strategies.
- Groundnut, horse-gram and soybean are routed pea-family examples; cicada, frog hopper and pond skater are insects; poisonous species occur in multiple animal groups, so a broad group name does not prove toxicity. Cassava is a woody shrub, ginger a rhizomatous herb, Malabar spinach a climbing vine, mint a herb and papaya a large herb; mushrooms can have species-specific edible, medicinal, psychoactive, insect-pathogenic or bioluminescent properties.
- Cellulose is a structural plant polysaccharide; decomposer fungi and bacteria secrete cellulases that hydrolyse it into smaller sugars, after which microbial metabolism returns carbon through soil and food-web processes. Moisture, temperature, oxygen, lignin and community composition constrain the rate, and natural decomposition is not proof of an industrially viable biofuel process.

Qualified conclusion: Claim: Plant physiology transport and stress response. **Named evidence/example:**

Photosynthesis stores light energy in chemical form while plants also respire continuously; xylem conducts water and minerals and phloem translocates organic food. Stomata regulate gas exchange and water loss, transpiration contributes to upward water movement, and auxin, gibberellin, cytokinin, abscisic acid and ethylene have distinct growth or stress roles; C3, C4 and CAM are different carbon-fixation strategies. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication.

Qualification: The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Claim: Routed taxonomy and organism-form distinctions. **Named**

evidence/example: Groundnut, horse-gram and soybean are routed pea-family examples; cicada, frog hopper and pond skater are insects; poisonous species occur in multiple animal groups, so a broad group name does not prove toxicity. Cassava is a woody shrub, ginger a rhizomatous herb, Malabar spinach a climbing vine, mint a herb and papaya a large herb; mushrooms can have species-specific edible, medicinal, psychoactive, insect-pathogenic or bioluminescent properties. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication.

Qualification: The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Cellulose decomposition route. **Named evidence/example:** Cellulose is a structural plant polysaccharide; decomposer fungi and bacteria secrete cellulases that hydrolyse it into smaller sugars, after which microbial metabolism returns carbon through soil and food-web processes. Moisture, temperature, oxygen, lignin and community composition constrain the rate, and natural decomposition is not proof of an industrially viable biofuel process. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication.

Qualification: The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Demand decoding: The directive **discuss** requires a direct position on "Discuss plant physiology, organism classification and cellulose decomposition while...", each clause, the exact technical mechanism, named Indian institution, dated status, application, risk and qualification.

Detailed examiner-grade model answer:

Introduction and thesis: Claim: Plant physiology transport and stress response. **Named evidence/example:**

Photosynthesis stores light energy in chemical form while plants also respire continuously; xylem conducts water and minerals and phloem translocates organic food. Stomata regulate gas exchange and water loss, transpiration contributes to upward water movement, and auxin, gibberellin, cytokinin, abscisic acid and ethylene have distinct growth or stress roles; C3, C4 and CAM are different carbon-fixation strategies. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication.

Qualification: The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Routed taxonomy and organism-form distinctions. **Named**

evidence/example: Groundnut, horse-gram and soybean are routed pea-family examples; cicada, frog hopper and pond skater are insects; poisonous species occur in multiple animal groups, so a broad group name does not prove

toxicity. Cassava is a woody shrub, ginger a rhizomatous herb, Malabar spinach a climbing vine, mint a herb and papaya a large herb; mushrooms can have species-specific edible, medicinal, psychoactive, insect-pathogenic or bioluminescent properties. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Cellulose decomposition route. **Named evidence/example:** Cellulose is a structural plant polysaccharide; decomposer fungi and bacteria secrete cellulases that hydrolyse it into smaller sugars, after which microbial metabolism returns carbon through soil and food-web processes. Moisture, temperature, oxygen, lignin and community composition constrain the rate, and natural decomposition is not proof of an industrially viable biofuel process. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Analytical body:

1. **Claim and named evidence:** Photosynthesis stores light energy in chemical form while plants also respire continuously; xylem conducts water and minerals and phloem translocates organic food. Stomata regulate gas exchange and water loss, transpiration contributes to upward water movement, and auxin, gibberellin, cytokinin, abscisic acid and ethylene have distinct growth or stress roles; C3, C4 and CAM are different carbon-fixation strategies. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
2. **Claim and named evidence:** Groundnut, horse-gram and soybean are routed pea-family examples; cicada, frog hopper and pond skater are insects; poisonous species occur in multiple animal groups, so a broad group name does not prove toxicity. Cassava is a woody shrub, ginger a rhizomatous herb, Malabar spinach a climbing vine, mint a herb and papaya a large herb; mushrooms can have species-specific edible, medicinal, psychoactive, insect-pathogenic or bioluminescent properties. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
3. **Claim and named evidence:** Cellulose is a structural plant polysaccharide; decomposer fungi and bacteria secrete cellulases that hydrolyse it into smaller sugars, after which microbial metabolism returns carbon through soil and food-web processes. Moisture, temperature, oxygen, lignin and community composition constrain the rate, and natural decomposition is not proof of an industrially viable biofuel process. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

Counter-position / limit: A launch, test, target, budget, capacity, approval, contract, dataset, benchmark or prototype cannot alone establish deployment, induction, commercial operation, safety, inclusion or observed outcome; test mechanism, status, scale and evidence.

Qualified conclusion: **Claim:** Plant physiology transport and stress response. **Named evidence/example:** Photosynthesis stores light energy in chemical form while plants also respire continuously; xylem conducts water and minerals and phloem translocates organic food. Stomata regulate gas exchange and water loss, transpiration contributes to upward water movement, and auxin, gibberellin, cytokinin, abscisic acid and ethylene have distinct growth or stress roles; C3, C4 and CAM are different carbon-fixation strategies. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Routed taxonomy and organism-form distinctions. **Named evidence/example:** Groundnut, horse-gram and soybean are routed pea-family examples; cicada, frog hopper and pond skater are insects; poisonous species occur in multiple animal groups, so a broad group name does not prove toxicity. Cassava is a woody shrub, ginger a rhizomatous herb, Malabar spinach a climbing vine, mint a herb and papaya a large herb; mushrooms can have species-specific edible, medicinal, psychoactive, insect-pathogenic or bioluminescent properties. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Cellulose

decomposition route. **Named evidence/example:** Cellulose is a structural plant polysaccharide; decomposer fungi and bacteria secrete cellulases that hydrolyse it into smaller sugars, after which microbial metabolism returns carbon through soil and food-web processes. Moisture, temperature, oxygen, lignin and community composition constrain the rate, and natural decomposition is not proof of an industrially viable biofuel process. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Executable exam-length answer / compression plan: For 20 marks, spend one-sixth of the time decoding the directive and drawing definition -> mechanism -> system/institution -> application/status -> risk/outcome; write four to seven claim -> named evidence -> analysis -> qualification points; reserve the final minute for unit, date, actor, platform, approval/test/deployment rung and residual-risk checks.

Why this earns marks: The answer obeys the directive, explains science rather than listing schemes, uses India-centric evidence and preserves technical, institutional, regulatory and maturity distinctions.

How to improve this answer: For "Discuss plant physiology, organism classification and cellulose decomposition while...", replace the weakest catalogue point with one exact mechanism, named mission/institution/platform, dated status, measurable boundary, implementation constraint and answer-specific qualification.